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(45) **Date of Patent:** Sep. 9, 2025

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|--------------|------|--------|---------------|-------------------------|
| 6,988,425 | B2 | 1/2006 | Nagai et al. | |
| 10,321,964 | B2 | 6/2019 | Grover et al. | |
| 2004/0193146 | A1 | 9/2004 | Lee et al. | |
| 2007/0032742 | A1 * | 2/2007 | Monson | A61B 10/0266
600/568 |

(Continued)

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FOREIGN PATENT DOCUMENTS

- | | | | |
|----|------------|----|---------|
| WO | 2015167808 | A1 | 11/2015 |
| WO | 2016043845 | A1 | 3/2016 |

(Continued)

- (*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 874 days.

OTHER PUBLICATIONS

- Written Opinion and International Search Report from corresponding PCT Application No. PCT/EP2021/066792 mailed Feb. 7, 2022.

(Continued)

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(74) *Attorney, Agent, or Firm* — Vorys, Sater, Seymour
and Pease LLP

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A61B 17/00 (2006.01)
B25J 19/06 (2006.01)

- (52) **U.S. Cl.**
CPC *A61B 17/00* (2013.01); *B25J 19/068*
(2013.01)

- (58) **Field of Classification Search**
CPC A61B 17/00
USPC 606/1
See application file for complete search history.

(57) **ABSTRACT**

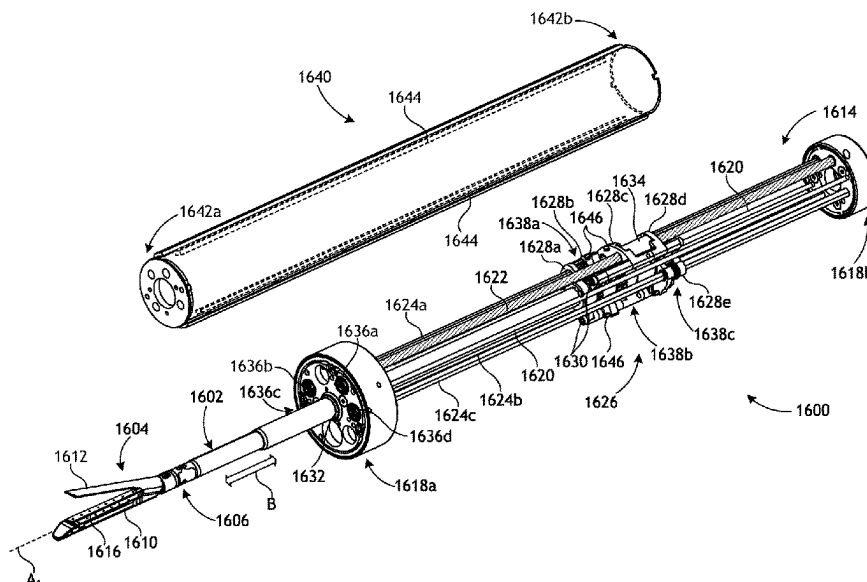
A surgical tool includes a handle having opposing first and second ends, a lead screw and a spline are rotatably coupled to and extend between the opposing ends. The spline is arranged a distance from the lead screw such that the lead screw assumes torsional loading generated by the spline. Another surgical tool includes first and second splines with a first activating mechanism coupled to the first spline and a second activating mechanism coupled to the second spline. Torsional loading generated by the first and second splines to operate the first and second activating mechanisms are balanced by rotating the first and second splines in opposite directions. A method of minimizing torsional loads on components of a robotic surgical tool includes actuating the surgical tool and assuming a torsional force acting on a carriage of the surgical tool with a load balancing member included in the surgical tool.

- (56)
- References Cited**

U.S. PATENT DOCUMENTS

- 6,010,476 A * 1/2000 Saadat A61B 17/3207
606/180
6,264,665 B1 7/2001 Yu et al.

18 Claims, 26 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2007/0179401	A1 *	8/2007	Hibner	A61B 10/0275	600/568
2008/0027423	A1 *	1/2008	Choi	A61B 17/2202	606/40
2012/0071895	A1 *	3/2012	Stahler	A61B 34/20	606/130
2012/0179215	A1 *	7/2012	Soubeiran	A61B 17/7216	606/86 R
2012/0215220	A1 *	8/2012	Manzo	A61B 34/30	606/46
2012/0330314	A1 *	12/2012	Schaller	A61B 17/16	606/79
2013/0296886	A1	11/2013	Green et al.		
2014/0005718	A1	1/2014	Shelton, IV et al.		
2014/0276594	A1 *	9/2014	Tanner	A61B 90/06	604/95.04
2014/0277333	A1 *	9/2014	Lewis	A61B 34/30	623/1.11
2014/0309666	A1	10/2014	Shelton, IV et al.		
2015/0119637	A1	4/2015	Alvarez et al.		
2015/0173755	A1 *	6/2015	Baxter, III	A61B 17/07207	227/180.1
2015/0313674	A1 *	11/2015	DeFreitas	A61B 90/11	600/564
2017/0072561	A1	3/2017	Schlegel et al.		
2018/0116741	A1 *	5/2018	Garcia Kilroy	B25J 13/085	
2018/0177473	A1 *	6/2018	Gregerson	A61B 6/0407	
2018/0228557	A1	8/2018	Darisse et al.		
2019/0142537	A1 *	5/2019	Covington	A61B 34/30	606/108
2019/0261991	A1 *	8/2019	Beckman	A61B 17/1155	
2019/0298464	A1	10/2019	Abbott		
2020/0107898	A1 *	4/2020	Kim	A61B 1/018	
2020/0315730	A1 *	10/2020	Zemlok	B25J 9/0021	
2020/0405375	A1 *	12/2020	Shelton, IV	A61B 18/1815	
2020/0405403	A1 *	12/2020	Shelton, IV	A61B 17/3421	
2020/0405405	A1 *	12/2020	Shelton, IV	A61B 34/20	
2021/0015572	A1	1/2021	Gomez et al.		
2021/0022815	A1	1/2021	Abbott		
2021/0106421	A1 *	4/2021	Montgomery	A61F 2/2418	
2021/0169457	A1 *	6/2021	Traina	A61B 34/35	
2021/0177528	A1 *	6/2021	Cappelleri	A61B 34/30	
2021/0393341	A1 *	12/2021	Beckman	A61B 34/30	
2021/0393342	A1	12/2021	Beckman et al.		
2021/0393345	A1 *	12/2021	Beckman	A61B 1/00133	
2021/0393346	A1	12/2021	Beckman et al.		
2021/0393354	A1 *	12/2021	Beckman	A61B 34/74	
2021/0393355	A1 *	12/2021	Beckman	A61B 34/30	
2021/0393357	A1 *	12/2021	Beckman	A61B 34/71	
2022/0250242	A1 *	8/2022	Suresh	B25J 9/1689	
2022/0273396	A1	9/2022	Bozung et al.		

FOREIGN PATENT DOCUMENTS

WO	2018053305	A1	3/2018
WO	2019191420	A1	10/2019
WO	2021255276	A1	12/2021

OTHER PUBLICATIONS

CN Search Report pertaining to CN Application No. 2021800443648;
Date of Mailing: May 7, 2025.

* cited by examiner

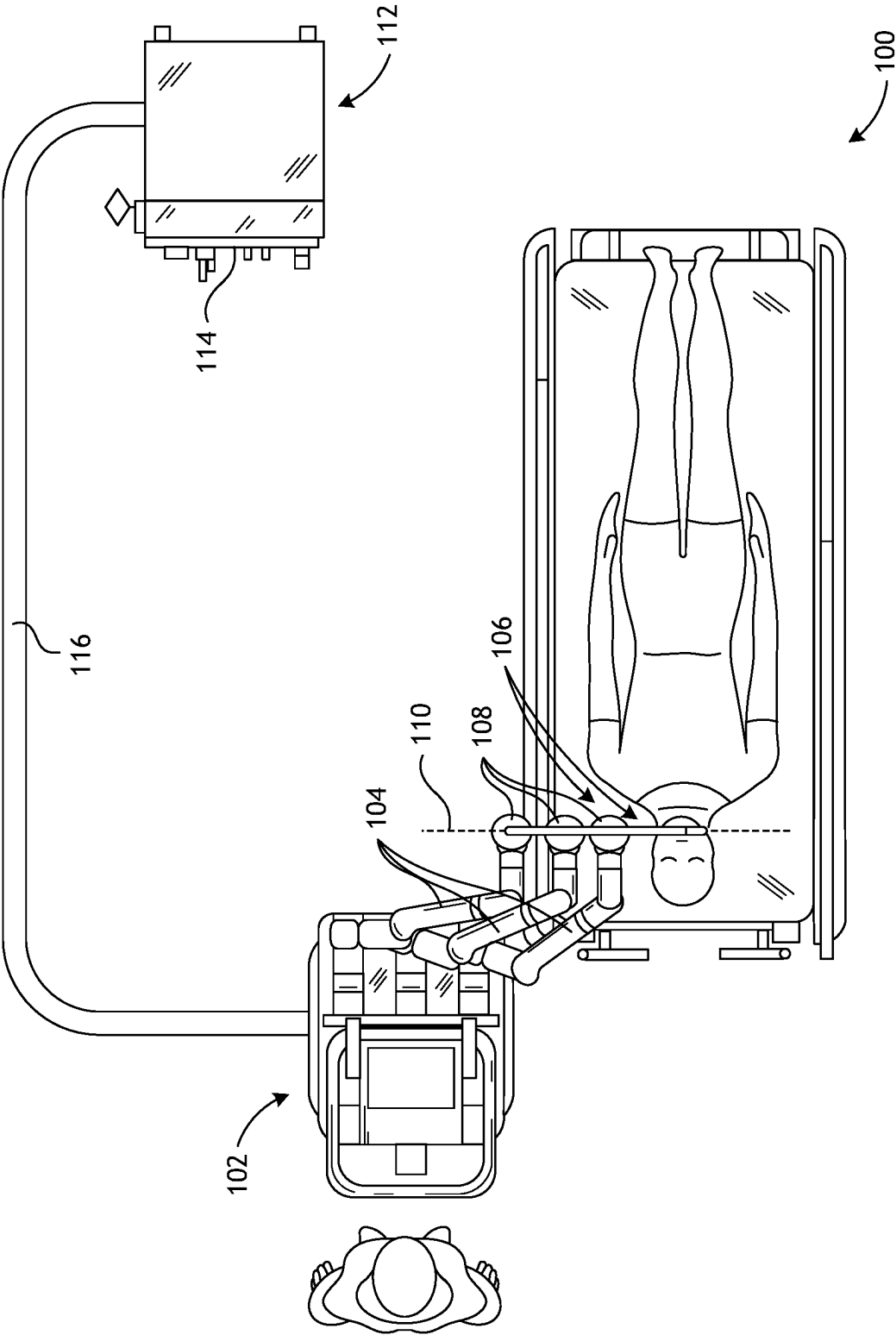


FIG. 1

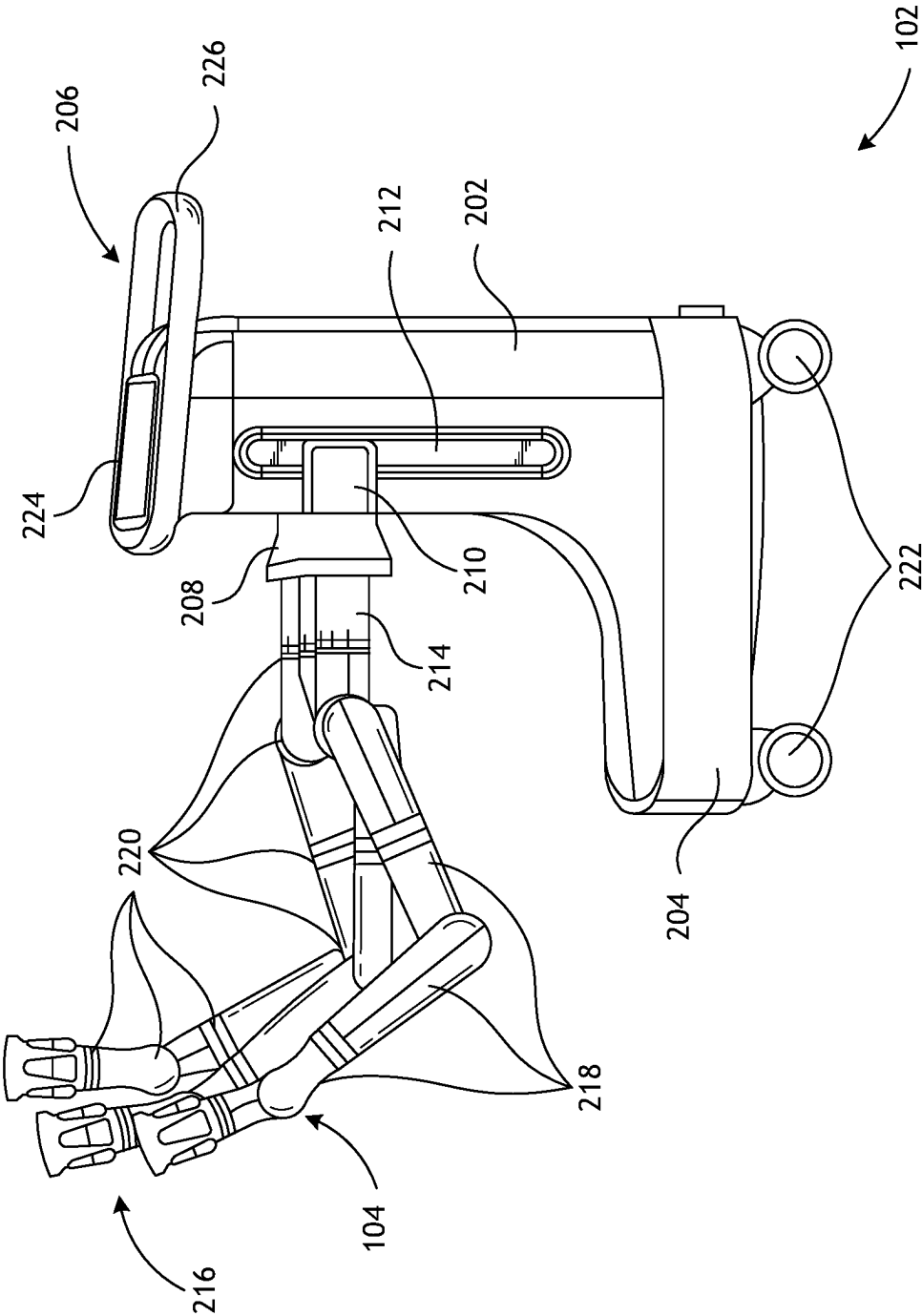


FIG. 2

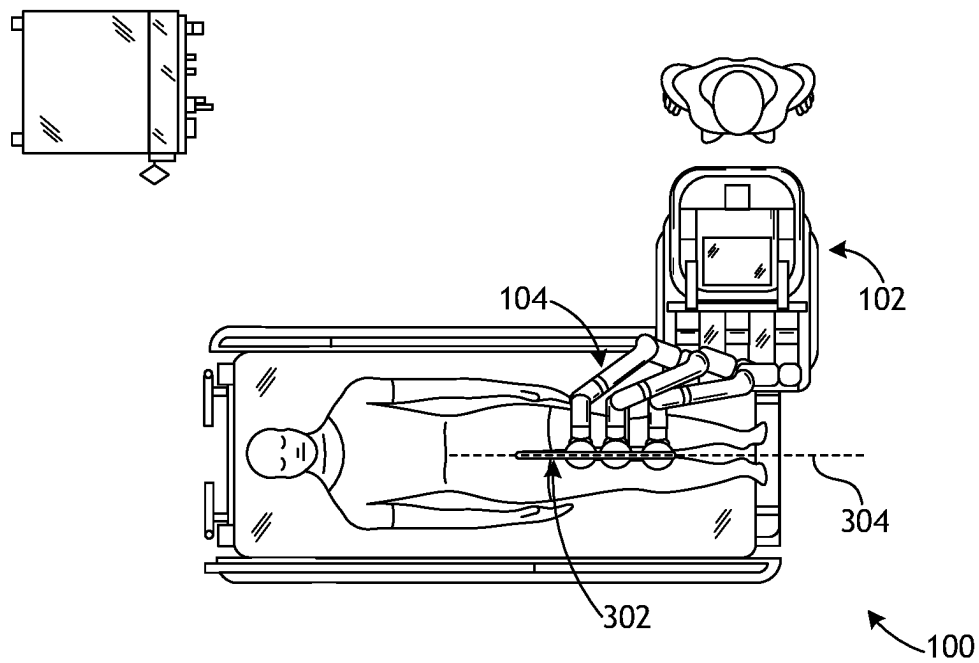


FIG. 3A

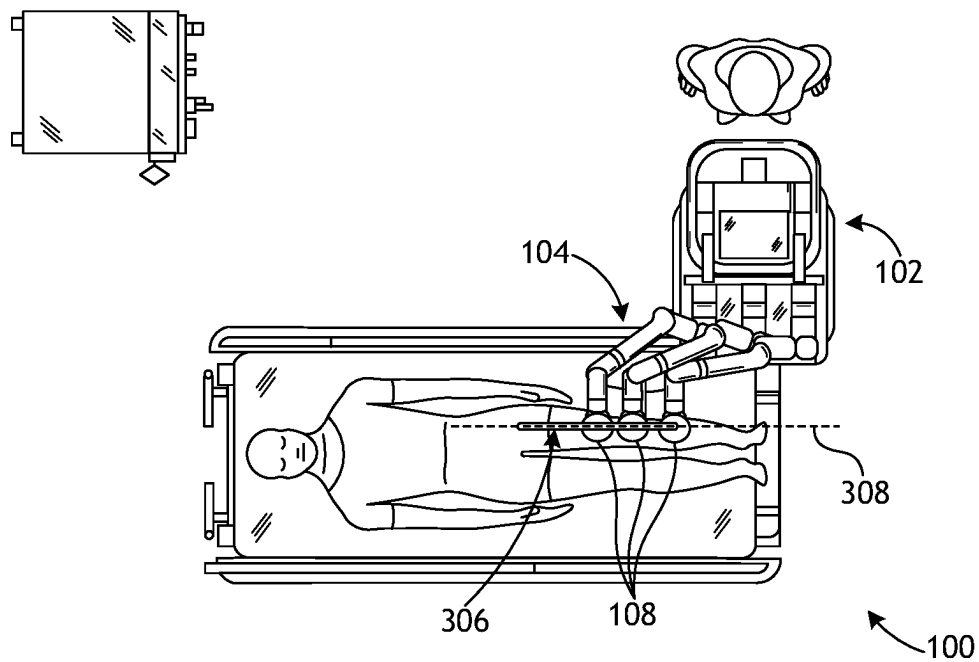


FIG. 3B

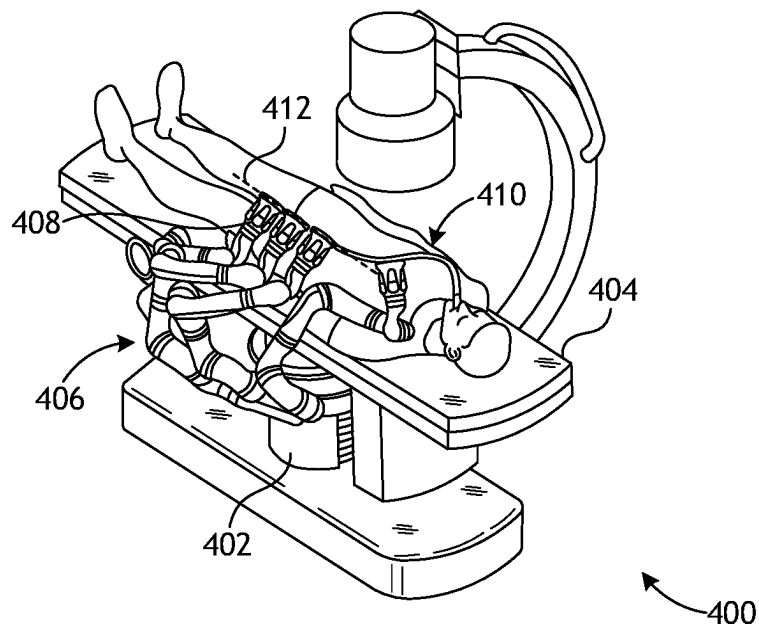


FIG. 4

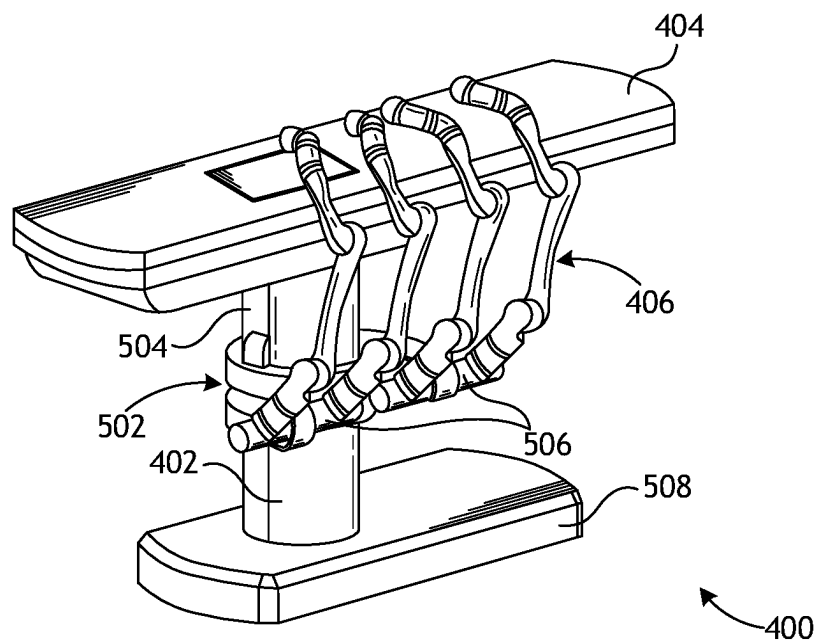


FIG. 5

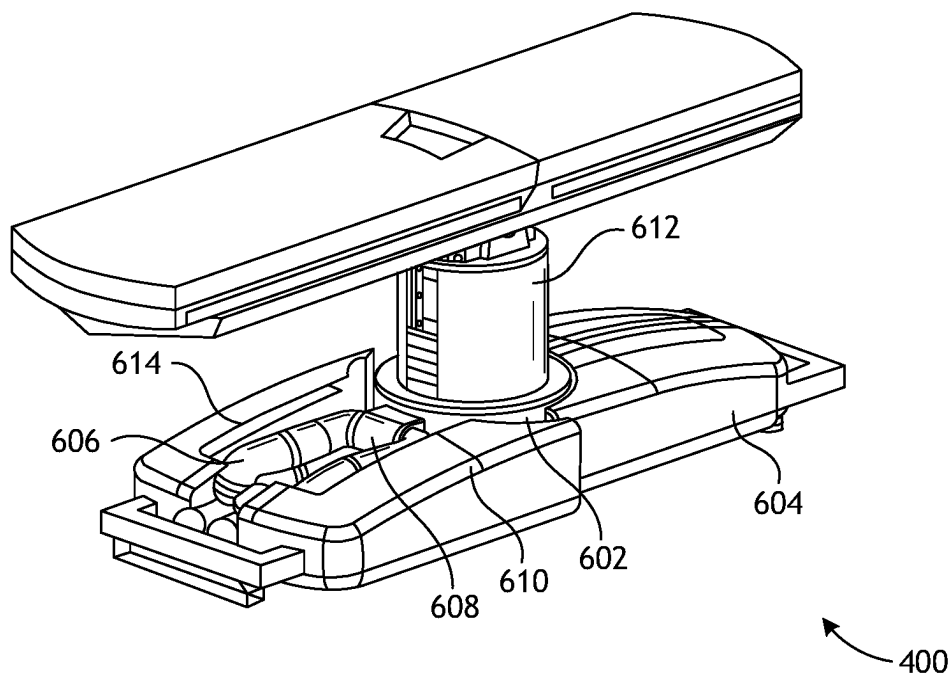


FIG. 6

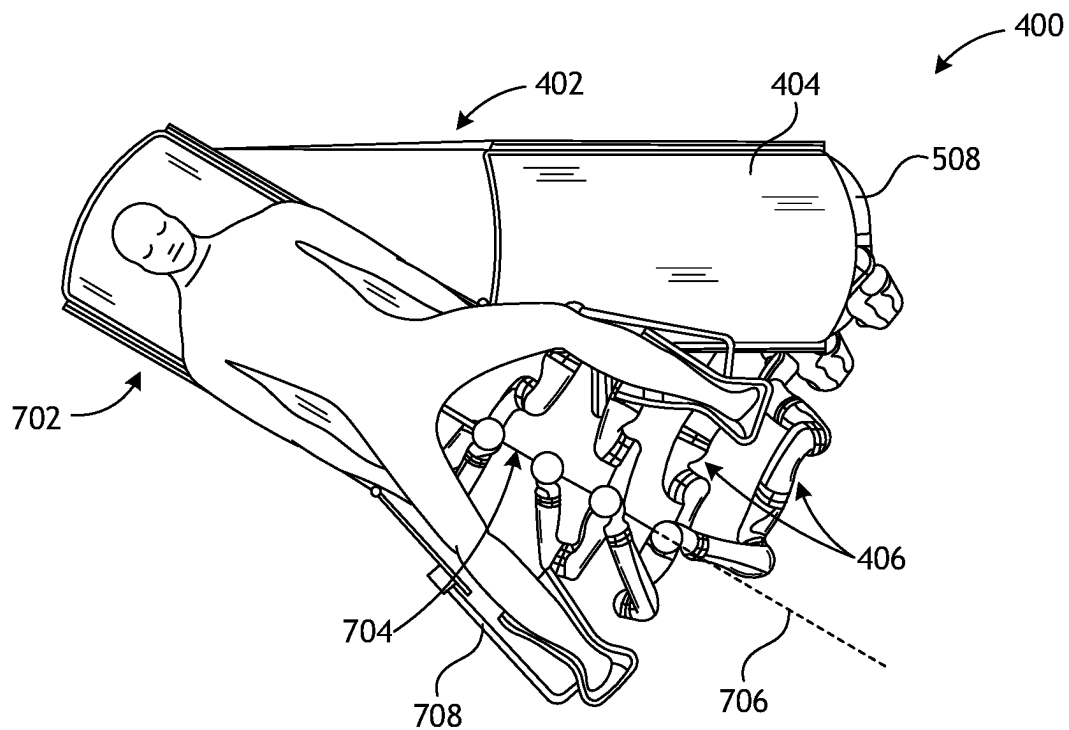


FIG. 7A

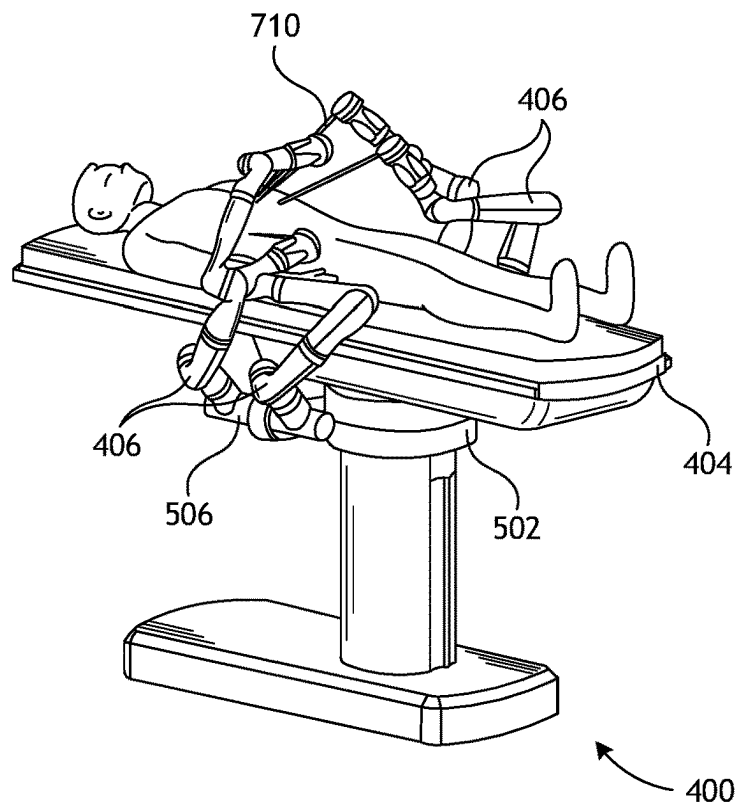


FIG. 7B

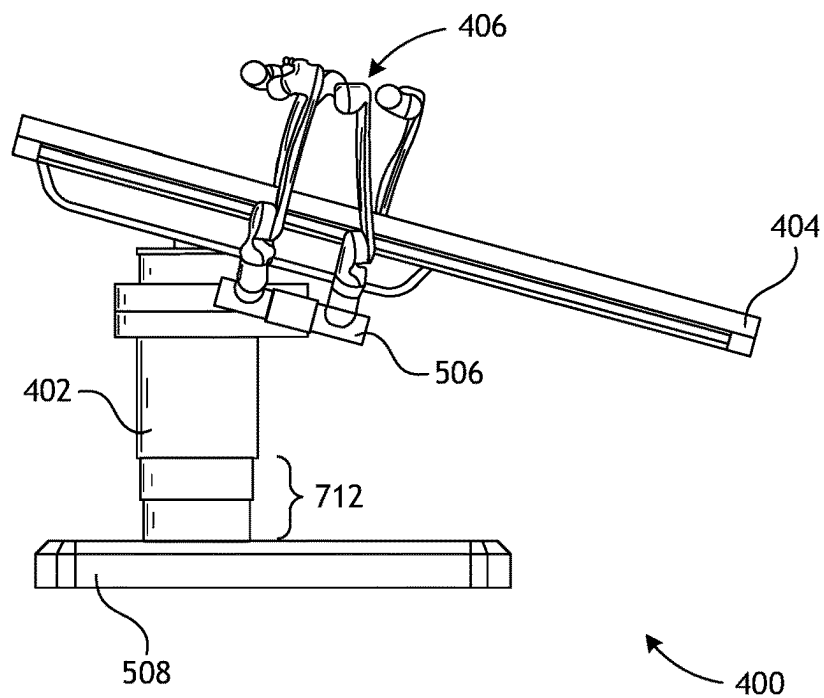


FIG. 7C

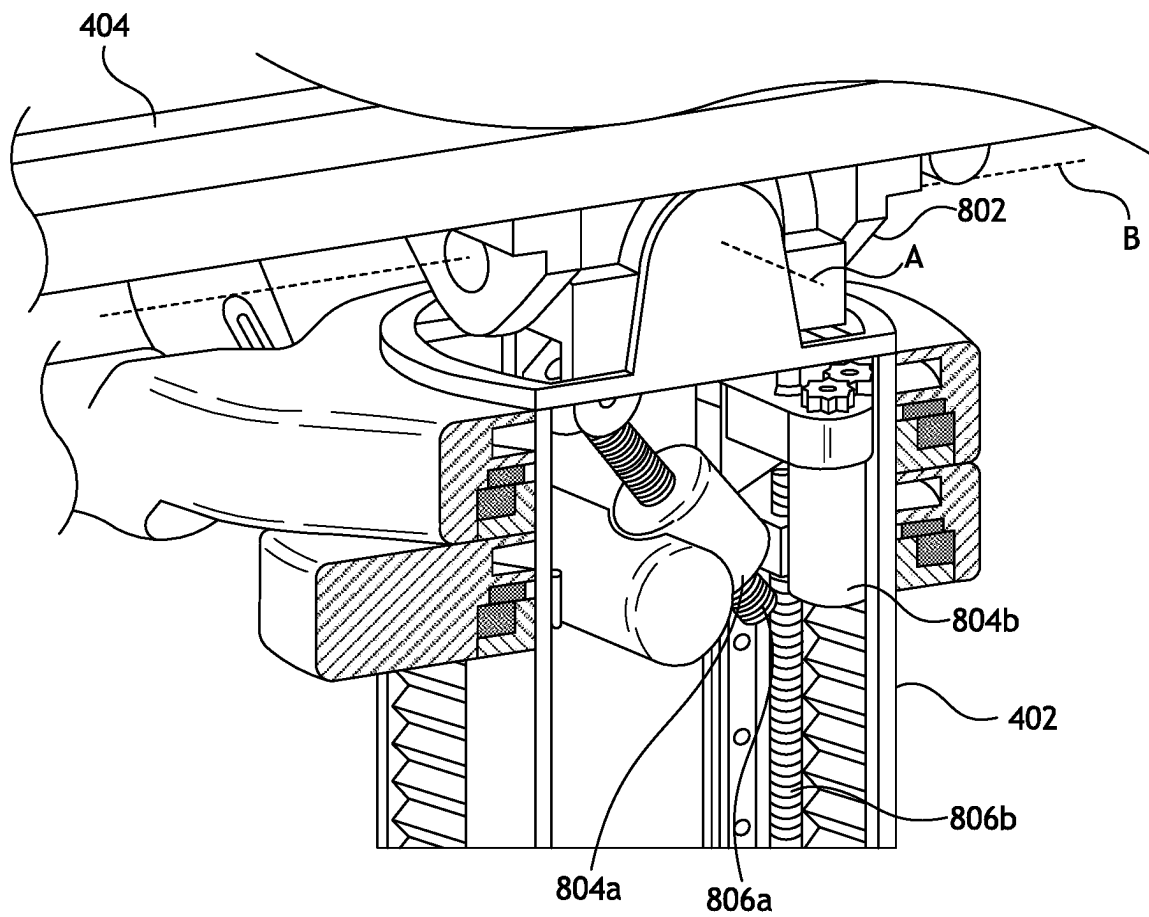


FIG. 8

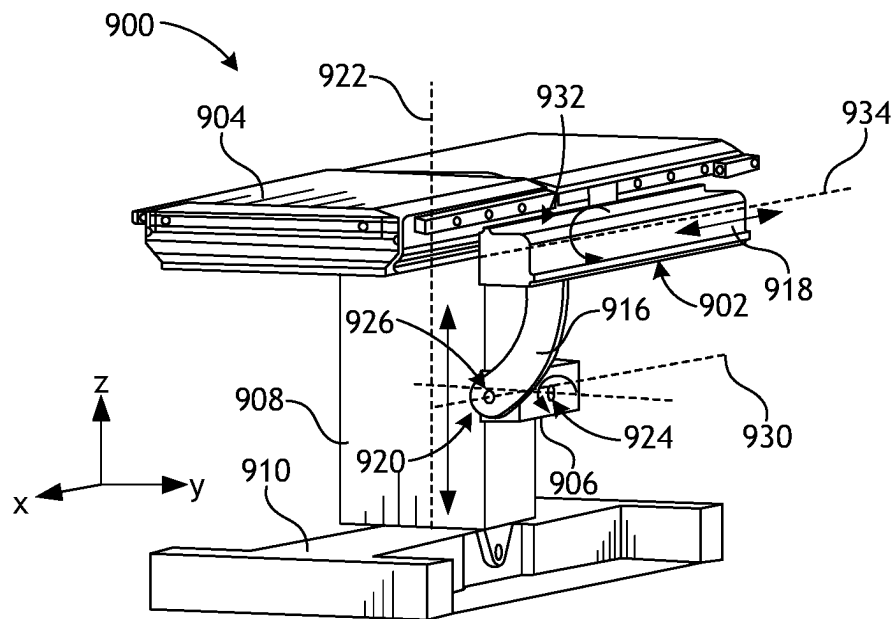


FIG. 9A

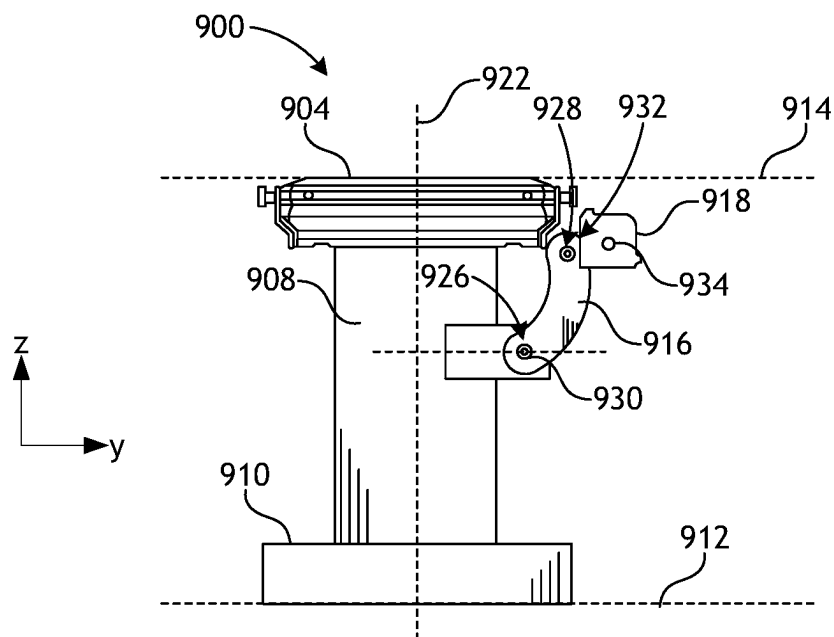


FIG. 9B

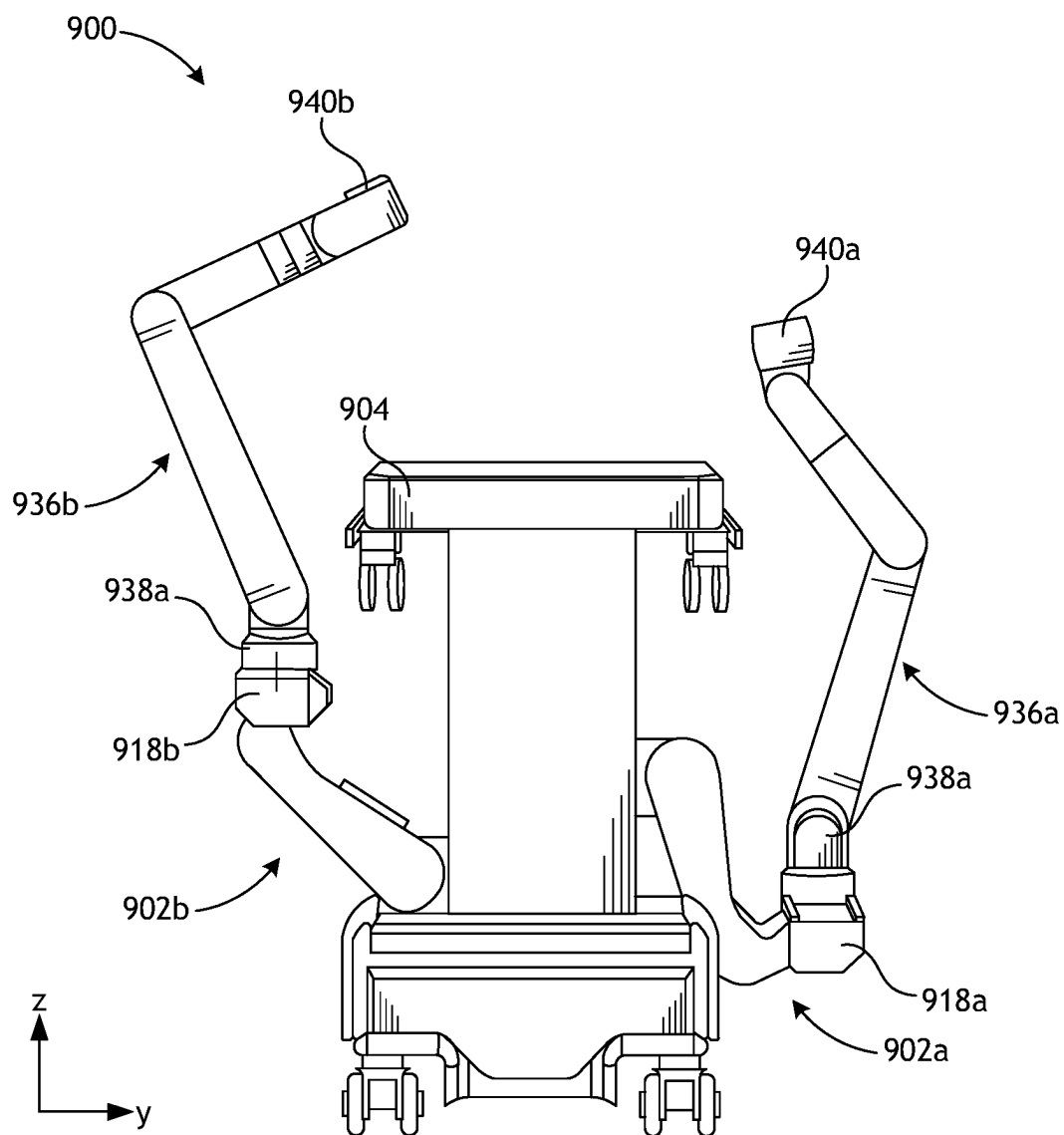


FIG. 9C

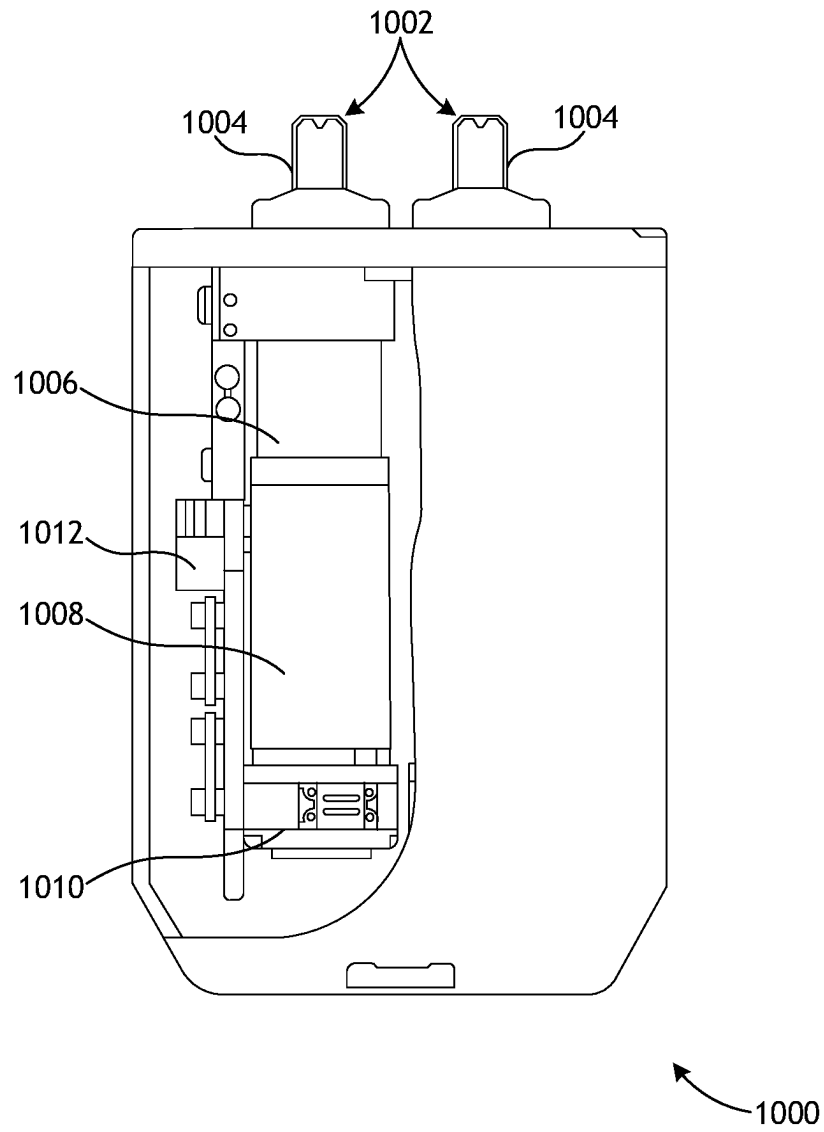


FIG. 10

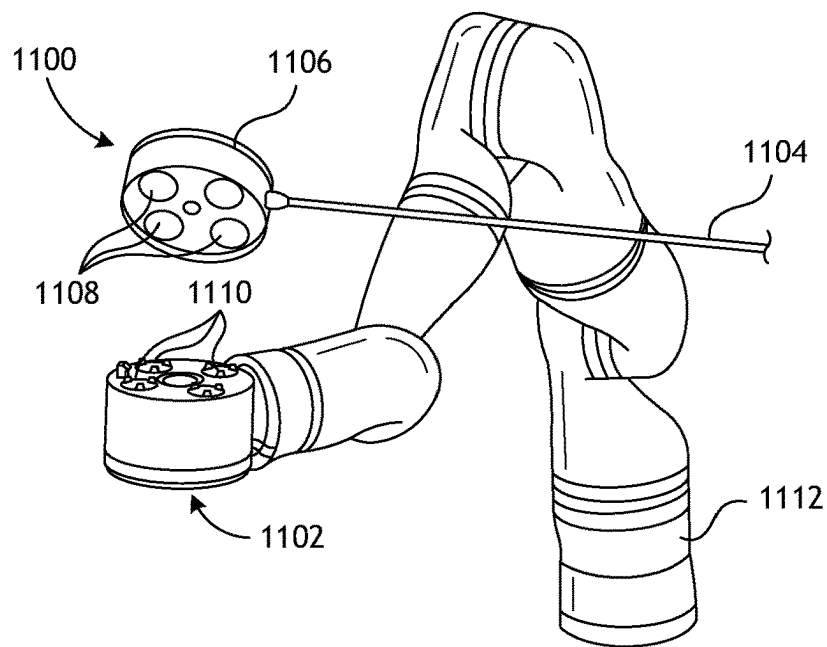


FIG. 11

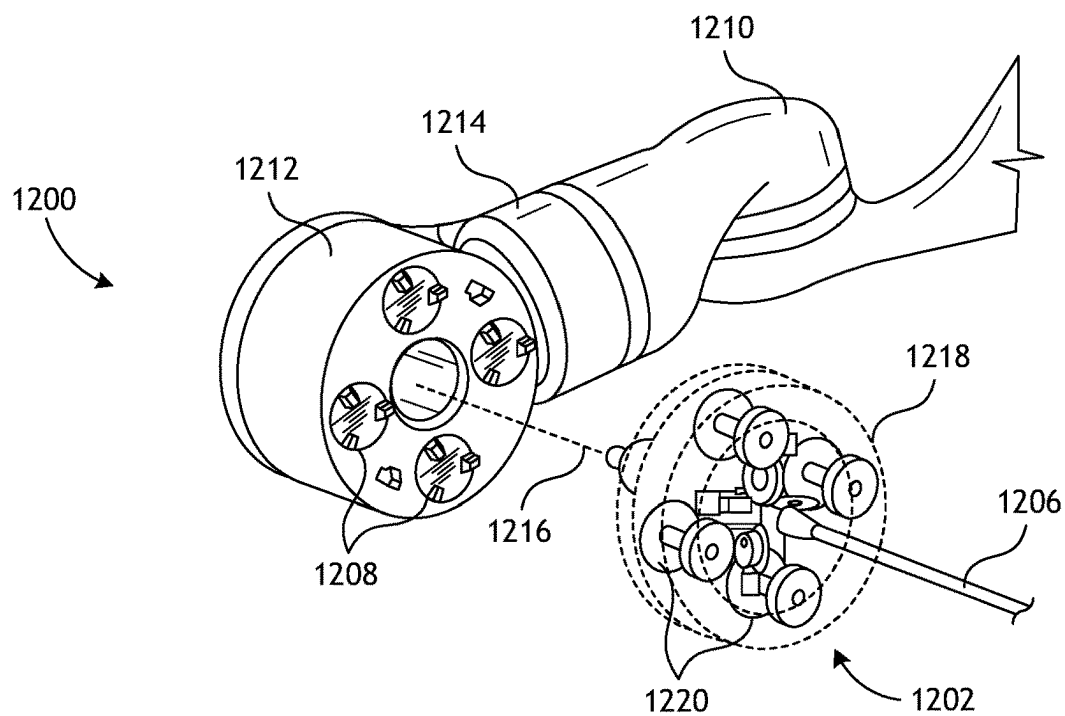


FIG. 12

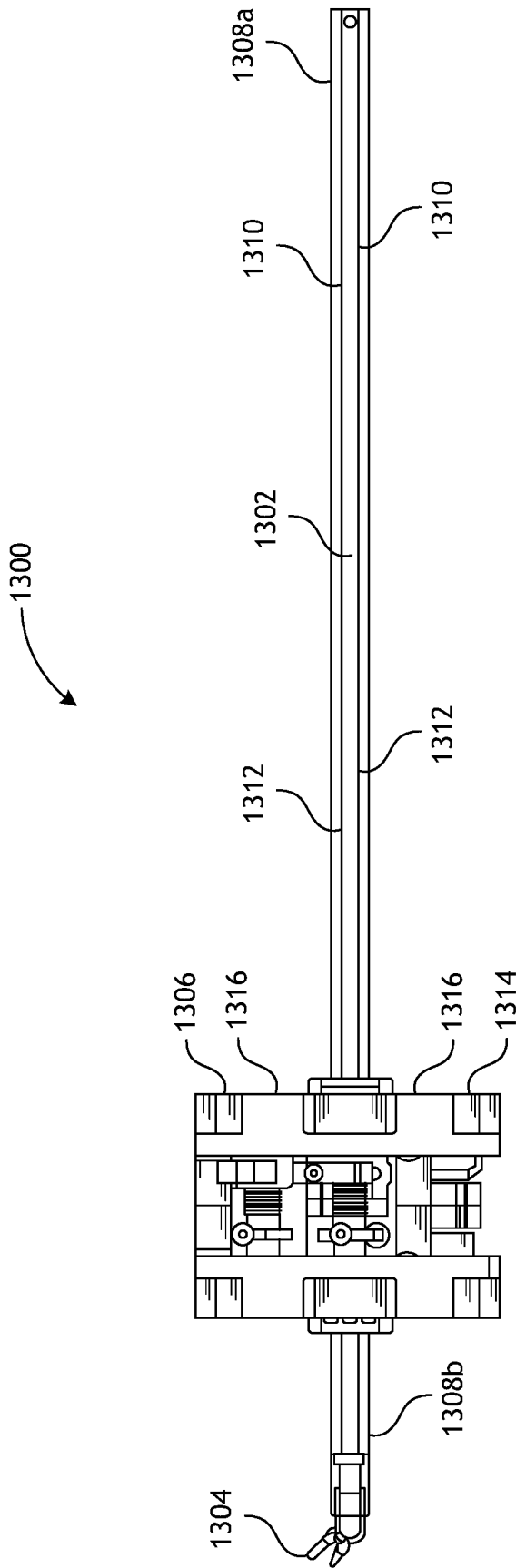


FIG. 13

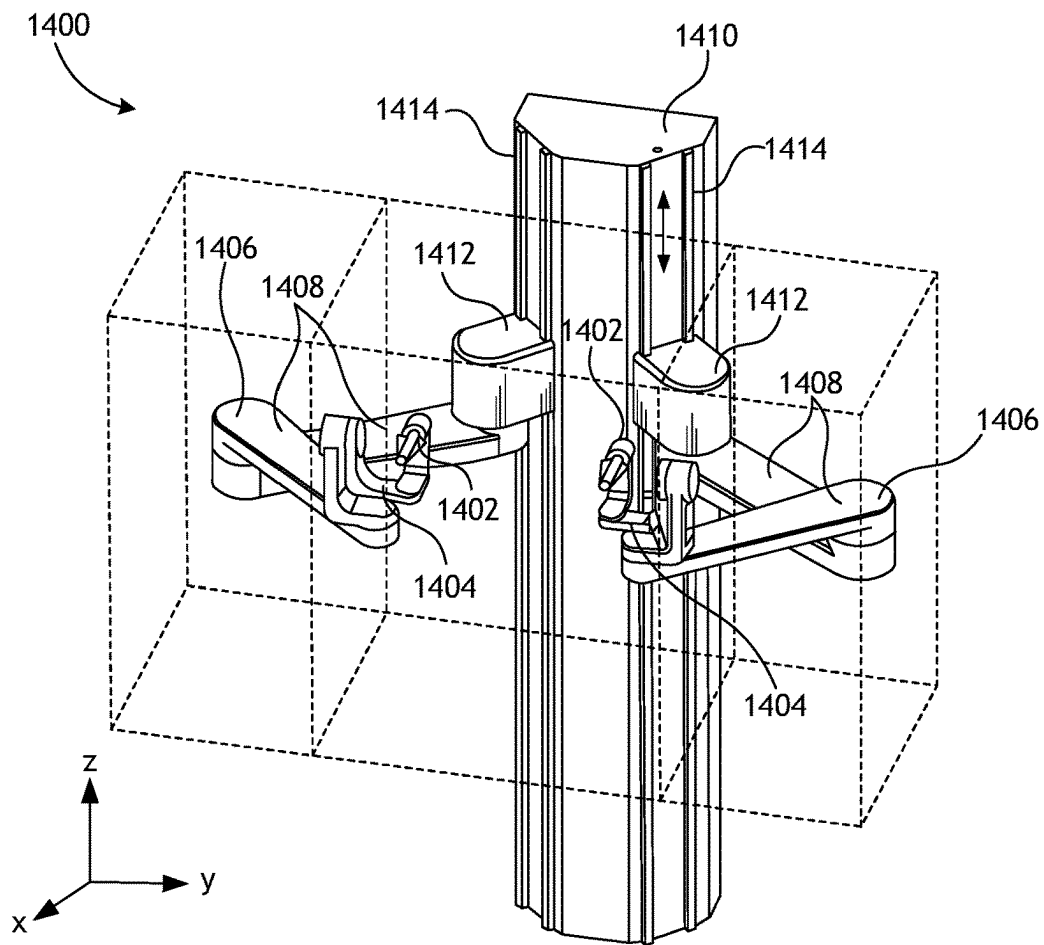


FIG. 14

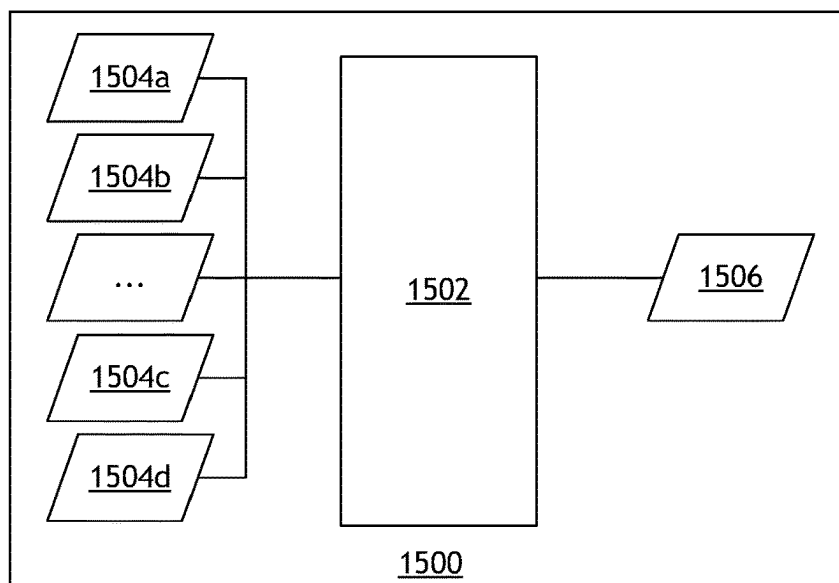


FIG. 15

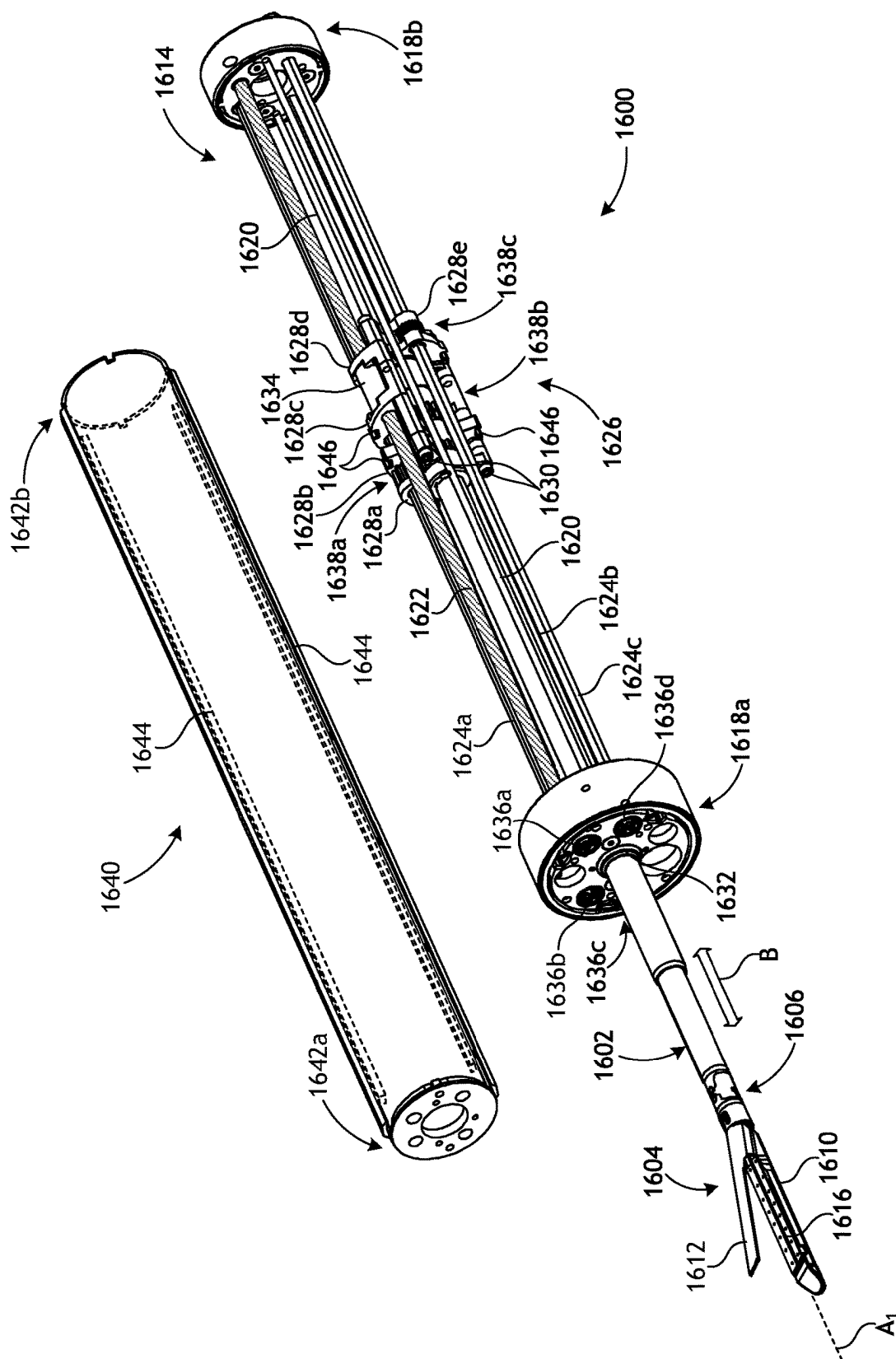


FIG. 16A

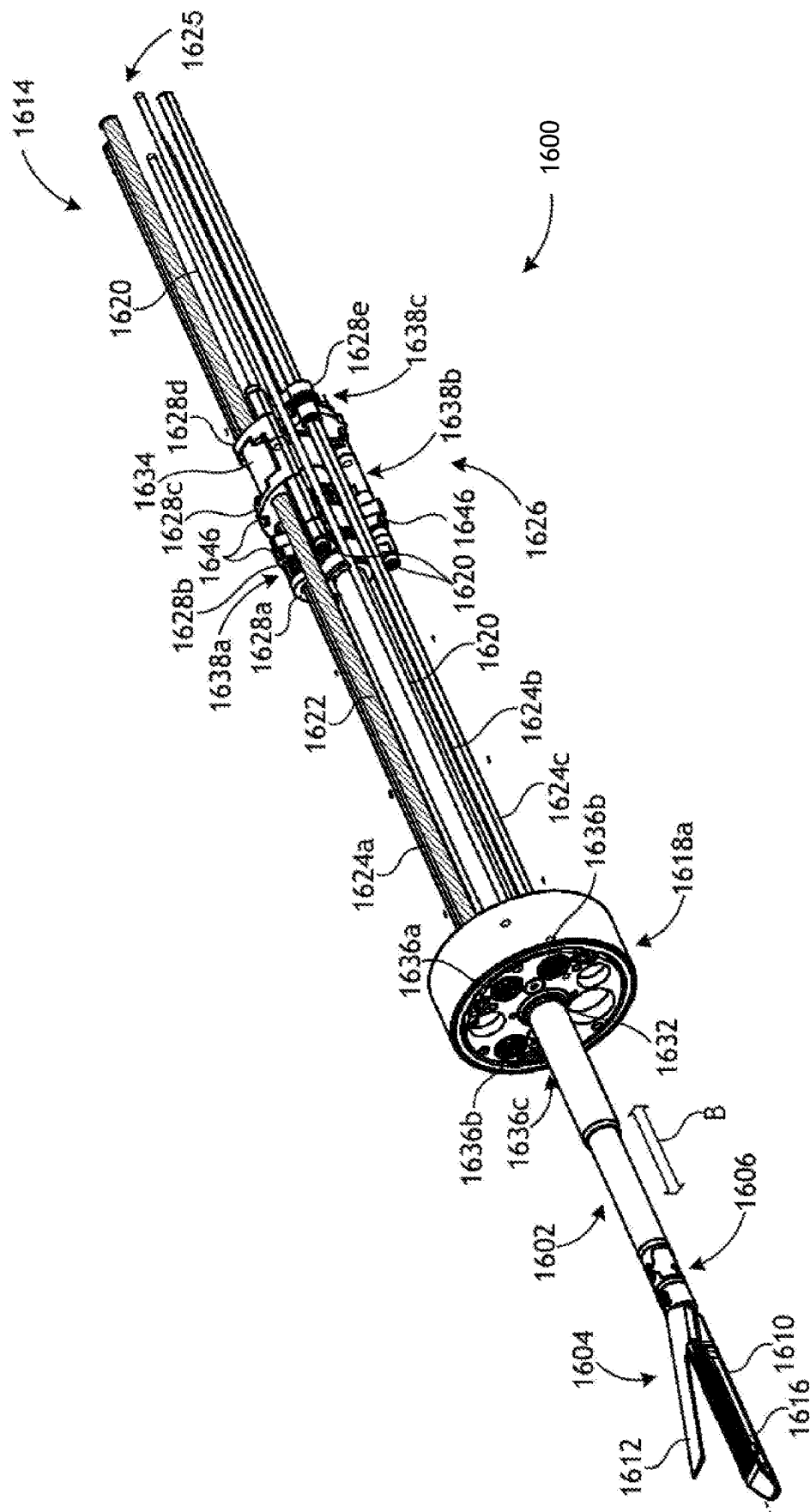


FIG. 16B

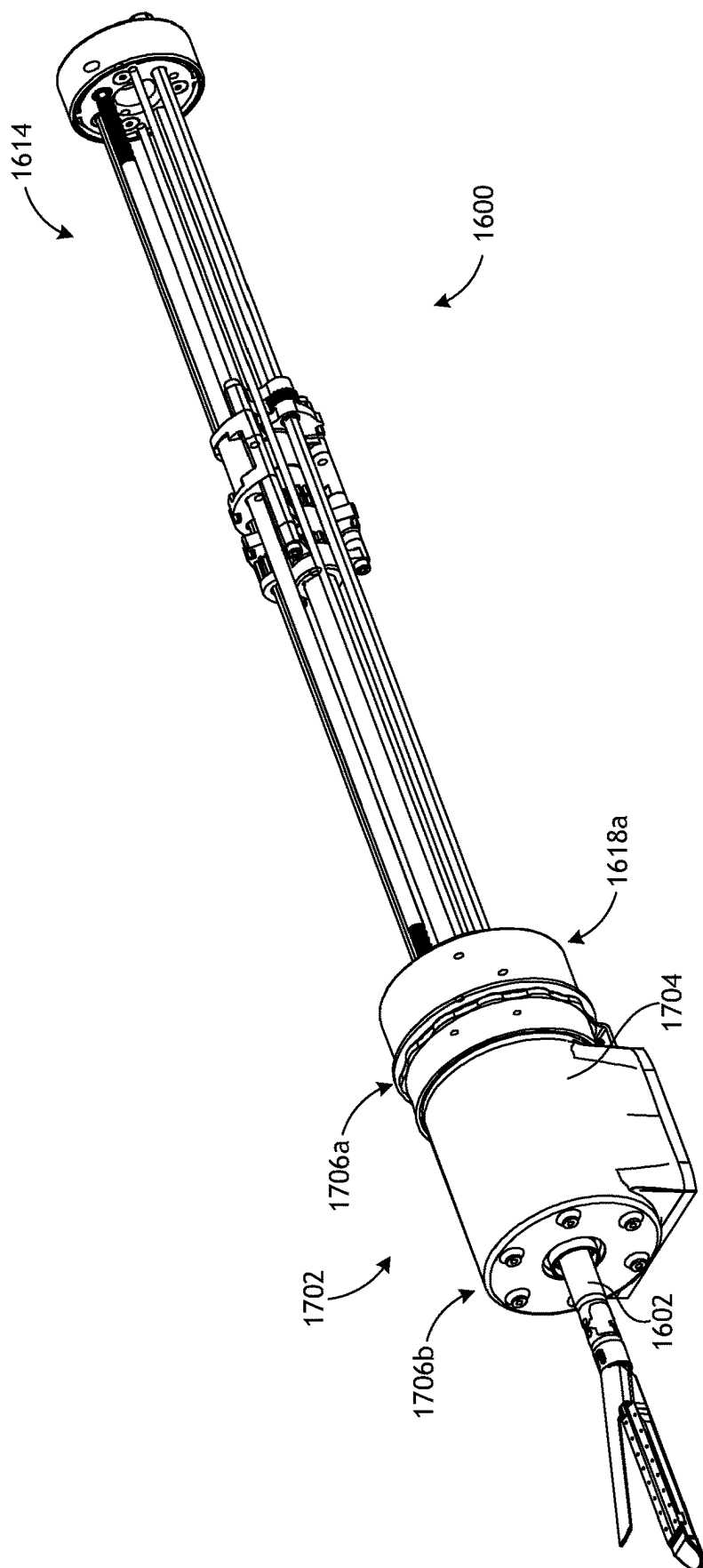


FIG. 17A

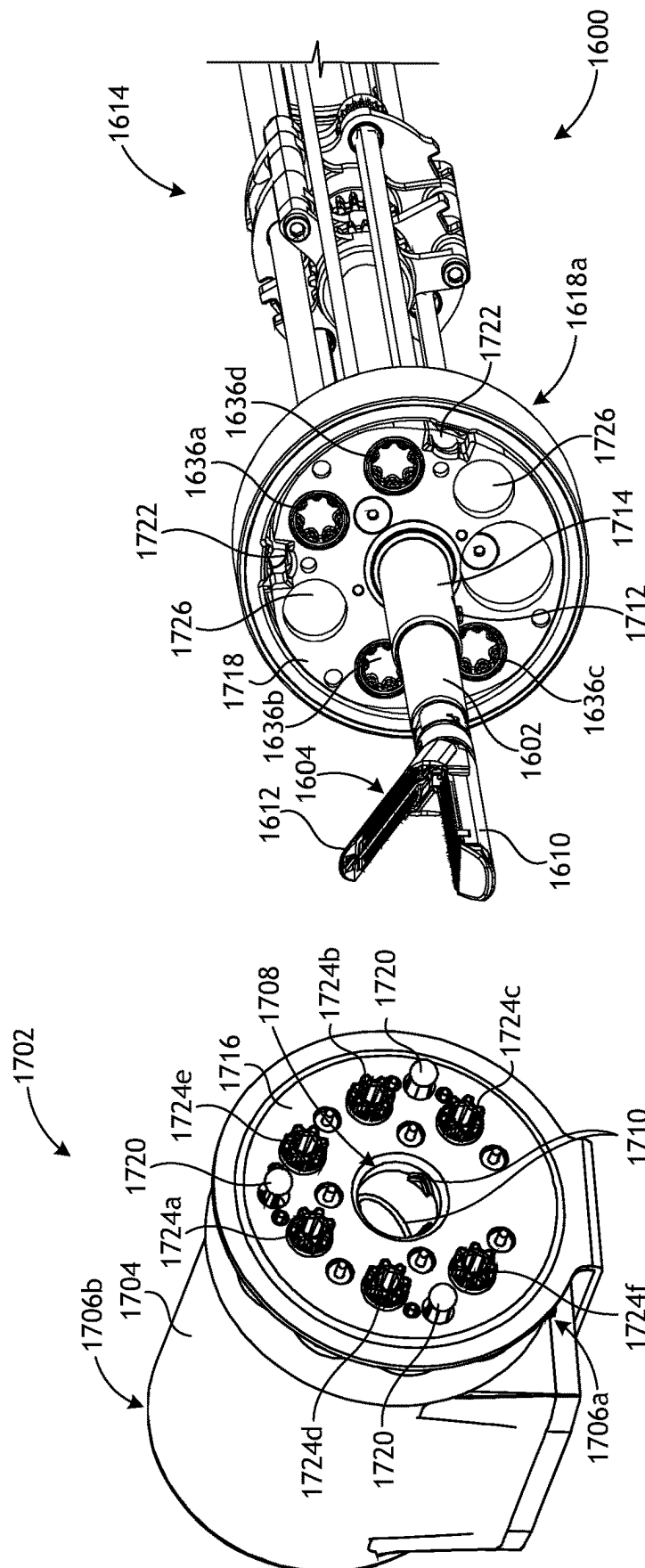


FIG. 17B

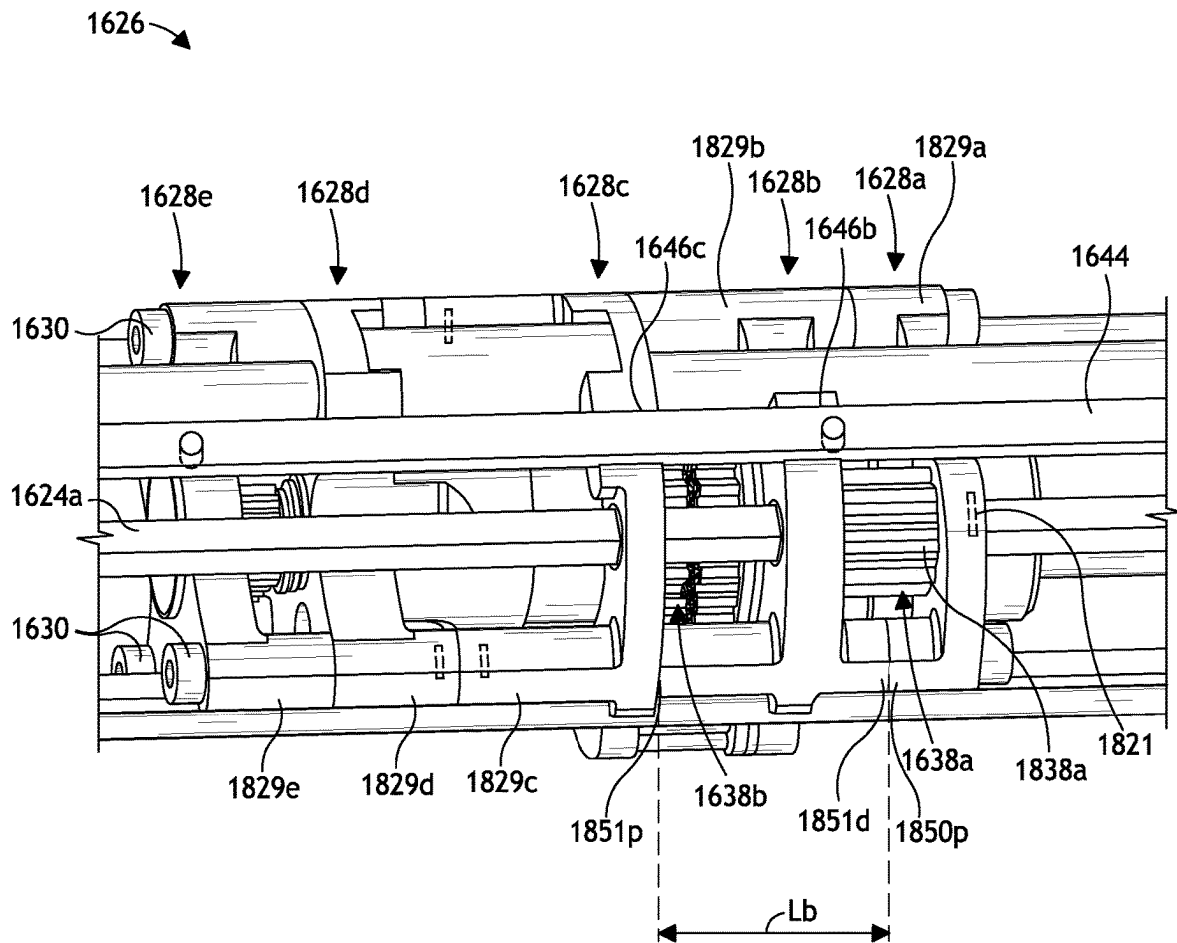


FIG. 18

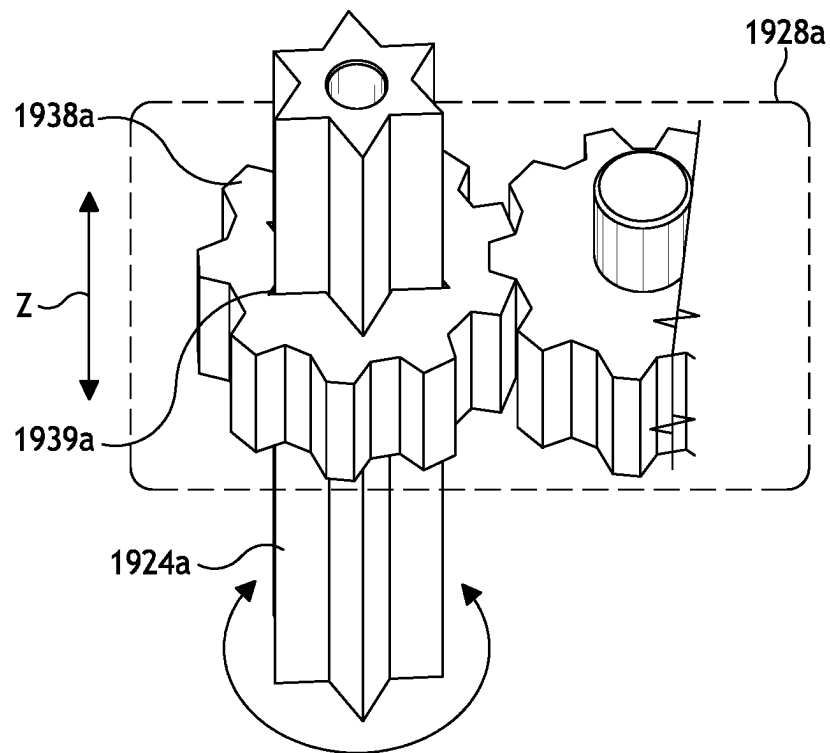


FIG. 19A

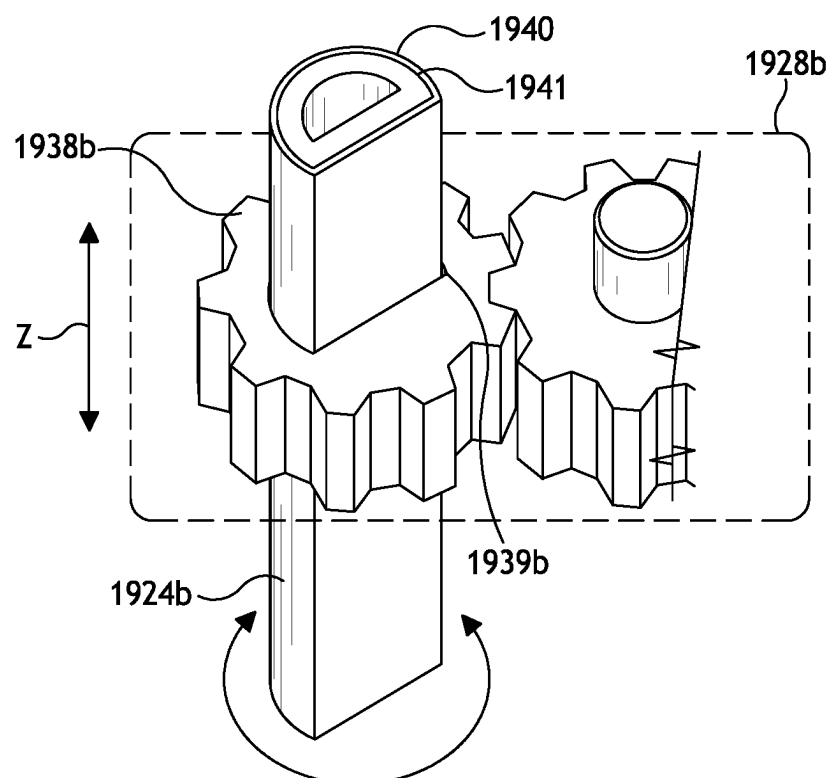


FIG. 19B

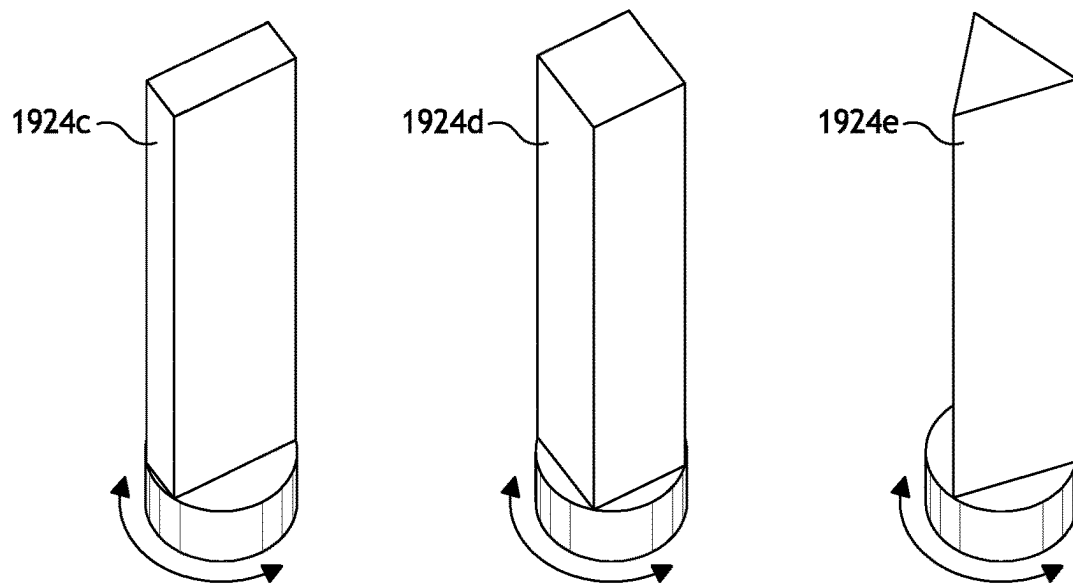


FIG. 19C

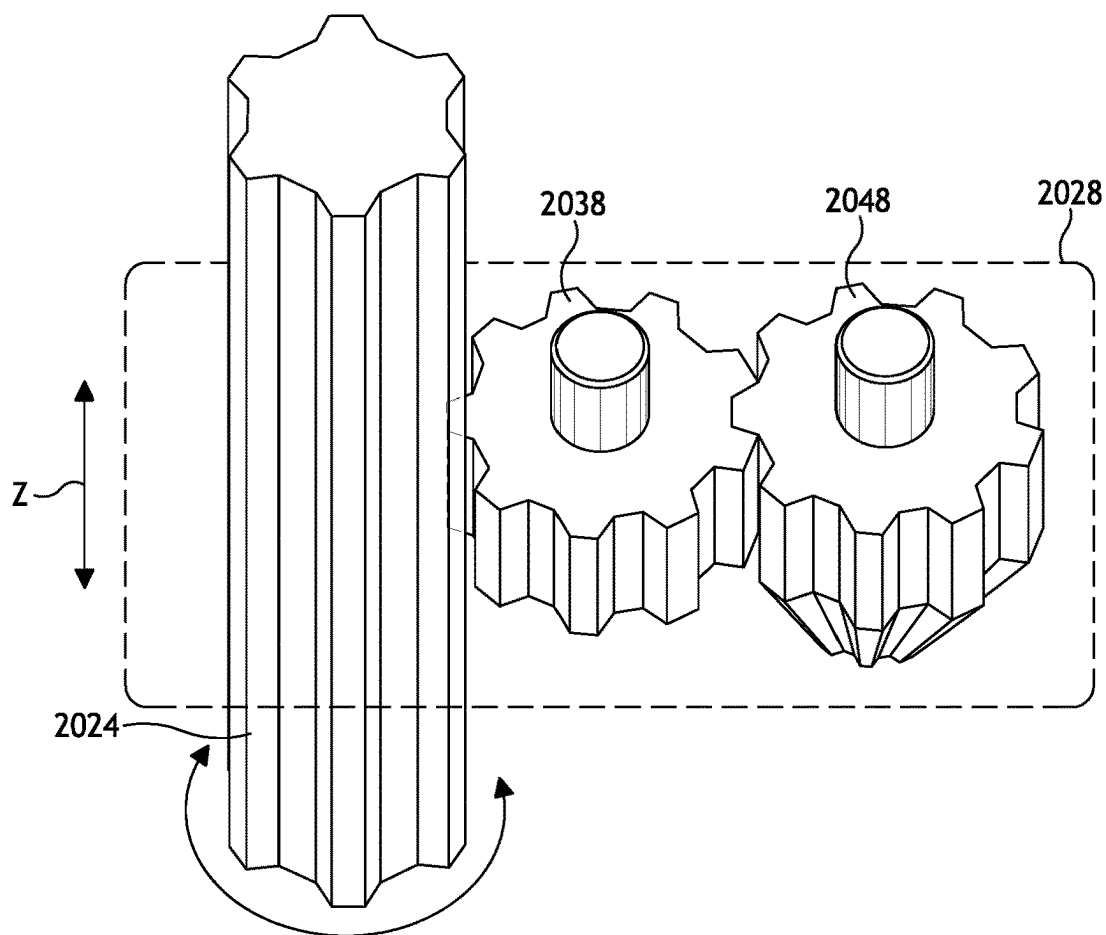


FIG. 20

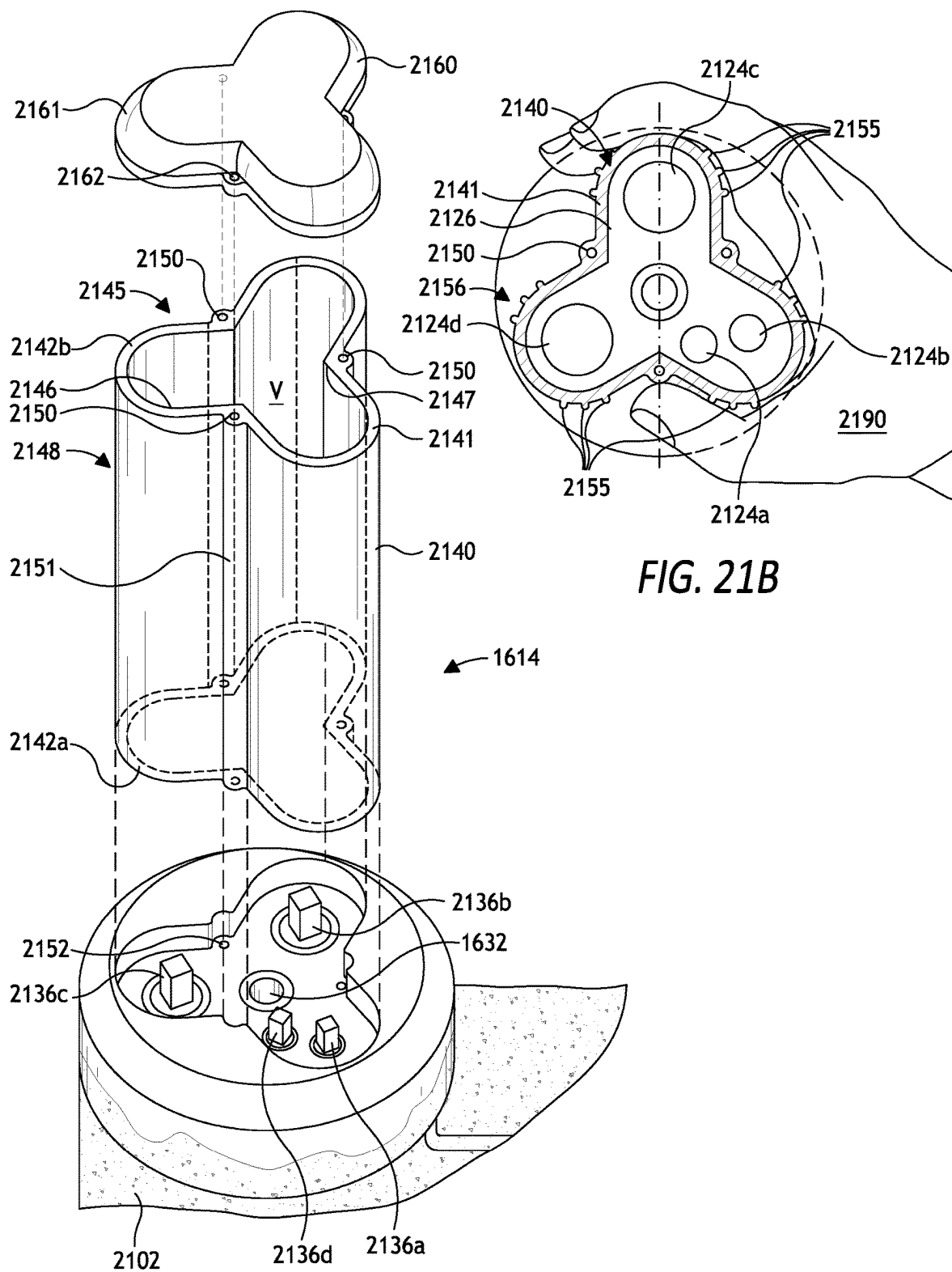


FIG. 21B

FIG. 21A

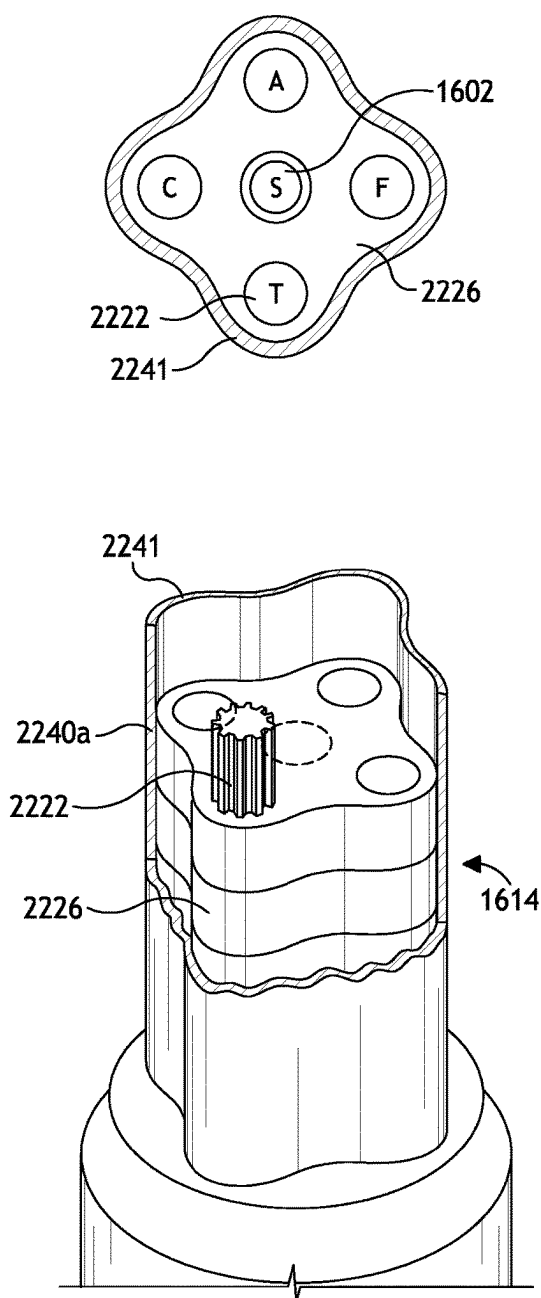


FIG. 22A

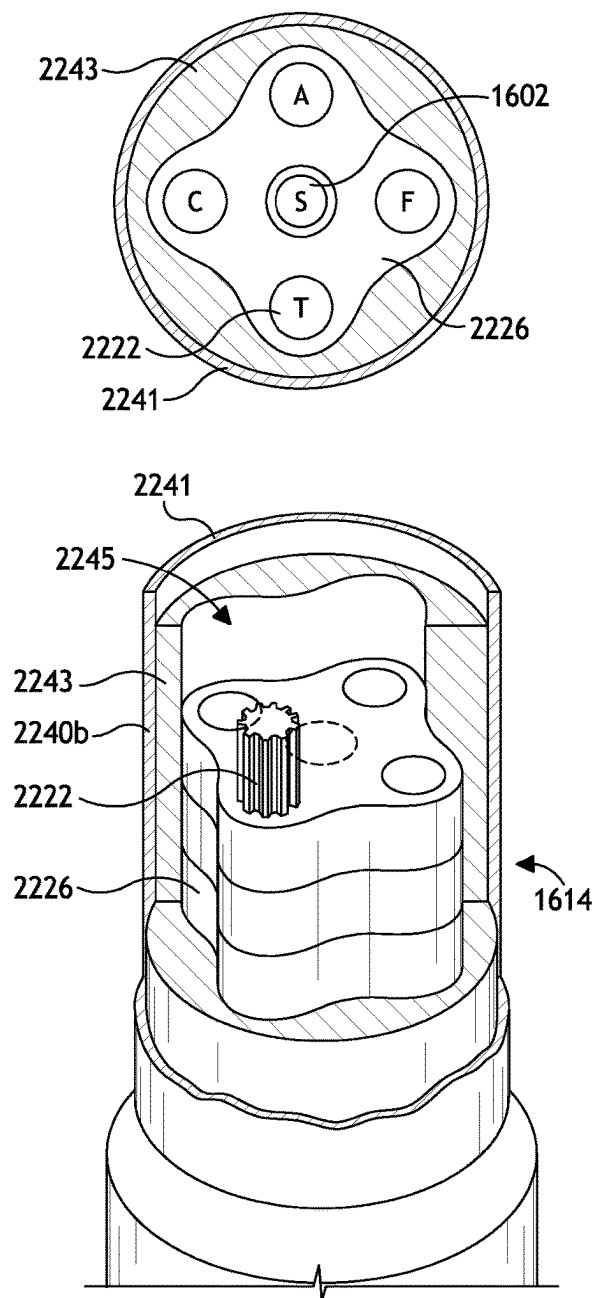


FIG. 22B

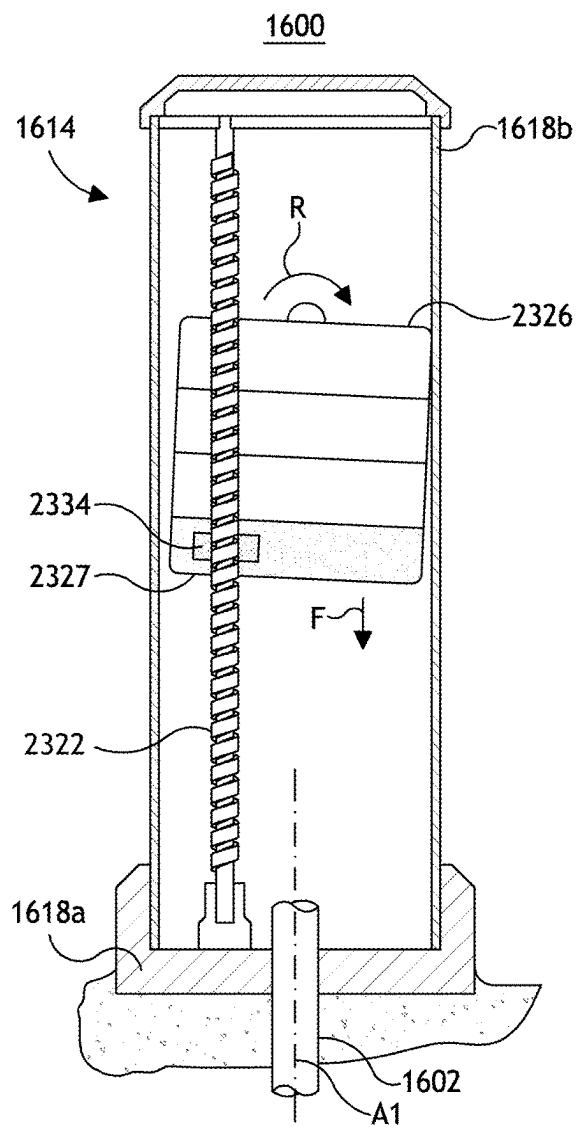


FIG. 23A

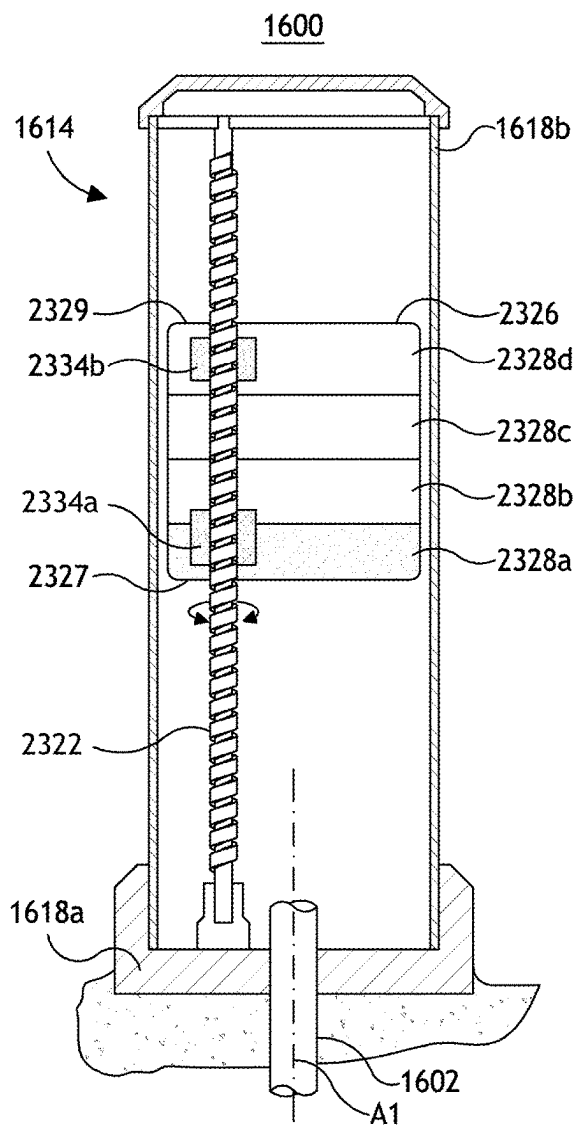


FIG. 23B

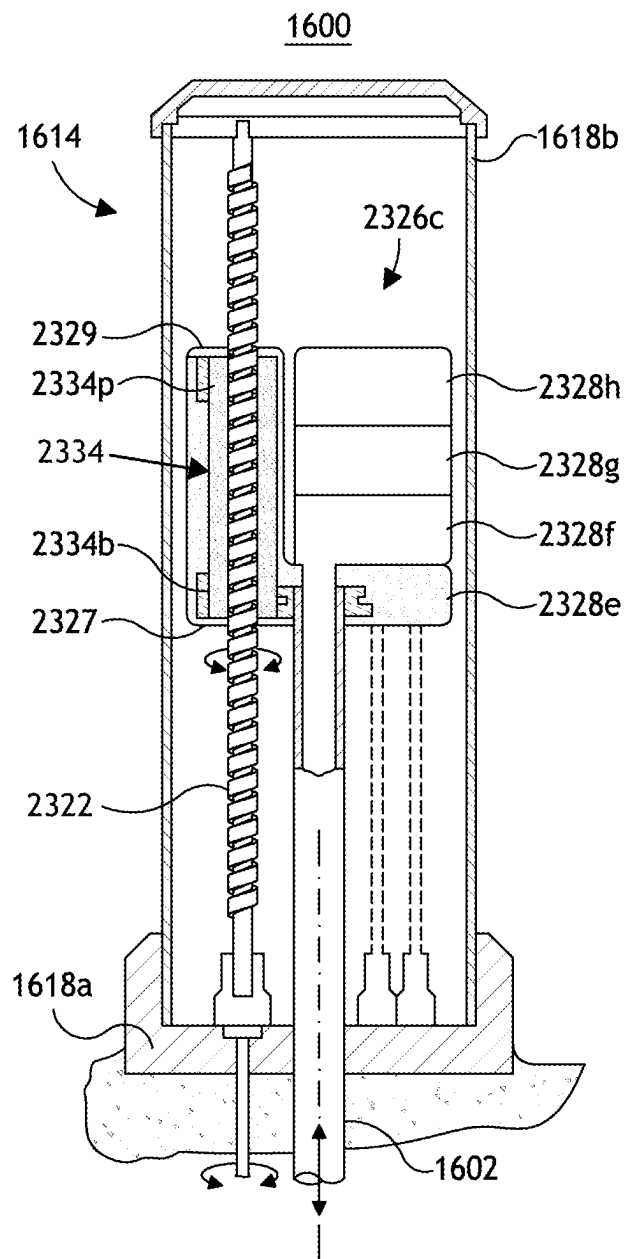


FIG. 23C

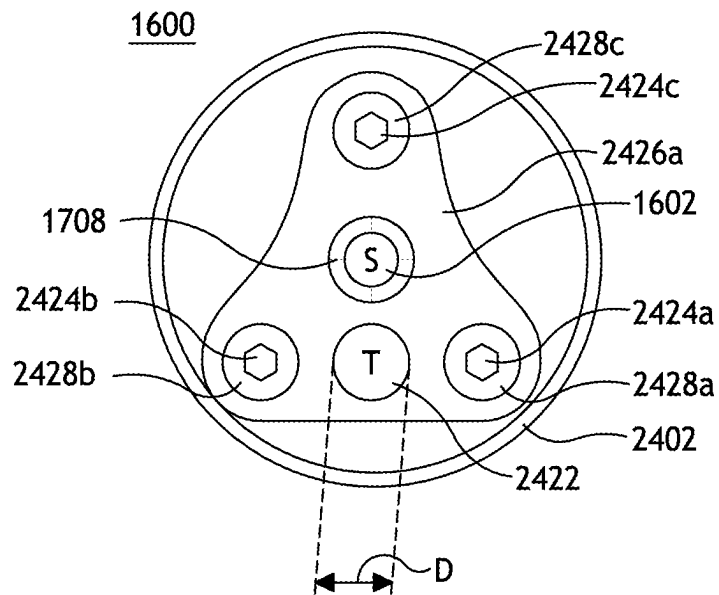


FIG. 24A

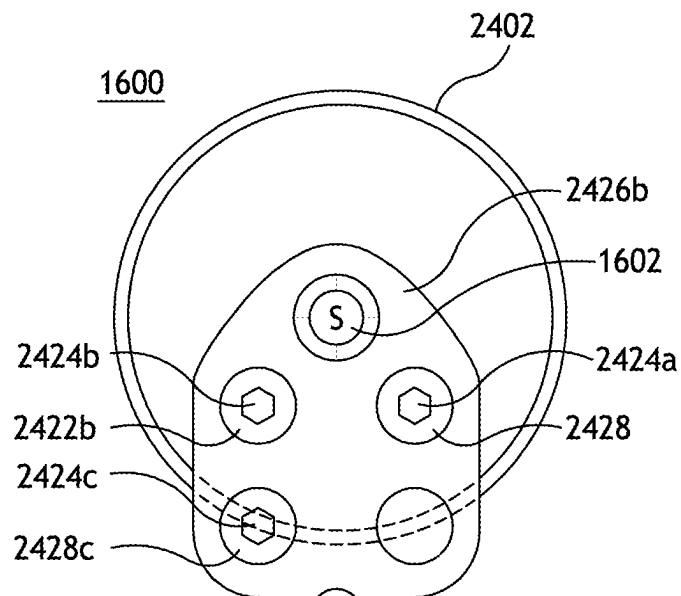


FIG. 24B

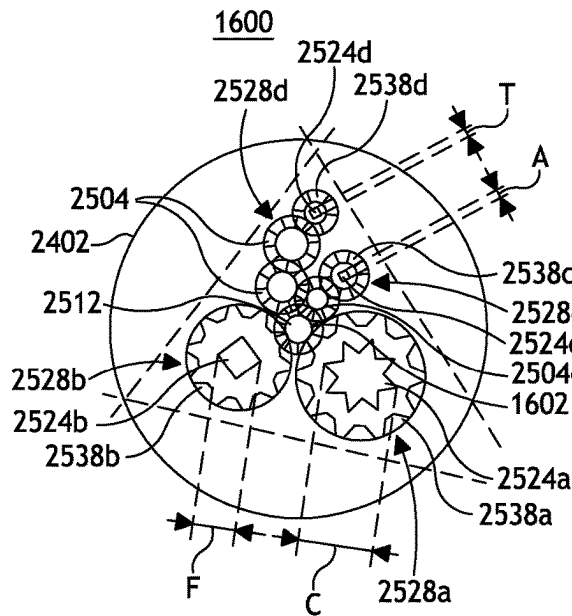


FIG. 25A

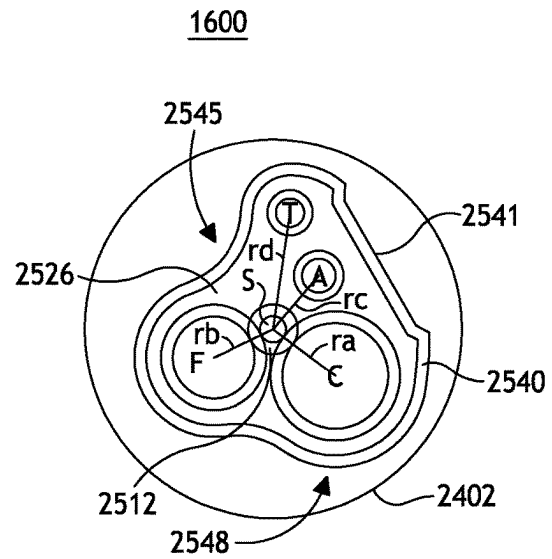


FIG. 25B

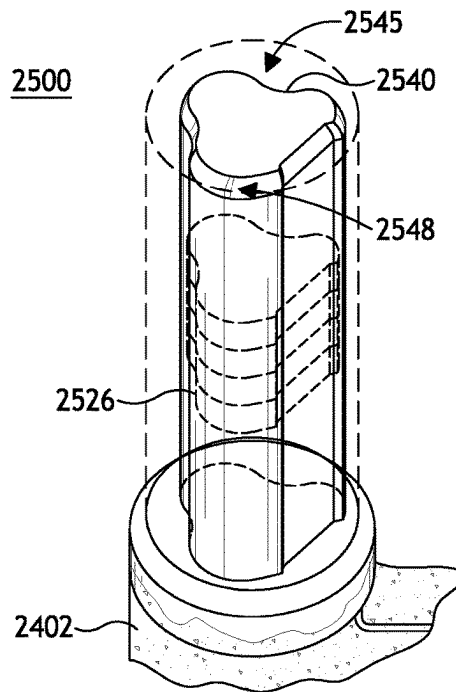


FIG. 25C

1

DRIVE PATTERNS AND SPLINE ARRANGEMENT FOR ROBOTIC SURGICAL TOOLS

TECHNICAL FIELD

The systems and methods disclosed herein are directed to robotic surgical systems and, more particularly to, surgical tools optimized for torque transfer and assuming operational torsional loading.

BACKGROUND

Minimally invasive surgical (MIS) instruments are often preferred over traditional open surgical devices due to the reduced post-operative recovery time and minimal scarring. The most common MIS procedure may be endoscopy, and the most common form of endoscopy is laparoscopy, in which one or more small incisions are formed in the abdomen of a patient and a trocar is inserted through the incision to form a pathway that provides access to the abdominal cavity. The cannula and sealing system of the trocar is used to introduce various instruments and tools into the abdominal cavity, as well as to provide insufflation to elevate the abdominal wall above the organs. The instruments can be used to engage and/or treat tissue in a number of ways to achieve a diagnostic or therapeutic effect.

Each surgical tool typically includes an end effector arranged at its distal end. Example end effectors include clamps, graspers, scissors, staplers, suction irrigators, blades (i.e., RF) and needle holders, and are similar to those used in conventional (open) surgery except that the end effector of each tool is separated from its handle by an approximately 12-inch long shaft. A camera or image capture device, such as an endoscope, is also commonly introduced into the abdominal cavity to enable the surgeon to view the surgical field and the operation of the end effectors during operation. The surgeon is able to view the procedure in real-time by means of a visual display in communication with the image capture device.

Various robotic systems have recently been developed to assist in MIS procedures. Robotic systems can allow for more intuitive hand movements by maintaining natural eye-hand axis. Robotic systems can also allow for more degrees of freedom in movement by including a "wrist" joint that creates a more natural hand-like articulation and allows for access to hard to reach spaces. The instrument's end effector can be articulated (moved) using motors and actuators forming part of a computerized motion system. A user (e.g., a surgeon) is able to remotely operate an instrument's end effector by grasping and manipulating in space one or more controllers that communicate with an instrument driver coupled to the surgical instrument. User inputs are processed by a computer system incorporated into the robotic surgical system and the instrument driver responds by actuating the motors and actuators of the motion system. Moving the drive cables and/or other mechanical mechanisms to manipulate the end effector to desired positions and configurations.

Improvements to robotically-enabled medical systems will provide physicians with the ability to perform endoscopic and laparoscopic procedures more effectively and with improved ease.

SUMMARY OF DISCLOSURE

Various details of the present disclosure are hereinafter summarized to provide a basic understanding. This sum-

2

mary is not an extensive overview of the disclosure and is neither intended to identify certain elements of the disclosure, nor to delineate the scope thereof. Rather, the primary purpose of this summary is to present some concepts of the disclosure in a simplified form prior to the more detailed description that is presented hereinafter.

Embodiments disclosed herein include a robotic surgical tool that includes a handle having a first end and a second end, a lead screw rotatably coupled to the handle and extending between the first and second ends, a spline rotatably coupled to the handle and extending between the first and second ends and carriage movably mounted to the lead screw for translation between the first and second ends. The surgical tool also includes at least one activating mechanism supported by the carriage and coupled to the spline, wherein the spline is arranged at a distance from the lead screw such that the lead screw assumes torsional loading generated by the spline. In a further embodiment, the surgical tool further includes an elongate shaft extending from the carriage and penetrating the first end, the shaft having an end effector arranged at a distal end thereof. In another further embodiment, a stiffness of the lead screw is greater than a stiffness of the spline. In another further embodiment, the spline is a first spline and the distance is a first distance, the surgical tool further including a second spline rotatably coupled to the handle and extending between the first and second ends, the second spline being coupled to at least one activating mechanism supported by the carriage, wherein the second spline is arranged at a second distance from the lead screw, such that the lead screw assumes torsional loading generated by the second spline. In another further embodiment, the first spline has a greater torsional loading than the second spline and the first distance from the lead screw is less than the second distance from the lead screw. In another further embodiment, the first spline rotates in a first direction and the second spline rotates in a second direction opposite the first direction, and wherein a first torsional load applied by the first spline is substantially equal to a second torsional load applied by the second spline. In another further embodiment, the spline is one of a plurality of splines, wherein a location of a center line of each spline of the plurality of splines is at a radius from the lead screw, the radius being determined by an input torque associated with each spline, and wherein the greater the input torque of a given spline of the plurality of splines, the shorter the radius from the lead screw. In another further embodiment, the plurality of splines comprise splines of at least two different cross-sectional areas. In another further embodiment, each spline of the plurality of splines comprises a different cross-sectional area and the cross-sectional area of each spline is determined based on the input torque associated with the spline.

Embodiments disclosed herein may further include a robotic surgical tool that includes a handle having a first end and a second end, a lead screw rotatably coupled to the handle and extending between the first and second ends, first and second splines rotatably coupled to the handle and extending between the first and second ends and a carriage movably mounted to the lead screw for translation between the first and second ends. The robotic surgical tool also includes a first activating mechanism supported by the carriage and coupled to the first spline and a second activating mechanism supported by the carriage and coupled to the second spline where a first torsional load generated by the first spline rotating in a first direction is balanced by a second torsional load generated by the second spline rotating in a second direction opposite the first direction. In a further

3

embodiment, the first spline is arranged at a first distance from the lead screw and the second spline is arranged at a second distance from the lead screw, and wherein the lead screw assumes torsional loading generated by the first and second splines. In another further embodiment, the first distance is equal to the second distance. In another further embodiment, the first distance is different from the second distance. In another further embodiment, the first spline generates a first torsional load greater than a second torsional load generated by the second spline, and wherein the first distance is less than the second distance from the lead screw. In another further embodiment, the first spline has a first cross-sectional area and the second spline has a second cross-sectional area smaller than the first cross-sectional area. In another further embodiment, the first spline has a first cross-sectional area and the second spline has a second cross-sectional area, the first cross-sectional area is about equal to the second cross sectional area.

Embodiments disclosed herein may further include a method that includes actuating a robotic surgical tool, the robotic surgical tool including a handle having a first end and a second end, at least one spline rotatably coupled to the handle and extending between the first and second ends, and a lead screw rotatably coupled to the handle and extending between the first and second ends and located in a screw location, wherein one of the lead screw and one of the at least one spline comprises a load balancing member. A carriage is movably mounted to the lead screw and an elongate shaft extends from the carriage and penetrates the first end, the shaft having an end effector arranged at a distal end thereof. The surgical tool includes an activating mechanism supported by the carriage. The method including assuming a torsional force acting on the carriage generated by the at least one spline with the load balancing member. In a further embodiment, the at least one spline comprises first and second splines, the method further includes rotating the first spline in a first direction and rotating the second spline simultaneously in a second direction different from the first direction. In another further embodiment, the at least one spline comprises a plurality of splines and each spline exhibits a different cross-sectional area determined based on an input torque associated with each spline. In another further embodiment, the lead screw is the load balancing member and a location of a center line of the at least one spline is at a radius from the lead screw, the radius being determined by an input torque associated with each spline, and where the greater the input torque of the at least one spline, the shorter the radius from the lead screw.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosed aspects will hereinafter be described in conjunction with the appended drawings, provided to illustrate and not to limit the disclosed aspects, wherein like designations denote like elements.

FIG. 1 illustrates an embodiment of a cart-based robotic system arranged for diagnostic and/or therapeutic bronchoscopy procedure(s).

FIG. 2 depicts further aspects of the robotic system of FIG. 1.

FIG. 3A illustrates an embodiment of the robotic system of FIG. 1 arranged for ureteroscopy.

FIG. 3B illustrates an embodiment of the robotic system of FIG. 1 arranged for a vascular procedure.

FIG. 4 illustrates an embodiment of a table-based robotic system arranged for a bronchoscopy procedure.

4

FIG. 5 provides an alternative view of the robotic system of FIG. 4.

FIG. 6 illustrates an example system configured to stow robotic arm(s).

FIG. 7A illustrates an embodiment of a table-based robotic system configured for an ureteroscopy procedure.

FIG. 7B illustrates an embodiment of a table-based robotic system configured for a laparoscopic procedure.

FIG. 7C illustrates an embodiment of the table-based robotic system of

FIGS. 4-7B with pitch or tilt adjustment.

FIG. 8 provides a detailed illustration of the interface between the table and the column of the table-based robotic system of FIGS. 4-7.

FIG. 9A illustrates an alternative embodiment of a table-based robotic system.

FIG. 9B illustrates an end view of the table-based robotic system of FIG. 9A.

FIG. 9C illustrates an end view of a table-based robotic system with robotic arms attached thereto.

FIG. 10 illustrates an exemplary instrument driver.

FIG. 11 illustrates an exemplary medical instrument with a paired instrument driver.

FIG. 12 illustrates an alternative design for an instrument driver and instrument where the axes of the drive units are parallel to the axis of the elongated shaft of the instrument.

FIG. 13 illustrates an instrument having an instrument-based insertion architecture.

FIG. 14 illustrates an exemplary controller.

FIG. 15 depicts a block diagram illustrating a localization system that estimates a location of one or more elements of the robotic systems of FIGS. 1-7C, such as the location of the instrument of FIGS. 11-13, in accordance to an example embodiment.

FIG. 16A is an isometric side view of an example surgical tool that may incorporate some or all of the principles of the present disclosure.

FIG. 16B is an isometric side view of an example surgical tool that may incorporate some or all of the principles of the present disclosure.

FIG. 17A is an isometric view of the surgical tool of FIG. 16A releasably coupled to an example instrument driver, according to one or more embodiments.

FIG. 17B provides separated isometric end views of the instrument driver of FIG. 17A and the surgical tool of FIG. 16A.

FIG. 18 is an isometric side view of an example carriage that may incorporate some or all of the principles of the present disclosure.

FIG. 19A illustrates an exemplary spline coupled to an activating mechanism that may incorporate some or all of the principles of the present disclosure.

FIG. 19B illustrates another example spline coupled to an activating mechanism that may incorporate some or all of the principles of the present disclosure.

FIG. 19C illustrates an example spline that may incorporate some or all of the principles of the present disclosure.

FIG. 20 illustrates an example spline coupled to an activating mechanism that may incorporate some or all of the principles of the present disclosure.

FIG. 21A illustrates an example surgical tool exoskeleton releasably coupled to an example instrument driver, according to one or more embodiments.

FIG. 21B illustrates a top view of the surgical tool shroud of FIG. 21A.

5

FIG. 22A illustrates an isometric side view of an example carriage and exoskeleton that may incorporate some or all of the principles of the present disclosure.

FIG. 22B illustrates an isometric side view of another example carriage and exoskeleton that may incorporate some or all of the principles of the present disclosure

FIG. 23A illustrates a cross-sectional view of an example carriage mounted to a lead screw that may incorporate some or all of the principles of the present disclosure.

FIG. 23B illustrates a cross-sectional view of another example carriage mounted to a lead screw that may incorporate some or all of the principles of the present disclosure.

FIG. 23C illustrates a cross-sectional view of another example carriage mounted to a lead screw that may incorporate some or all of the principles of the present disclosure.

FIG. 24A illustrates a top view of an example carriage operatively coupled to a plurality of splines.

FIG. 24B illustrates a top view of another example carriage operatively coupled to a plurality of splines.

FIG. 25A illustrates a top view of a plurality of activating mechanisms coupled to a plurality of splines.

FIG. 25B illustrates a top view of an example carriage operatively coupled to the activating mechanisms and splines of FIG. 25A.

FIG. 25C illustrates an isometric view of a surgical tool handle and exoskeleton configured to house the carriage of FIG. 25B.

DETAILED DESCRIPTION

1. Overview.

Aspects of the present disclosure may be integrated into a robotically-enabled medical system capable of performing a variety of medical procedures, including both minimally invasive (e.g., laparoscopy) and non-invasive (e.g., endoscopy) procedures. Among endoscopy procedures, the system may be capable of performing bronchoscopy, ureteroscopy, gastroscopy, etc.

In addition to performing the breadth of procedures, the system may provide additional benefits, such as enhanced imaging and guidance to assist the physician. Additionally, the system may provide the physician with the ability to perform the procedure from an ergonomic position without the need for awkward arm motions and positions. Still further, the system may provide the physician with the ability to perform the procedure with improved ease of use such that one or more of the instruments of the system can be controlled by a single user.

Various embodiments will be described below in conjunction with the drawings for purposes of illustration. It should be appreciated that many other implementations of the disclosed concepts are possible, and various advantages can be achieved with the disclosed implementations. Headings are included herein for reference and to aid in locating various sections. These headings are not intended to limit the scope of the concepts described with respect thereto. Such concepts may have applicability throughout the entire specification.

A. Robotic System—Cart.

The robotically-enabled medical system may be configured in a variety of ways depending on the particular procedure. FIG. 1 illustrates an embodiment of a cart-based robotically-enabled system 100 arranged for a diagnostic and/or therapeutic bronchoscopy procedure. For a bronchoscopy procedure, the robotic system 100 may include a cart 102 having one or more robotic arms 104 (three shown) to deliver a medical instrument (alternately referred to as a

6

“surgical tool”), such as a steerable endoscope 106 (e.g., a procedure-specific bronchoscope for bronchoscopy), to a natural orifice access point (i.e., the mouth of the patient) to deliver diagnostic and/or therapeutic tools. As shown, the cart 102 may be positioned proximate to the patient’s upper torso in order to provide access to the access point. Similarly, the robotic arms 104 may be actuated to position the bronchoscope relative to the access point. The arrangement in FIG. 1 may also be utilized when performing a gastrointestinal (GI) procedure with a gastroscope, a specialized endoscope for GI procedures.

Once the cart 102 is properly positioned adjacent the patient, the robotic arms 104 are operated to insert the steerable endoscope 106 into the patient robotically, manually, or a combination thereof. The steerable endoscope 106 may comprise at least two telescoping parts, such as an inner leader portion and an outer sheath portion, where each portion is coupled to a separate instrument driver of a set of instrument drivers 108. As illustrated, each instrument driver 108 is coupled to the distal end of a corresponding one of the robotic arms 104. This linear arrangement of the instrument drivers 108, which facilitates coaxially aligning the leader portion with the sheath portion, creates a “virtual rail” 110 that may be repositioned in space by manipulating the robotic arms 104 into different angles and/or positions. Translation of the instrument drivers 108 along the virtual rail 110 telescopes the inner leader portion relative to the outer sheath portion, thus effectively advancing or retracting the endoscope 106 relative to the patient.

As illustrated, the virtual rail 110 (and other virtual rails described herein) is depicted in the drawings using dashed lines, thus not constituting any physical structure of the system 100. The angle of the virtual rail 110 may be adjusted, translated, and pivoted based on clinical application or physician preference. For example, in bronchoscopy, the angle and position of the virtual rail 110 as shown represents a compromise between providing physician access to the endoscope 106 while minimizing friction that results from bending the endoscope 106 into the patient’s mouth.

After insertion into the patient’s mouth, the endoscope 106 may be directed down the patient’s trachea and lungs using precise commands from the robotic system 100 until reaching a target destination or operative site. In order to enhance navigation through the patient’s lung network and/or reach the desired target, the endoscope 106 may be manipulated to telescopically extend the inner leader portion from the outer sheath portion to obtain enhanced articulation and greater bend radius. The use of separate instrument drivers 108 also allows the leader portion and sheath portion to be driven independent of each other.

For example, the endoscope 106 may be directed to deliver a biopsy needle to a target, such as, for example, a lesion or nodule within the lungs of a patient. The needle may be deployed down a working channel that runs the length of the endoscope 106 to obtain a tissue sample to be analyzed by a pathologist. Depending on the pathology results, additional tools may be deployed down the working channel of the endoscope for additional biopsies. After identifying a tissue sample to be malignant, the endoscope 106 may endoscopically deliver tools to resect the potentially cancerous tissue. In some instances, diagnostic and therapeutic treatments can be delivered in separate procedures. In those circumstances, the endoscope 106 may also be used to deliver a fiducial marker to “mark” the location

of a target nodule as well. In other instances, diagnostic and therapeutic treatments may be delivered during the same procedure.

The system **100** may also include a movable tower **112**, which may be connected via support cables to the cart **102** to provide support for controls, electronics, fluidics, optics, sensors, and/or power to the cart **102**. Placing such functionality in the tower **112** allows for a smaller form factor cart **102** that may be more easily adjusted and/or re-positioned by an operating physician and his/her staff. Additionally, the division of functionality between the cart/table and the support tower **112** reduces operating room clutter and facilitates improving clinical workflow. While the cart **102** may be positioned close to the patient, the tower **112** may alternatively be stowed in a remote location to stay out of the way during a procedure.

In support of the robotic systems described above, the tower **112** may include component(s) of a computer-based control system that stores computer program instructions, for example, within a non-transitory computer-readable storage medium such as a persistent magnetic storage drive, solid state drive, etc. The execution of those instructions, whether the execution occurs in the tower **112** or the cart **102**, may control the entire system or sub-system(s) thereof. For example, when executed by a processor of the computer system, the instructions may cause the components of the robotics system to actuate the relevant carriages and arm mounts, actuate the robotics arms, and control the medical instruments. For example, in response to receiving the control signal, motors in the joints of the robotic arms **104** may position the arms into a certain posture or angular orientation.

The tower **112** may also include one or more of a pump, flow meter, valve control, and/or fluid access in order to provide controlled irrigation and aspiration capabilities to the system **100** that may be deployed through the endoscope **106**. These components may also be controlled using the computer system of the tower **112**. In some embodiments, irrigation and aspiration capabilities may be delivered directly to the endoscope **106** through separate cable(s).

The tower **112** may include a voltage and surge protector designed to provide filtered and protected electrical power to the cart **102**, thereby avoiding placement of a power transformer and other auxiliary power components in the cart **102**, resulting in a smaller, more moveable cart **102**.

The tower **112** may also include support equipment for sensors deployed throughout the robotic system **100**. For example, the tower **112** may include opto-electronics equipment for detecting, receiving, and processing data received from optical sensors or cameras throughout the robotic system **100**. In combination with the control system, such opto-electronics equipment may be used to generate real-time images for display in any number of consoles deployed throughout the system, including in the tower **112**. Similarly, the tower **112** may also include an electronic subsystem for receiving and processing signals received from deployed electromagnetic (EM) sensors. The tower **112** may also be used to house and position an EM field generator for detection by EM sensors in or on the medical instrument.

The tower **112** may also include a console **114** in addition to other consoles available in the rest of the system, e.g., a console mounted to the cart **102**. The console **114** may include a user interface and a display screen (e.g., a touchscreen) for the physician operator. Consoles in the system **100** are generally designed to provide both robotic controls as well as pre-operative and real-time information of the procedure, such as navigational and localization information

of the endoscope **106**. When the console **114** is not the only console available to the physician, it may be used by a second operator, such as a nurse, to monitor the health or vitals of the patient and the operation of system, as well as provide procedure-specific data, such as navigational and localization information. In other embodiments, the console **114** may be housed in a body separate from the tower **112**.

The tower **112** may be coupled to the cart **102** and endoscope **106** through one or more cables **116** connections. In some embodiments, support functionality from the tower **112** may be provided through a single cable **116** extending to the cart **102**, thus simplifying and de-cluttering the operating room. In other embodiments, specific functionality may be coupled in separate cabling and connections. For example, while power may be provided through a single power cable to the cart **102**, support for controls, optics, fluidics, and/or navigation may be provided through one or more separate cables.

FIG. 2 provides a detailed illustration of an embodiment of the cart **102** from the cart-based robotically-enabled system **100** of FIG. 1. The cart **102** generally includes an elongated support structure **202** (also referred to as a “column”), a cart base **204**, and a console **206** at the top of the column **202**. The column **202** may include one or more carriages, such as a carriage **208** (alternatively “arm support”) for supporting the deployment of the robotic arms **104**. The carriage **208** may include individually configurable arm mounts that rotate along a perpendicular axis to adjust the base **214** of the robotic arms **104** for better positioning relative to the patient. The carriage **208** also includes a carriage interface **210** that allows the carriage **208** to vertically translate along the column **202**.

The carriage interface **210** is connected to the column **202** through slots, such as slot **212**, that are positioned on opposite sides of the column **202** to guide the vertical translation of the carriage **208**. The slot **212** contains a vertical translation interface to position and hold the carriage **208** at various vertical heights relative to the cart base **204**. Vertical translation of the carriage **208** allows the cart **102** to adjust the reach of the robotic arms **104** to meet a variety of table heights, patient sizes, and physician preferences. Similarly, the individually configurable arm mounts on the carriage **208** allow a base **214** of the robotic arms **104** to be angled in a variety of configurations.

In some embodiments, the slot **212** may be supplemented with slot covers (not shown) that are flush and parallel to the slot surface to prevent dirt and fluid ingress into the internal chambers of the column **202** and the vertical translation interface as the carriage **208** vertically translates. The slot covers may be deployed through pairs of spring spools positioned near the vertical top and bottom of the slot **212**. The covers are coiled within the spools until deployed to extend and retract from their coiled state as the carriage **208** vertically translates up and down. The spring-loading of the spools provides force to retract the cover into a spool when carriage **208** translates towards the spool, while also maintaining a tight seal when the carriage **208** translates away from the spool. The covers may be connected to the carriage **208** using, for example, brackets in the carriage interface **210** to ensure proper extension and retraction of the cover as the carriage **208** translates.

The column **202** may internally comprise mechanisms, such as gears and motors, that are designed to use a vertically aligned lead screw to translate the carriage **208** in a mechanized fashion in response to control signals generated in response to user inputs, e.g., inputs from the console **206**.

The robotic arms **104** may generally comprise robotic arm bases **214** and end effectors **216** (three shown), separated by a series of linkages **218** connected by a corresponding series of joints **220**, each joint **220** including an independent actuator, and each actuator including an independently controllable motor. Each independently controllable joint **220** represents an independent degree of freedom available to the corresponding robotic arm **104**. In the illustrated embodiment, each arm **104** has seven joints **220**, thus providing seven degrees of freedom. A multitude of joints **220** result in a multitude of degrees of freedom, allowing for “redundant” degrees of freedom. Redundant degrees of freedom allow the robotic arms **104** to position their respective end effectors **216** at a specific position, orientation, and trajectory in space using different linkage positions and joint angles. This allows for the system **100** to position and direct a medical instrument from a desired point in space while allowing the physician to move the arm joints **220** into a clinically advantageous position away from the patient to create greater access, while avoiding arm collisions.

The cart base **204** balances the weight of the column **202**, carriage **208**, and arms **104** over the floor. Accordingly, the cart base **204** houses heavier components, such as electronics, motors, power supply, as well as components that either enable movement and/or immobilize the cart. For example, the cart base **204** includes rolling casters **222** that allow for the cart to easily move around the room prior to a procedure. After reaching an appropriate position, the casters **222** may be immobilized using wheel locks to hold the cart **102** in place during the procedure.

Positioned at the vertical end of the column **202**, the console **206** allows for both a user interface for receiving user input and a display screen (or a dual-purpose device such as, for example, a touchscreen **224**) to provide the physician user with both pre-operative and intra-operative data. Potential pre-operative data on the touchscreen **224** may include pre-operative plans, navigation and mapping data derived from pre-operative computerized tomography (CT) scans, and/or notes from pre-operative patient interviews. Intra-operative data on the touchscreen **224** may include optical information provided from the tool, sensor and coordinate information from sensors, as well as vital patient statistics, such as respiration, heart rate, and/or pulse. The console **206** may be positioned and tilted to allow a physician to access the console from the side of the column **202** opposite carriage **208**. From this position, the physician may view the console **206**, the robotic arms **104**, and the patient while operating the console **206** from behind the cart **102**. As shown, the console **206** also includes a handle **226** to assist with maneuvering and stabilizing cart **102**.

FIG. 3A illustrates an embodiment of the system **100** of FIG. 1 arranged for ureteroscopy. In a ureteroscopic procedure, the cart **102** may be positioned to deliver a ureteroscope **302**, a procedure-specific endoscope designed to traverse a patient's urethra and ureter, to the lower abdominal area of the patient. In ureteroscopy, it may be desirable for the ureteroscope **302** to be directly aligned with the patient's urethra to reduce friction and forces on the sensitive anatomy. As shown, the cart **102** may be aligned at the foot of the table to allow the robotic arms **104** to position the ureteroscope **302** for direct linear access to the patient's urethra. From the foot of the table, the robotic arms **104** may insert the ureteroscope **302** along a virtual rail **304** directly into the patient's lower abdomen through the urethra.

After insertion into the urethra, using similar control techniques as in bronchoscopy, the ureteroscope **302** may be navigated into the bladder, ureters, and/or kidneys for diag-

nostic and/or therapeutic applications. For example, the ureteroscope **302** may be directed into the ureter and kidneys to break up kidney stone build-up using a laser or ultrasonic lithotripsy device deployed down a working channel of the ureteroscope **302**. After lithotripsy is complete, the resulting stone fragments may be removed using baskets deployed down the working channel of the ureteroscope **302**.

FIG. 3B illustrates another embodiment of the system **100** of FIG. 1 arranged for a vascular procedure. In a vascular procedure, the system **100** may be configured such that the cart **102** may deliver a medical instrument **306**, such as a steerable catheter, to an access point in the femoral artery in the patient's leg. The femoral artery presents both a larger diameter for navigation as well as a relatively less circuitous and tortuous path to the patient's heart, which simplifies navigation. As in a ureteroscopic procedure, the cart **102** may be positioned towards the patient's legs and lower abdomen to allow the robotic arms **104** to provide a virtual rail **308** with direct linear access to the femoral artery access point in the patient's thigh/hip region. After insertion into the artery, the medical instrument **306** may be directed and advanced by translating the instrument drivers **108**. Alternatively, the cart **102** may be positioned around the patient's upper abdomen in order to reach alternative vascular access points, such as, for example, the carotid and brachial arteries near the patient's shoulder and wrist.

B. Robotic System—Table.

Embodiments of the robotically-enabled medical system may also incorporate the patient's table. Incorporation of the table reduces the amount of capital equipment within the operating room by removing the cart, which allows greater access to the patient. FIG. 4 illustrates an embodiment of such a robotically-enabled system **400** arranged for a bronchoscopy procedure. As illustrated, the system **400** includes a support structure or column **402** for supporting platform **404** (shown as a “table” or “bed”) over the floor. Much like in the cart-based systems, the end effectors of the robotic arms **406** of the system **400** comprise instrument drivers **408** that are designed to manipulate an elongated medical instrument, such as a bronchoscope **410**, through or along a virtual rail **412** formed from the linear alignment of the instrument drivers **408**. In practice, a C-arm for providing fluoroscopic imaging may be positioned over the patient's upper abdominal area by placing the emitter and detector around the table **404**.

FIG. 5 provides an alternative view of the system **400** without the patient and medical instrument for discussion purposes. As shown, the column **402** may include one or more carriages **502** shown as ring-shaped in the system **400**, from which the one or more robotic arms **406** may be based. The carriages **502** may translate along a vertical column interface **504** that runs the length (height) of the column **402** to provide different vantage points from which the robotic arms **406** may be positioned to reach the patient. The carriage(s) **502** may rotate around the column **402** using a mechanical motor positioned within the column **402** to allow the robotic arms **406** to have access to multiples sides of the table **404**, such as, for example, both sides of the patient. In embodiments with multiple carriages **502**, the carriages **502** may be individually positioned on the column **402** and may translate and/or rotate independent of the other carriages **502**. While carriages **502** need not surround the column **402** or even be circular, the ring-shape as shown facilitates rotation of the carriages **502** around the column **402** while maintaining structural balance. Rotation and translation of the carriages **502** allows the system **400** to align medical

11

instruments, such as endoscopes and laparoscopes, into different access points on the patient.

In other embodiments (discussed in greater detail below with respect to FIG. 9A), the system 400 can include a patient table or bed with adjustable arm supports in the form of bars or rails extending alongside it. One or more robotic arms 406 (e.g., via a shoulder with an elbow joint) can be attached to the adjustable arm supports, which can be vertically adjusted. By providing vertical adjustment, the robotic arms 406 are advantageously capable of being stowed compactly beneath the patient table or bed, and subsequently raised during a procedure.

The arms 406 may be mounted on the carriages 502 through a set of arm mounts 506 comprising a series of joints that may individually rotate and/or telescopically extend to provide additional configurability to the robotic arms 406. Additionally, the arm mounts 506 may be positioned on the carriages 502 such that when the carriages 502 are appropriately rotated, the arm mounts 506 may be positioned on either the same side of the table 404 (as shown in FIG. 5), on opposite sides of table 404 (as shown in FIG. 7B), or on adjacent sides of the table 404 (not shown).

The column 402 structurally provides support for the table 404, and a path for vertical translation of the carriages 502. Internally, the column 402 may be equipped with lead screws for guiding vertical translation of the carriages, and motors to mechanize the translation of said carriages based the lead screws. The column 402 may also convey power and control signals to the carriage 502 and robotic arms 406 mounted thereon.

A table base 508 serves a similar function as the cart base 204 of the cart 102 shown in FIG. 2, housing heavier components to balance the table/bed 404, the column 402, the carriages 502, and the robotic arms 406. The table base 508 may also incorporate rigid casters to provide stability during procedures. Deployed from the bottom of the table base 508, the casters may extend in opposite directions on both sides of the base 508 and retract when the system 400 needs to be moved.

In some embodiments, the system 400 may also include a tower (not shown) that divides the functionality of system 400 between table and tower to reduce the form factor and bulk of the table 404. As in earlier disclosed embodiments, the tower may provide a variety of support functionalities to the table 404, such as processing, computing, and control capabilities, power, fluidics, and/or optical and sensor processing. The tower may also be movable to be positioned away from the patient to improve physician access and de-clutter the operating room. Additionally, placing components in the tower allows for more storage space in the table base 508 for potential stowage of the robotic arms 406. The tower may also include a master controller or console that provides both a user interface for user input, such as keyboard and/or pendant, as well as a display screen (or touchscreen) for pre-operative and intra-operative information, such as real-time imaging, navigation, and tracking information. In some embodiments, the tower may also contain holders for gas tanks to be used for insufflation.

In some embodiments, a table base may stow and store the robotic arms when not in use. FIG. 6 illustrates an embodiment of the system 400 that is configured to stow robotic arms in an embodiment of the table-based system. In the system 400, one or more carriages 602 (one shown) may be vertically translated into a base 604 to stow one or more robotic arms 606, one or more arm mounts 608, and the carriages 602 within the base 604. Base covers 610 may be translated and retracted open to deploy the carriages 602, the

12

arm mounts 608, and the arms 606 around the column 612, and closed to stow and protect them when not in use. The base covers 610 may be sealed with a membrane 614 along the edges of its opening to prevent dirt and fluid ingress when closed.

FIG. 7A illustrates an embodiment of the robotically-enabled table-based system 400 configured for a ureteroscopy procedure. In ureteroscopy, the table 404 may include a swivel portion 702 for positioning a patient off-angle from the column 402 and the table base 508. The swivel portion 702 may rotate or pivot around a pivot point (e.g., located below the patient's head) in order to position the bottom portion of the swivel portion 702 away from the column 402. For example, the pivoting of the swivel portion 702 allows a C-arm (not shown) to be positioned over the patient's lower abdomen without competing for space with the column (not shown) below table 404. By rotating the carriage (not shown) around the column 402, the robotic arms 406 may directly insert a ureteroscope 704 along a virtual rail 706 into the patient's groin area to reach the urethra. In ureteroscopy, stirrups 708 may also be fixed to the swivel portion 702 of the table 404 to support the position of the patient's legs during the procedure and allow clear access to the patient's groin area.

FIG. 7B illustrates an embodiment of the system 400 configured for a laparoscopic procedure. In a laparoscopic procedure, through small incision(s) in the patient's abdominal wall, minimally invasive instruments may be inserted into the patient's anatomy. In some embodiments, the minimally invasive instruments comprise an elongated rigid member, such as a shaft, which is used to access anatomy within the patient. After inflation of the patient's abdominal cavity, the instruments may be directed to perform surgical or medical tasks, such as grasping, cutting, ablating, suturing, etc. In some embodiments, the instruments can comprise a scope, such as a laparoscope. As shown in FIG. 7B, the carriages 502 of the system 400 may be rotated and vertically adjusted to position pairs of the robotic arms 406 on opposite sides of the table 404, such that an instrument 710 may be positioned using the arm mounts 506 to be passed through minimal incisions on both sides of the patient to reach his/her abdominal cavity.

To accommodate laparoscopic procedures, the system 400 may also tilt the platform to a desired angle. FIG. 7C illustrates an embodiment of the system 400 with pitch or tilt adjustment. As shown in FIG. 7C, the system 400 may accommodate tilt of the table 404 to position one portion of the table 404 at a greater distance from the floor than the other. Additionally, the arm mounts 506 may rotate to match the tilt such that the arms 406 maintain the same planar relationship with table 404. To accommodate steeper angles, the column 402 may also include telescoping portions 712 that allow vertical extension of the column 402 to keep the table 404 from touching the floor or colliding with the base 508.

FIG. 8 provides a detailed illustration of the interface between the table 404 and the column 402. Pitch rotation mechanism 802 may be configured to alter the pitch angle of the table 404 relative to the column 402 in multiple degrees of freedom. The pitch rotation mechanism 802 may be enabled by the positioning of orthogonal axes A and B at the column-table interface, each axis actuated by a separate motor 804a and 804b responsive to an electrical pitch angle command. Rotation along one screw 806a would enable tilt adjustments in one axis A, while rotation along another screw 806b would enable tilt adjustments along the other axis B. In some embodiments, a ball joint can be used to alter

the pitch angle of the table **404** relative to the column **402** in multiple degrees of freedom.

For example, pitch adjustments are particularly useful when trying to position the table in a Trendelenburg position, i.e., position the patient's lower abdomen at a higher position from the floor than the patient's lower abdomen, for lower abdominal surgery. The Trendelenburg position causes the patient's internal organs to slide towards his/her upper abdomen through the force of gravity, clearing out the abdominal cavity for minimally invasive tools to enter and perform lower abdominal surgical or medical procedures, such as laparoscopic prostatectomy.

FIGS. **9A** and **9B** illustrate isometric and end views, respectively, of an alternative embodiment of a table-based surgical robotics system **900**. The surgical robotics system **900** includes one or more adjustable arm supports **902** that can be configured to support one or more robotic arms (see, for example, FIG. **9C**) relative to a table **904**. In the illustrated embodiment, a single adjustable arm support **902** is shown, though an additional arm support can be provided on an opposite side of the table **904**. The adjustable arm support **902** can be configured so that it can move relative to the table **904** to adjust and/or vary the position of the adjustable arm support **902** and/or any robotic arms mounted thereto relative to the table **904**. For example, the adjustable arm support **902** may be adjusted in one or more degrees of freedom relative to the table **904**. The adjustable arm support **902** provides high versatility to the system **900**, including the ability to easily stow the one or more adjustable arm supports **902** and any robotics arms attached thereto beneath the table **904**. The adjustable arm support **902** can be elevated from the stowed position to a position below an upper surface of the table **904**. In other embodiments, the adjustable arm support **902** can be elevated from the stowed position to a position above an upper surface of the table **904**.

The adjustable arm support **902** can provide several degrees of freedom, including lift, lateral translation, tilt, etc. In the illustrated embodiment of FIGS. **9A** and **9B**, the arm support **902** is configured with four degrees of freedom, which are illustrated with arrows in FIG. **9A**. A first degree of freedom allows for adjustment of the adjustable arm support **902** in the z-direction ("Z-lift"). For example, the adjustable arm support **902** can include a carriage **906** configured to move up or down along or relative to a column **908** supporting the table **904**. A second degree of freedom can allow the adjustable arm support **902** to tilt. For example, the adjustable arm support **902** can include a rotary joint, which can allow the adjustable arm support **902** to be aligned with the bed in a Trendelenburg position. A third degree of freedom can allow the adjustable arm support **902** to "pivot up," which can be used to adjust a distance between a side of the table **904** and the adjustable arm support **902**. A fourth degree of freedom can permit translation of the adjustable arm support **902** along a longitudinal length of the table.

The surgical robotics system **900** in FIGS. **9A** and **9B** can comprise a table **904** supported by a column **908** that is mounted to a base **910**. The base **910** and the column **908** support the table **904** relative to a support surface. A floor axis **912** and a support axis **914** are shown in FIG. **9B**.

The adjustable arm support **902** can be mounted to the column **908**. In other embodiments, the arm support **902** can be mounted to the table **904** or the base **910**. The adjustable arm support **902** can include a carriage **906**, a bar or rail connector **916** and a bar or rail **918**. In some embodiments,

one or more robotic arms mounted to the rail **918** can translate and move relative to one another.

The carriage **906** can be attached to the column **908** by a first joint **920**, which allows the carriage **906** to move relative to the column **908** (e.g., such as up and down a first or vertical axis **922**). The first joint **920** can provide the first degree of freedom ("Z-lift") to the adjustable arm support **902**. The adjustable arm support **902** can include a second joint **924**, which provides the second degree of freedom (tilt) for the adjustable arm support **902**. The adjustable arm support **902** can include a third joint **926**, which can provide the third degree of freedom ("pivot up") for the adjustable arm support **902**. An additional joint **928** (shown in FIG. **9B**) can be provided that mechanically constrains the third joint **926** to maintain an orientation of the rail **918** as the rail connector **916** is rotated about a third axis **930**. The adjustable arm support **902** can include a fourth joint **932**, which can provide a fourth degree of freedom (translation) for the adjustable arm support **902** along a fourth axis **934**.

FIG. **9C** illustrates an end view of the surgical robotics system **900** with two adjustable arm supports **902a** and **902b** mounted on opposite sides of the table **904**. A first robotic arm **936a** is attached to the first bar or rail **918a** of the first adjustable arm support **902a**. The first robotic arm **936a** includes a base **938a** attached to the first rail **918a**. The distal end of the first robotic arm **936a** includes an instrument drive mechanism or input **940a** that can attach to one or more robotic medical instruments or tools. Similarly, the second robotic arm **936b** includes a base **938a** attached to the second rail **918b**. The distal end of the second robotic arm **936b** includes an instrument drive mechanism or input **940b** configured to attach to one or more robotic medical instruments or tools.

In some embodiments, one or more of the robotic arms **936a,b** comprises an arm with seven or more degrees of freedom. In some embodiments, one or more of the robotic arms **936a,b** can include eight degrees of freedom, including an insertion axis (1-degree of freedom including insertion), a wrist (3-degrees of freedom including wrist pitch, yaw and roll), an elbow (1-degree of freedom including elbow pitch), a shoulder (2-degrees of freedom including shoulder pitch and yaw), and base **938a,b** (1-degree of freedom including translation). In some embodiments, the insertion degree of freedom can be provided by the robotic arm **936a,b**, while in other embodiments, the instrument itself provides insertion via an instrument-based insertion architecture.

C. Instrument Driver & Interface.

The end effectors of a system's robotic arms comprise (i) an instrument driver (alternatively referred to as "instrument drive mechanism," "instrument device manipulator," and "drive input") that incorporate electro-mechanical means for actuating the medical instrument, and (ii) a removable or detachable medical instrument, which may be devoid of any electro-mechanical components, such as motors. This dichotomy may be driven by the need to sterilize medical instruments used in medical procedures, and the inability to adequately sterilize expensive capital equipment due to their intricate mechanical assemblies and sensitive electronics. Accordingly, the medical instruments may be designed to be detached, removed, and interchanged from the instrument driver (and thus the system) for individual sterilization or disposal by the physician or the physician's staff. In contrast, the instrument drivers need not be changed or sterilized, and may be draped for protection.

FIG. **10** illustrates an example instrument driver **1000**, according to one or more embodiments. Positioned at the distal end of a robotic arm, the instrument driver **1000**

comprises of one or more drive outputs **1002** arranged with parallel axes to provide controlled torque to a medical instrument via corresponding drive shafts **1004**. Each drive output **1002** comprises an individual drive shaft **1004** for interacting with the instrument, a gear head **1006** for converting the motor shaft rotation to a desired torque, a motor **1008** for generating the drive torque, and an encoder **1010** to measure the speed of the motor shaft and provide feedback to control circuitry **1012**, which can also be used for receiving control signals and actuating the drive output **1002**. Each drive output **1002** being independent controlled and motorized, the instrument driver **1000** may provide multiple (at least two shown in FIG. 10) independent drive outputs to the medical instrument. In operation, the control circuitry **1012** receives a control signal, transmits a motor signal to the motor **1008**, compares the resulting motor speed as measured by the encoder **1010** with the desired speed, and modulates the motor signal to generate the desired torque.

For procedures that require a sterile environment, the robotic system may incorporate a drive interface, such as a sterile adapter connected to a sterile drape, that sits between the instrument driver and the medical instrument. The chief purpose of the sterile adapter is to transfer angular motion from the drive shafts of the instrument driver to the drive inputs of the instrument while maintaining physical separation, and thus sterility, between the drive shafts and drive inputs. Accordingly, an example sterile adapter may comprise of a series of rotational inputs and outputs intended to be mated with the drive shafts of the instrument driver and drive inputs on the instrument. Connected to the sterile adapter, the sterile drape, comprised of a thin, flexible material such as transparent or translucent plastic, is designed to cover the capital equipment, such as the instrument driver, robotic arm, and cart (in a cart-based system) or table (in a table-based system). Use of the drape would allow the capital equipment to be positioned proximate to the patient while still being located in an area not requiring sterilization (i.e., non-sterile field). On the other side of the sterile drape, the medical instrument may interface with the patient in an area requiring sterilization (i.e., sterile field). D. Medical Instrument.

FIG. 11 illustrates an example medical instrument **1100** with a paired instrument driver **1102**. Like other instruments designed for use with a robotic system, the medical instrument **1100** (alternately referred to as a "surgical tool") comprises an elongated shaft **1104** (or elongate body) and an instrument base **1106**. The instrument base **1106**, also referred to as an "instrument handle" due to its intended design for manual interaction by the physician, may generally comprise rotatable drive inputs **1108**, e.g., receptacles, pulleys or spools, that are designed to be mated with drive outputs **1110** that extend through a drive interface on the instrument driver **1102** at the distal end of a robotic arm **1112**. When physically connected, latched, and/or coupled, the mated drive inputs **1108** of the instrument base **1106** may share axes of rotation with the drive outputs **1110** in the instrument driver **1102** to allow the transfer of torque from the drive outputs **1110** to the drive inputs **1108**. In some embodiments, the drive outputs **1110** may comprise splines that are designed to mate with receptacles on the drive inputs **1108**.

The elongated shaft **1104** is designed to be delivered through either an anatomical opening or lumen, e.g., as in endoscopy, or a minimally invasive incision, e.g., as in laparoscopy. The elongated shaft **1104** may be either flexible (e.g., having properties similar to an endoscope) or rigid

(e.g., having properties similar to a laparoscope) or contain a customized combination of both flexible and rigid portions. When designed for laparoscopy, the distal end of the shaft **1104** may be connected to an end effector extending from a jointed wrist formed from a clevis with at least one degree of freedom and a surgical tool or medical instrument, such as, for example, a grasper or scissors, that may be actuated based on force from the tendons as the drive inputs **1008** rotate in response to torque received from the drive outputs **1110** of the instrument driver **1102**. When designed for endoscopy, the distal end of the flexible elongated shaft **1104** may include a steerable or controllable bending section that may be articulated and bent based on torque received from the drive outputs **1110** of the instrument driver **1102**.

The elongated shaft **1104** is designed to be delivered through either an anatomical opening or lumen, e.g., as in endoscopy, or a minimally invasive incision, e.g., as in laparoscopy. The elongated shaft **1104** may be either flexible (e.g., having properties similar to an endoscope) or rigid (e.g., having properties similar to a laparoscope) or contain a customized combination of both flexible and rigid portions. When designed for laparoscopy, the distal end of the shaft **1104** may be connected to an end effector extending from a jointed wrist formed from a clevis with at least one degree of freedom and a surgical tool or medical instrument, such as, for example, a grasper or scissors, that may be actuated based on force from the tendons as the drive inputs **1008** rotate in response to torque received from the drive outputs **1110** of the instrument driver **1102**. When designed for endoscopy, the distal end of the flexible elongated shaft **1104** may include a steerable or controllable bending section that may be articulated and bent based on torque received from the drive outputs **1110** of the instrument driver **1102**.

In endoscopy, the tendons may be coupled to a bending or articulating section positioned along the elongated shaft **1104** (e.g., at the distal end) via adhesive, control ring, or other mechanical fixation. When fixedly attached to the distal end of a bending section, torque exerted on drive inputs **1108** would be transmitted down the tendons, causing the softer, bending section (sometimes referred to as the articulable section or region) to bend or articulate. Along the non-bending sections, it may be advantageous to spiral or helix the individual pull lumens that direct the individual tendons along (or inside) the walls of the endoscope shaft to balance the radial forces that result from tension in the pull wires. The angle of the spiraling and/or spacing there between may be altered or engineered for specific purposes, wherein tighter spiraling exhibits lesser shaft compression under load forces, while lower amounts of spiraling results in greater shaft compression under load forces, but also exhibits limits bending. On the other end of the spectrum, the pull lumens may be directed parallel to the longitudinal axis of the elongated shaft **1104** to allow for controlled articulation in the desired bending or articulable sections.

In endoscopy, the elongated shaft **1104** houses a number of components to assist with the robotic procedure. The shaft may comprise of a working channel for deploying surgical tools (or medical instruments), irrigation, and/or aspiration to the operative region at the distal end of the shaft **1104**. The shaft **1104** may also accommodate wires and/or optical fibers to transfer signals to/from an optical assembly at the distal tip, which may include of an optical camera. The shaft **1104** may also accommodate optical fibers to carry light from proximally-located light sources, such as light emitting diodes, to the distal end of the shaft.

At the distal end of the instrument **1100**, the distal tip may also comprise the opening of a working channel for deliv-

17

ering tools for diagnostic and/or therapy, irrigation, and aspiration to an operative site. The distal tip may also include a port for a camera, such as a fiberscope or a digital camera, to capture images of an internal anatomical space. Relatedly, the distal tip may also include ports for light sources for illuminating the anatomical space when using the camera.

In the example of FIG. 11, the drive shaft axes, and thus the drive input axes, are orthogonal to the axis of the elongated shaft. This arrangement, however, complicates roll capabilities for the elongated shaft 1104. Rolling the elongated shaft 1104 along its axis while keeping the drive inputs 1108 static results in undesirable tangling of the tendons as they extend off the drive inputs 1108 and enter pull lumens within the elongated shaft 1104. The resulting entanglement of such tendons may disrupt any control algorithms intended to predict movement of the flexible elongated shaft during an endoscopic procedure.

FIG. 12 illustrates an alternative design for a circular instrument driver 1200 and corresponding instrument 1202 (alternately referred to as a “surgical tool”) where the axes of the drive units are parallel to the axis of the elongated shaft 1206 of the instrument 1202. As shown, the instrument driver 1200 comprises four drive units with corresponding drive outputs 1208 aligned in parallel at the end of a robotic arm 1210. The drive units and their respective drive outputs 1208 are housed in a rotational assembly 1212 of the instrument driver 1200 that is driven by one of the drive units within the assembly 1212. In response to torque provided by the rotational drive unit, the rotational assembly 1212 rotates along a circular bearing that connects the rotational assembly 1212 to a non-rotational portion 1214 of the instrument driver 1200. Power and control signals may be communicated from the non-rotational portion 1214 of the instrument driver 1200 to the rotational assembly 1212 through electrical contacts maintained through rotation by a brushed slip ring connection (not shown). In other embodiments, the rotational assembly 1212 may be responsive to a separate drive unit that is integrated into the non-rotatable portion 1214, and thus not in parallel with the other drive units. The rotational assembly 1212 allows the instrument driver 1200 to rotate the drive units and their respective drive outputs 1208 as a single unit around an instrument driver axis 1216.

Like earlier disclosed embodiments, the instrument 1202 may include an elongated shaft 1206 and an instrument base 1218 (shown in phantom) including a plurality of drive inputs 1220 (such as receptacles, pulleys, and spools) that are configured to mate with the drive outputs 1208 of the instrument driver 1200. Unlike prior disclosed embodiments, the instrument shaft 1206 extends from the center of the instrument base 1218 with an axis substantially parallel to the axes of the drive inputs 1220, rather than orthogonal as in the design of FIG. 11.

When coupled to the rotational assembly 1212 of the instrument driver 1200, the medical instrument 1202, comprising instrument base 1218 and instrument shaft 1206, rotates in combination with the rotational assembly 1212 about the instrument driver axis 1216. Since the instrument shaft 1206 is positioned at the center of the instrument base 1218, the instrument shaft 1206 is coaxial with the instrument driver axis 1216 when attached. Thus, rotation of the rotational assembly 1212 causes the instrument shaft 1206 to rotate about its own longitudinal axis. Moreover, as the instrument base 1218 rotates with the instrument shaft 1206, any tendons connected to the drive inputs 1220 in the instrument base 1218 are not tangled during rotation.

18

Accordingly, the parallelism of the axes of the drive outputs 1208, the drive inputs 1220, and the instrument shaft 1206 allows for the shaft rotation without tangling any control tendons.

FIG. 13 illustrates a medical instrument 1300 having an instrument based insertion architecture in accordance with some embodiments. The instrument 1300 (alternately referred to as a “surgical tool”) can be coupled to any of the instrument drivers discussed herein above and, as illustrated, can include an elongated shaft 1302, an end effector 1304 connected to the shaft 1302, and a handle 1306 coupled to the shaft 1302. The elongated shaft 1302 comprises a tubular member having a proximal portion 1308a and a distal portion 1308b. The elongated shaft 1302 comprises one or more channels or grooves 1310 along its outer surface and configured to receive one or more wires or cables 1312 therethrough. One or more cables 1312 thus run along an outer surface of the elongated shaft 1302. In other embodiments, the cables 1312 can also run through the elongated shaft 1302. Manipulation of the cables 1312 (e.g., via an instrument driver) results in actuation of the end effector 1304.

The instrument handle 1306, which may also be referred to as an instrument base, may generally comprise an attachment interface 1314 having one or more mechanical inputs 1316, e.g., receptacles, pulleys or spools, that are designed to be reciprocally mated with one or more drive outputs on an attachment surface of an instrument driver.

In some embodiments, the instrument 1300 comprises a series of pulleys or cables that enable the elongated shaft 1302 to translate relative to the handle 1306. In other words, the instrument 1300 itself comprises an instrument-based insertion architecture that accommodates insertion of the instrument, thereby minimizing the reliance on a robot arm to provide insertion of the instrument 1300. In other embodiments, a robotic arm can be largely responsible for instrument insertion.

E. Controller.

Any of the robotic systems described herein can include an input device or controller for manipulating an instrument attached to a robotic arm. In some embodiments, the controller can be coupled (e.g., communicatively, electronically, electrically, wirelessly and/or mechanically) with an instrument such that manipulation of the controller causes a corresponding manipulation of the instrument e.g., via master slave control.

FIG. 14 is a perspective view of an embodiment of a controller 1400. In the present embodiment, the controller 1400 comprises a hybrid controller that can have both impedance and admittance control. In other embodiments, the controller 1400 can utilize just impedance or passive control. In other embodiments, the controller 1400 can utilize just admittance control. By being a hybrid controller, the controller 1400 advantageously can have a lower perceived inertia while in use.

In the illustrated embodiment, the controller 1400 is configured to allow manipulation of two medical instruments, and includes two handles 1402. Each of the handles 1402 is connected to a gimbal 1404, and each gimbal 1404 is connected to a positioning platform 1406.

As shown in FIG. 14, each positioning platform 1406 includes a selective compliance assembly robot arm (SCARA) 1408 coupled to a column 1410 by a prismatic joint 1412. The prismatic joints 1412 are configured to translate along the column 1410 (e.g., along rails 1414) to allow each of the handles 1402 to be translated in the z-direction, providing a first degree of freedom. The

SCARA arm **1408** is configured to allow motion of the handle **1402** in an x-y plane, providing two additional degrees of freedom.

In some embodiments, one or more load cells are positioned in the controller **1400**. For example, in some embodiments, a load cell (not shown) is positioned in the body of each of the gimbals **1404**. By providing a load cell, portions of the controller **1400** are capable of operating under admittance control, thereby advantageously reducing the perceived inertia of the controller **1400** while in use. In some embodiments, the positioning platform **1406** is configured for admittance control, while the gimbal **1404** is configured for impedance control. In other embodiments, the gimbal **1404** is configured for admittance control, while the positioning platform **1406** is configured for impedance control. Accordingly, for some embodiments, the translational or positional degrees of freedom of the positioning platform **1406** can rely on admittance control, while the rotational degrees of freedom of the gimbal **1404** rely on impedance control.

F. Navigation and Control.

Traditional endoscopy may involve the use of fluoroscopy (e.g., as may be delivered through a C-arm) and other forms of radiation-based imaging modalities to provide endoluminal guidance to an operator physician. In contrast, the robotic systems contemplated by this disclosure can provide for non-radiation-based navigational and localization means to reduce physician exposure to radiation and reduce the amount of equipment within the operating room. As used herein, the term “localization” may refer to determining and/or monitoring the position of objects in a reference coordinate system. Technologies such as pre-operative mapping, computer vision, real-time EM tracking, and robot command data may be used individually or in combination to achieve a radiation-free operating environment. In other cases, where radiation-based imaging modalities are still used, the pre-operative mapping, computer vision, real-time EM tracking, and robot command data may be used individually or in combination to improve upon the information obtained solely through radiation-based imaging modalities.

FIG. **15** is a block diagram illustrating a localization system **1500** that estimates a location of one or more elements of the robotic system, such as the location of the instrument, in accordance to an example embodiment. The localization system **1500** may be a set of one or more computer devices configured to execute one or more instructions. The computer devices may be embodied by a processor (or processors) and computer-readable memory in one or more components discussed above. By way of example and not limitation, the computer devices may be in the tower **112** shown in FIG. **1**, the cart **102** shown in FIGS. **1-3B**, the beds shown in FIGS. **4-9**, etc.

As shown in FIG. **15**, the localization system **1500** may include a localization module **1502** that processes input data **1504a**, **1504b**, **1504c**, and **1504d** to generate location data **1506** for the distal tip of a medical instrument. The location data **1506** may be data or logic that represents a location and/or orientation of the distal end of the instrument relative to a frame of reference. The frame of reference can be a frame of reference relative to the anatomy of the patient or to a known object, such as an EM field generator (see discussion below for the EM field generator).

The various input data **1504a-d** are now described in greater detail. Pre-operative mapping may be accomplished through the use of the collection of low dose CT scans. Pre-operative CT scans are reconstructed into three-dimensional images, which are visualized, e.g. as “slices” of a

cutaway view of the patient’s internal anatomy. When analyzed in the aggregate, image-based models for anatomical cavities, spaces and structures of the patient’s anatomy, such as a patient lung network, may be generated. Techniques such as center-line geometry may be determined and approximated from the CT images to develop a three-dimensional volume of the patient’s anatomy, referred to as model data **1504a** (also referred to as “preoperative model data” when generated using only preoperative CT scans). The use of center-line geometry is discussed in U.S. patent application Ser. No. 14/523,760, the contents of which are herein incorporated in its entirety. Network topological models may also be derived from the CT-images, and are particularly appropriate for bronchoscopy.

In some embodiments, the instrument may be equipped with a camera to provide vision data **1504b**. The localization module **1502** may process the vision data **1504b** to enable one or more vision-based location tracking. For example, the preoperative model data may be used in conjunction with the vision data **1504b** to enable computer vision-based tracking of the medical instrument (e.g., an endoscope or an instrument advance through a working channel of the endoscope). For example, using the preoperative model data **1504a**, the robotic system may generate a library of expected endoscopic images from the model based on the expected path of travel of the endoscope, each image linked to a location within the model. Intra-operatively, this library may be referenced by the robotic system in order to compare real-time images captured at the camera (e.g., a camera at a distal end of the endoscope) to those in the image library to assist localization.

Other computer vision-based tracking techniques use feature tracking to determine motion of the camera, and thus the endoscope. Some features of the localization module **1502** may identify circular geometries in the preoperative model data **1504a** that correspond to anatomical lumens and track the change of those geometries to determine which anatomical lumen was selected, as well as the relative rotational and/or translational motion of the camera. Use of a topological map may further enhance vision-based algorithms or techniques.

Optical flow, another computer vision-based technique, may analyze the displacement and translation of image pixels in a video sequence in the vision data **1504b** to infer camera movement. Examples of optical flow techniques may include motion detection, object segmentation calculations, luminance, motion compensated encoding, stereo disparity measurement, etc. Through the comparison of multiple frames over multiple iterations, movement and location of the camera (and thus the endoscope) may be determined.

The localization module **1502** may use real-time EM tracking to generate a real-time location of the endoscope in a global coordinate system that may be registered to the patient’s anatomy, represented by the preoperative model. In EM tracking, an EM sensor (or tracker) comprising of one or more sensor coils embedded in one or more locations and orientations in a medical instrument (e.g., an endoscopic tool) measures the variation in the EM field created by one or more static EM field generators positioned at a known location. The location information detected by the EM sensors is stored as EM data **1504c**. The EM field generator (or transmitter), may be placed close to the patient to create a low intensity magnetic field that the embedded sensor may detect. The magnetic field induces small currents in the sensor coils of the EM sensor, which may be analyzed to determine the distance and angle between the EM sensor and the EM field generator. These distances and orientations may

be intra-operatively “registered” to the patient anatomy (e.g., the preoperative model) in order to determine the geometric transformation that aligns a single location in the coordinate system with a position in the pre-operative model of the patient’s anatomy. Once registered, an embedded EM tracker in one or more positions of the medical instrument (e.g., the distal tip of an endoscope) may provide real-time indications of the progression of the medical instrument through the patient’s anatomy.

Robotic command and kinematics data **1504d** may also be used by the localization module **1502** to provide localization data **1506** for the robotic system. Device pitch and yaw resulting from articulation commands may be determined during pre-operative calibration. Intra-operatively, these calibration measurements may be used in combination with known insertion depth information to estimate the position of the instrument. Alternatively, these calculations may be analyzed in combination with EM, vision, and/or topological modeling to estimate the position of the medical instrument within the network.

As FIG. 15 shows, a number of other input data can be used by the localization module **1502**. For example, although not shown in FIG. 15, an instrument utilizing shape-sensing fiber can provide shape data that the localization module **1502** can use to determine the location and shape of the instrument.

The localization module **1502** may use the input data **1504a-d** in combination(s). In some cases, such a combination may use a probabilistic approach where the localization module **1502** assigns a confidence weight to the location determined from each of the input data **1504a-d**. Thus, where the EM data **1504c** may not be reliable (as may be the case where there is EM interference) the confidence of the location determined by the EM data **1504c** can be decrease and the localization module **1502** may rely more heavily on the vision data **1504b** and/or the robotic command and kinematics data **1504d**.

As discussed above, the robotic systems discussed herein may be designed to incorporate a combination of one or more of the technologies above. The robotic system’s computer-based control system, based in the tower, bed and/or cart, may store computer program instructions, for example, within a non-transitory computer-readable storage medium such as a persistent magnetic storage drive, solid state drive, or the like, that, upon execution, cause the system to receive and analyze sensor data and user commands, generate control signals throughout the system, and display the navigational and localization data, such as the position of the instrument within the global coordinate system, anatomical map, etc.

2. Introduction.

Embodiments of the disclosure relate to systems and techniques for optimizing torque transfer and assuming torsional loading of a surgical tool. The surgical tool may include a handle having a first end and a second end, a lead screw rotatably coupled to the handle and extending between the first and second ends, and a spline rotatably coupled to the handle and extending between the first and second ends. The spline is arranged at a distance from the lead screw such that the lead screw assumes torsional loading generated by the spline. A surgical tool with first and second splines with a first activating mechanism supported by the carriage and coupled to the first spline and a second activating mechanism supported by the carriage and coupled to the second spline may balance a first torsional load generated by the first spline rotating in a first direction with

a second torsional load generated by the second spline rotating in a second direction opposite the first direction.

3. Description.

FIG. 16A is an isometric side view of an example surgical tool **1600** that may incorporate some or all of the principles of the present disclosure. The surgical tool **1600** may be similar in some respects to any of the medical instruments described above with reference to FIGS. 11-13 and, therefore, may be used in conjunction with a robotic surgical system, such as the robotically-enabled systems **100**, **400**, and **900** of FIGS. 1-13. As illustrated, the surgical tool **1600** includes an elongated shaft **1602**, an end effector **1604** arranged at the distal end of the shaft **1602**, and an articulating wrist **1606** (alternately referred to as a “wrist joint”) that interposes and couples the end effector **1604** to the distal end of the shaft **1602**.

The terms “proximal” and “distal” are defined herein relative to a robotic surgical system having an interface configured to mechanically and electrically couple the surgical tool **1600** to a robotic manipulator. The term “proximal” refers to the position of an element closer to the robotic manipulator and the term “distal” refers to the position of an element closer to the end effector **1604** and thus closer to the patient during operation. Moreover, the use of directional terms such as above, below, upper, lower, upward, downward, left, right, and the like are used in relation to the illustrative embodiments as they are depicted in the figures, the upward or upper direction being toward the top of the corresponding figure and the downward or lower direction being toward the bottom of the corresponding figure.

The surgical tool **1600** can have any of a variety of configurations capable of performing one or more surgical functions. In the illustrated embodiment, the end effector **1604** comprises a surgical stapler, alternately referred to as an “endocutter,” configured to cut and staple (fasten) tissue. As illustrated, the end effector **1604** includes opposing jaws **1610**, **1612** configured to move (articulate) between open and closed positions. Alternatively, the end effector **1604** may comprise other types of instruments requiring opposing jaws such as, but not limited to, tissue graspers, surgical scissors, advanced energy vessel sealers, clip applicators, needle drivers, a babcock including a pair of opposed grasping jaws, bipolar jaws (e.g., bipolar Maryland grasper, forceps, a fenestrated grasper, etc.), etc. In other embodiments, the end effector **1604** may instead comprise any end effector or instrument capable of being operated in conjunction with the presently disclosed robotic surgical systems and methods. Such end effectors or instruments include, but are not limited to, a suction irrigator, an endoscope (e.g., a camera), or any combination thereof.

One or both of the jaws **1610**, **1612** may be configured to pivot to actuate the end effector **1604** between open and closed positions. In the illustrated example, the second jaw **1612** is rotatable (pivotable) relative to the first jaw **1610** to move between an open, unclamped position and a closed, clamped position. In other embodiments, however, the first jaw **1610** may move (rotate) relative to the second jaw **1612**, without departing from the scope of the disclosure. In yet other embodiments, both jaws **1610**, **1612** may move to actuate the end effector **1604** between open and closed positions.

In the illustrated example, the first jaw **1610** is referred to as a “cartridge” or “channel” jaw, and the second jaw **1612** is referred to as an “anvil” jaw. The first jaw **1610** may include a frame that houses or supports a staple cartridge, and the second jaw **1612** is pivotally supported relative to

the first jaw **1610** and defines a surface that operates as an anvil to deform staples ejected from the staple cartridge during operation.

The wrist **1606** enables the end effector **1604** to articulate (pivot) relative to the shaft **1602** and thereby position the end effector **1604** at various desired orientations and locations relative to a surgical site. In the illustrated embodiment, the wrist **1606** is designed to allow the end effector **1604** to pivot (swivel) left and right relative to a longitudinal axis A_i of the shaft **1602**. In other embodiments, however, the wrist **1606** may be designed to provide multiple degrees of freedom, including one or more translational variables (i.e., surge, heave, and sway) and/or one or more rotational variables (i.e., Euler angles or roll, pitch, and yaw). The translational and rotational variables describe the position and orientation of a component of a surgical system (e.g., the end effector **1604**) with respect to a given reference Cartesian frame. “Surge” refers to forward and backward translational movement, “heave” refers to translational movement up and down, and “sway” refers to translational movement left and right. With regard to the rotational terms, “roll” refers to tilting side to side, “pitch” refers to tilting forward and backward, and “yaw” refers to turning left and right.

In the illustrated embodiment, the pivoting motion at the wrist **1606** is limited to movement in a single plane, e.g., only yaw movement relative to the longitudinal axis A_1 . The end effector **1604** is depicted in FIG. **16A** in the unarticulated position where the longitudinal axis of the end effector **1604** is substantially aligned with the longitudinal axis A_1 of the shaft **1602**, such that the end effector **1604** is at a substantially zero angle relative to the shaft **1602**. In the articulated position, the longitudinal axis of the end effector **1604** would be angularly offset from the longitudinal axis A_1 such that the end effector **1604** would be oriented at a non-zero angle relative to the shaft **1602**.

Still referring to FIG. **16A**, the surgical tool **1600** may include a drive housing or “handle” **1614** that operates as an actuation system designed to facilitate articulation of the wrist **1606** and actuation (operation) of the end effector **1604** (e.g., clamping, firing, rotation, articulation, energy delivery, etc.). As described in more detail below, the handle **1614** provides various coupling features that releasably couple the surgical tool **1600** to an instrument driver of a robotic surgical system.

The handle **1614** includes a plurality of drive members (observed in FIG. **16A**) that extend to the wrist **1606** and the end effector **1604**. Selective actuation of one or more of the drive members causes the end effector **1604** to articulate (pivot) relative to the shaft **1602** at the wrist **1606**. Selective actuation of one or more other drive members causes the end effector **1604** to actuate (operate). Actuating the end effector **1604** may include closing and/or opening the jaws, **1610**, **1612**, and thereby enabling the end effector **1604** to grasp (clamp) onto tissue. Once tissue is grasped or clamped between the opposing jaws **1610**, **1612**, actuating the end effector **1604** may further include “firing” the end effector **1604**, which may refer to causing a cutting element or knife (not visible) to advance distally within a slot **1616** defined in the first jaw **1610**. As it moves distally, the cutting element transects any tissue grasped between the opposing jaws **1610**, **1612**. Moreover, as the cutting element advances distally, a plurality of staples contained within the staple cartridge (e.g., housed within the first jaw **1610**) are urged (cammed) into deforming contact with corresponding anvil surfaces (e.g., pockets) provided on the second jaw **1612**. The deployed staples may form multiple rows of staples that seal opposing sides of the transected tissue.

As illustrated, the handle **1614** has a first or “distal” end **1618a** and a second or “proximal” end **1618b** opposite the first end **1618a**. In some embodiments, one or more struts **1620** (two shown) extend longitudinally between the first and second ends **1618a,b** to help fix the distance between the first and second ends **1618a,b**, provide structural stability to the handle **1614**, and secure the first end **1618a** to the second end **1618b**. In other embodiments, however, the struts **1620** may be omitted, without departing from the scope of the disclosure.

The handle **1614** may also include a lead screw **1622** and one or more splines **1624**, which also extend longitudinally between the first and second ends **1618a,b**. In the illustrated embodiment, the handle **1614** includes a first spline **1624a**, a second spline **1624b**, and a third spline **1624c**. While three splines **1624a-c** are depicted in the handle **1614**, more or less than three may be included, without departing from the scope of the disclosure. Unlike the struts **1620**, the lead screw **1622** and the splines **1624a-c** are rotatably mounted to the first and second ends **1618a,b**. As described in more detail below, selective rotation of the lead screw **1622** and the splines **1624a-c** causes various functions of the handle **1614** to transpire, such as translating the end effector **1604** along the longitudinal axis A_1 (e.g., z-axis translation) causing the end effector **1604** to articulate (pivot) at the wrist **1606**, causing the jaws **1610**, **1612** to open and close, and causing the end effector **1604** to fire (operate).

The handle **1614** further includes a carriage **1626** movably mounted along the lead screw **1622** and the splines **1624a-c** and housing various activating mechanisms configured to cause operation of specific functions of the end effector **1604**. The carriage **1626** may comprise two or more layers, shown in FIG. **16A** as a first layer **1628a**, a second layer **1628b**, a third layer **1628c**, a fourth layer **1628d**, and a fifth layer **1628e**. The lead screw **1622** and the splines **1624a-c** each extend through portions of one or more of the layers **1628a-e** to allow the carriage **1626** to translate along the longitudinal axis A_i with respect to the lead screw **1622** and the splines **1624a-c**. In some embodiments, the layers **1628a-e** may be secured to each other in series using one or more mechanical fasteners **1630** (two visible) extending between the first layer **1628a** and the fifth layer **1628e** and through coaxially aligned holes defined in some or all of the layers **1628a-e**. While five layers **1628a-e** are depicted, more or less than five may be included in the carriage **1626**, without departing from the scope of the disclosure.

The shaft **1602** is coupled to and extends distally from the carriage **1626** through the first end **1618a** of the handle **1614**. In the illustrated embodiment, for example, the shaft **1602** penetrates the first end **1618a** at a central aperture **1632** defined through the first end **1618a**. The carriage **1626** is movable between the first and second ends **1618a,b** along the longitudinal axis A_1 (e.g., z-axis translation) and is thereby able to advance or retract the end effector **1604** relative to the handle **1614**, as indicated by the arrows **B**. More specifically, in some embodiments, the carriage **1626** includes a carriage nut **1634** mounted to the lead screw **1622** and secured between the third and fourth layers **1628c,d**. The outer surface of the lead screw **1622** defines outer helical threading and the carriage nut **1634** defines corresponding internal helical threading (not shown) matable with the outer helical threading of the lead screw **1622**. As a result, rotation of the lead screw **1622** causes the carriage nut **1634** to advance or retract the carriage **1626** along the longitudinal axis A_i and correspondingly advance or retract the end effector **1604** relative to the handle **1614**.

As indicated above, the lead screw **1622** and the splines **1624a-c** are rotatably mounted to the first and second ends **1618a,b**. More specifically, the first end **1618a** of the handle **1614** may include one or more rotatable drive inputs actuable to independently drive (rotate) the lead screw **1622** and the splines **1624a-c**. In the illustrated embodiment, the handle **1614** includes a first drive input **1636a**, a second drive input **1636b**, a third drive input **1636c** (occluded by the shaft **1602**, see FIG. 17B), and a fourth drive input **1636d**. As described below, each drive input **1636a-d** may be matable with a corresponding drive output of an instrument driver such that movement (rotation) of a given drive output correspondingly moves (rotates) the associated drive input **1636a-d** and thereby rotates the mated lead screw **1622** or spline **1624a-c**. While only four drive inputs **1636a-d** are depicted, more or less than four may be included in the handle **1614**, depending on the application.

The first drive input **1636a** may be operatively coupled to the lead screw **1622** such that rotation of the first drive input **1636a** correspondingly rotates the lead screw **1622**, which causes the carriage nut **1634** and the carriage **1626** to advance or retract along the longitudinal axis A_1 , depending on the rotational direction of the lead screw **1622**. As used herein the phrase “operatively coupled” refers to a coupled engagement, either directly or indirectly, where movement of one component causes corresponding movement of another component. With respect to the first drive input **1636a** being operatively coupled to the lead screw **1622**, such operative coupling may be facilitated through intermeshed gears (not shown) arranged within the second end **1618a**, but could alternatively be facilitated through other mechanical means, such as cables, pulleys, drive rods, direct couplings, etc., without departing from the scope of the disclosure.

The second drive input **1636b** may be operatively coupled to the first spline **1624a** such that rotation of the second drive input **1636b** correspondingly rotates the first spline **1624a**. In some embodiments, the first spline **1624a** may be operatively coupled to a first activating mechanism **1638a** of the carriage **1626**, and the first activating mechanism **1638a** may be operable to open and close the jaws **1610**, **1612**. Accordingly, rotating the second drive input **1636b** will correspondingly actuate the first activating mechanism **1638a** and thereby open or close the jaws **1610**, **1612**, depending on the rotational direction of the first spline **1624a**.

The third drive input **1636c** may be operatively coupled to the second spline **1624b** such that rotation of the third drive input **1636c** correspondingly rotates the second spline **1624b**. In some embodiments, the second spline **1624b** may be operatively coupled to a second activating mechanism **1638b** of the carriage **1626**, and the second activating mechanism **1638b** may be operable to articulate the end effector **1604** at the wrist **1606**. Accordingly, rotating the third drive input **1636c** will correspondingly actuate the second activating mechanism **1638b** and thereby cause the wrist **1606** to articulate in at least one degree of freedom, depending on the rotational direction of the second spline **1624b**.

The fourth drive input **1636d** may be operatively coupled to the third spline **1624c** such that rotation of the fourth drive input **1636d** correspondingly rotates the third spline **1624c**. In some embodiments, the third spline **1624c** may be operatively coupled to a third activating mechanism **1638c** of the carriage **1626**, and the third activating mechanism **1638c** may be operable to fire the cutting element (knife) at the end effector **1604**. Accordingly, rotating the fourth drive input

1636d will correspondingly actuate the third activating mechanism **1638c** and thereby cause the knife to advance or retract, depending on the rotational direction of the third spline **1624c**.

In the illustrated embodiment, and as described in more detail below, the activating mechanisms **1838a-c** comprise intermeshed gearing assemblies including one or more drive gears driven by rotation of the corresponding spline **1624a-c** and configured to drive one or more corresponding driven gears that cause operation of specific functions of the end effector **1604**.

In some embodiments, the handle **1614** may include a shroud **1640** sized to receive and otherwise surround the carriage **1626**, the lead screw **1622**, and the splines **1624a-c**. In the illustrated embodiment, the shroud **1640** comprises a tubular or cylindrical structure having a first end **1642a** matable with the first end **1618a** of the handle **1614**, and a second end **1642b** matable with the second end **1618b** of the handle **1614**. The carriage **1626**, the lead screw **1622**, and the splines **1624a-c** can all be accommodated within the interior of the shroud **1640**, and the carriage **1626** may engage and traverse (ride on) one or more rails **1644** (shown in phantom) fixed to the shroud **1640**. The rails **1644** extend longitudinally and parallel to the lead screw **1622** and are sized to be received within corresponding notches **1646** defined on the outer periphery of the carriage **1626** and, more particularly, on the outer periphery of one or more of the carriage layers **1628a-e**. As the carriage **1626** translates along the longitudinal axis A_1 , the rails **1644** help maintain the angular position of the carriage **1626** and assume any torsional loading that might otherwise adversely affect movement or operation of the carriage **1626**.

FIG. 16B is an isometric view of another exemplary embodiment of the surgical tool **1600**, according to one or more additional embodiments. The surgical tool **1600** of FIG. 16B is substantially similar to the surgical tool **1600** of FIG. 16A, except the handle **1614** only has a first or “distal” end **1618a** to which the lead screw **1622** and the one or more splines **1624** are rotatably mounted and extend longitudinally therefrom. That is, there is no second end **1618b** to which the splines **1624** and lead screw **1622** also rotatably mount. Rather, the splines **1624** and lead screw **1622** are cantilevered into the air having a cantilevered proximal portion **1625**.

The carriage **1626** is movably mounted along the lead screw **1622** and the splines **1624a-c** and houses the various activating mechanisms configured to cause operation of specific functions of the end effector **1604**. The lead screw **1622** and the splines **1624a-c** each extend through portions of one or more of the layers **1628a-e** to allow the carriage **1626** to translate along the longitudinal axis A_1 with respect to the lead screw **1622** and the splines **1624a-c**. The carriage **1626** may translate from the distal end **1618a** of the handle to the cantilevered proximal portion **1625**. Without the added mass of the second end **1618b** the distribution of mass of the surgical tool **1600** is optimized more toward the first end **1618a** compared to the surgical tool **1600**.

FIG. 17A is an isometric view of the surgical tool **1600** of FIG. 16A releasably coupled to an example instrument driver **1702** according to one or more embodiments. The instrument driver **1702** may be similar in some respects to the instrument drivers **1102**, **1200** of FIGS. 11 and 12, respectively, and therefore may be best understood with reference thereto. Similar to the instrument drivers **1102**, **1200**, for example, the instrument driver **1702** may be mounted to or otherwise positioned at the end of a robotic arm (not shown) and is designed to provide the motive

forces required to operate the surgical tool 1600. Unlike the instrument drivers 1102, 1200, however, the shaft 1602 of the surgical tool 1600 extends through and penetrates the instrument driver 1702.

The instrument driver 1702 has a body 1704 having a first or “proximal” end 1706a and a second or “distal” end 1706b opposite the first end 1706a. In the illustrated embodiment, the first end 1706a of the instrument driver 1702 is matable with and releasably coupled to the first end 1618a of the handle 1614, and the shaft 1602 of the surgical tool 1600 extends through the body 1704 and distally from the second end 1706b.

FIG. 17B depicts separated isometric end views of the instrument driver 1702 and the surgical tool 1600 of FIG. 17A. With the jaws 1610, 1612 closed, the shaft 1602 and the end effector 1604 can penetrate the instrument driver 1702 by extending through a central aperture 1708 defined longitudinally through the body 1704 between the first and second ends 1706a,b. To align the surgical tool 1600 with the instrument driver 1702 in a proper angular orientation, one or more alignment guides 1710 may be provided or otherwise defined within the central aperture 1708 and configured to engage one or more corresponding alignment features 1712 provided on the surgical tool 1600. In the illustrated embodiment, the alignment feature 1712 comprises a protrusion or projection defined on or otherwise provided by an alignment nozzle 1714 extending distally from the first end 1618a of the handle 1614. In one or more embodiments, the alignment guide 1710 may comprise a curved or arcuate shoulder or lip configured to receive and guide the alignment feature 1712 as the alignment nozzle 1714 enters the central aperture 1708. As a result, the surgical tool 1600 is oriented to a proper angular alignment with the instrument driver 1702 as the alignment nozzle 1714 is advanced distally through the central aperture 1708. In other embodiments, the alignment nozzle 1714 may be omitted and the alignment feature 1712 may alternatively be provided on the shaft 1602, without departing from the scope of the disclosure.

As illustrated, a drive interface 1716 is provided at the first end 1706a of the instrument driver 1702, and a driven interface 1718 is provided at the first end 1618a of the handle 1614. The drive and driven interfaces 1716, 1718 may be configured to mechanically, magnetically, and/or electrically couple the handle 1614 to the instrument driver 1702. To accomplish this, the drive and driven interfaces 1716, 1718 may provide one or more matable locating features configured to secure the handle 1614 to the instrument driver 1702. In the illustrated embodiment, for example, the drive interface 1716 provides one or more interlocking features 1720 (three shown) configured to locate and mate with one or more complementary-shaped pockets 1722 (two shown, one occluded) provided on the driven interface 1718. In some embodiments, the features 1720 may be configured to align and mate with the pockets 1722 via an interference or snap fit engagement, for example.

The instrument driver 1702 also includes one or more drive outputs that extend through the drive interface 1716 to mate with the drive inputs 1636a-d provided at the first end 1618a of the handle 1614. More specifically, the instrument driver 1702 includes a first drive output 1724a matable with the first drive input 1636a, a second drive output 1724b matable with the second drive input 1636b, a third drive output 1724c matable with the third drive input 1636c, and a fourth drive output 1724d matable with the fourth drive input 1636d. In some embodiments, as illustrated, the drive

outputs 1724a-d may define splines or features designed to mate with corresponding splined receptacles of the drive inputs 1636a-d. Once properly mated, the drive inputs 1636a-d will share axes of rotation with the corresponding drive outputs 1724a-d to allow the transfer of rotational torque from the drive outputs 1724a-d to the corresponding drive inputs 1636a-d. In some embodiments, each drive output 1724a-d may be spring loaded and otherwise biased to spring outwards away from the drive interface 1716. Each drive output 1724a-d may be capable of partially or fully retracting into the drive interface 1716.

In some embodiments, the instrument driver 1702 may include additional drive outputs, depicted in FIG. 17B as a fifth drive output 1724e and a sixth drive output 1724f. The fifth and sixth drive outputs 1724e,f may be configured to mate with additional drive inputs (not shown) of the handle 1614 to help undertake one or more additional functions of the surgical tool 1600. In the illustrated embodiment, however, the handle 1614 does not include additional drive inputs matable with the fifth and sixth drive outputs 1724e,f. Instead, the driven interface 1718 defines corresponding recesses 1726 configured to receive the fifth and sixth drive outputs 1724e,f. In other applications, however, fifth and/or sixth drive inputs (not shown) could be included in the handle 1614 to mate with the fifth and sixth drive outputs 1724e,f, or the surgical tool 1600 might be replaced with another surgical tool having fifth and/or sixth drive inputs, which would be driven by the fifth and/or sixth drive outputs 1724e,f.

While not shown, in some embodiments, an instrument sterile adapter (ISA) may be placed at the interface between the instrument driver 1702 and the surgical tool 1600. In such applications, the interlocking features 1720 may operate as alignment features and possible latches for the ISA to be placed, stabilized, and secured. Stability of the ISA may be accomplished by a nose cone feature provided by the ISA and extending into the central aperture 1708 of the instrument driver 1702. Latching can occur either with the interlocking features 1720 or at other locations at the interface. In some cases, the ISA will provide the means to help align and facilitate the latching of the surgical tool 1600 to the ISA and simultaneously to the instrument driver 1702.

Layered Carriage

FIG. 18 is an enlarged side view of an exemplary embodiment of the carriage 1626 of FIG. 16A, according to one or more embodiments. In accordance with some aspects of the present disclosure, the carriage 1626 is configured to house various instrument specific functions that are independent of the insertion function (z-axis translation of the carriage 1626). As discussed above, the carriage 1626 may comprise two or more structural layers, operatively coupled to one another in series for cooperative z-axis translation, wherein each layer is associated with at least one function of the surgical tool 1600. In other words, the two or more structural layers operate functions of the end effector 1604 (FIG. 16A) decoupled from and translatable along the insertion axis of the surgical tool 1600.

The carriage 1626 includes structural layers 1628a-e that are stacked together to form the carriage framework. The layers 1628a-e contain or house one or more activating mechanisms that are operatively coupled to corresponding splines that extend through portions of the layers 1628a-e for performing separate functions of an end effector distally located on a tool shaft. The carriage 1626 and the structural layers 1628a-e are able to axially traverse the drive splines during instrument insertion (z-translation) without impacting the end effector functions (e.g., wristed motion, grasp-

ing, etc.). While the illustrated embodiment illustrates five structural layers **1628a-e**, it is to be understood that the number of layers is non-limiting and that the number of layers can vary, e.g., in relation to the number of functions desired for the associated surgical tool. For example, the number of layers **1628a-e** may equal the number of desired tool functions.

Each structural layer **1628a-e** includes a layer body **1829a-e**, respectively, having a distal surface and a spaced apart proximal surface. As illustrated in the exemplary embodiment of FIG. 18, each layer body **1829a-e** includes a substantially planar distal surface and a substantially planar proximal surface. For example, the second structural layer **1628b** having second layer body **1829b** includes a substantially planar distal surface **1851d** and a substantially planar proximal surface **1851p** opposite the distal surface **1851d**. The second layer body **1829b** has a length L_b defined as the distance between the two spaced apart surfaces **1851d**, **1851p**, and in the direction of the insertion axis of the surgical tool. In some embodiments, the second layer body **1829b** houses or otherwise helps secure a corresponding activating mechanism **1638b** within the length L_b .

The distal and proximal surfaces of each layer body **1829a-e** facilitate a stacking arrangement of the layers **1628a-e**. That is, the proximal surface of one layer, e.g., proximal surface **1850p** of layer **1628a**, is complementary in shape to abut the distal surface **1851d** of adjacent layer **1628b**. While substantially flat planar surfaces are illustrated in FIG. 18, it is to be appreciated that other complementary surface configurations may be employed, e.g., adjacent convex and concave surfaces. In other words, each structural layer is configured to abut at least one surface of an adjacent layer. For example, the first structural layer **1628a** includes a proximal abutment surface **1850p** that abuts a distal abutment surface **1851d** of the adjacent second structural layer **1628b**. While not illustrated, it is contemplated that a spacer or washer may be present between the entire surface and/or portions of the surface of adjacent layers, e.g., between surface **1850p** and **1851d**, wherein the layers are arranged in series and in substantial alignment along a shaft of the surgical tool.

As briefly mentioned above, the layers **1628a-e** can be removably secured to each other in series using one or more mechanical fasteners **1630** (three visible). Since each layer **1628a-e** houses or helps contain an activating mechanism, generally corresponding to a specific function of the associated surgical tool, functions of the surgical tool can be disabled by disconnecting the related layer and associated activating mechanism. The disabling feature may be advantageous in bailout situations, wherein the tool or certain functions need to be disabled. For example, if the robotic system is unable to perform a specific function, disconnecting the related layer allows the instrument to be removed from the instrument driver.

In FIG. 18, some or all of the layers **1628a-e** include coaxially aligned holes in each layer body **1829a-e** that are configured to receive the mechanical fastener **1630**. The mechanical fastener **1630** may couple two or more layers together in series, by extending through the coaxially aligned holes in each layer **1628a-e**. In this way, individual layers **1628a-e** may be disconnected from the carriage assembly **1626** by removing the corresponding fasteners **1630**. In other embodiments, the carriage **1626** is configured such that each individual layer **1628a-e** is mechanically coupled to adjacent layers via discrete fasteners. In this way, individual layers **1628a-e** may be disconnected piecewise. The mechanical fasteners **1630** may include any conven-

tional fastener including, but not limited to, screws, bolts, nuts, or by snap fit connectors and the like.

As briefly mentioned above, the carriage **1626** may define or otherwise provide one or more notches configured to engage the guide rails **1644** of the handle **1614** (FIG. 16A). Each guide rail **1644** may comprise an elongated structure that extends in a substantially longitudinal manner along the axial length of the handle **1614**, and may be configured to help maintain an angular position of the carriage **1626**. In the illustrated embodiment, the guide rail **1644** is received within notches **1646b,c** defined in the second and third layers **1628b,c**, respectively. More specifically, the notches **1646b** and **1646c** are defined in the outer periphery of the bodies **1829b,c** of the second and third structural layers **1628b,c**, respectively. The notches **1646b,c** are complementary in shape to the cross-section of the guide rail **1644** allowing for a sliding relationship between the carriage **1626** and guide rail **1644**.

In some embodiments, the carriage **1626** transverses the length of the handle **1614** guided by at least two carriage rails **1644** received within corresponding notches, which provide resistance to torsional loads experienced by the carriage **1626**. In some cases, a single layer includes two notches, each configured to engage a separate guide rail. In other cases, at least one layer includes a notch configured to engage a first rail and another connected layer includes a notch configured to engage a second rail, wherein the second rail opposes the first rail. For example, the third layer **1628c** includes a notch **1646c** capable of engaging the first rail **1644** while another connected layer, e.g., one or more of layers **1628a,b,d,e** includes a notch on an opposing side (occluded by carriage **1626**) configured to slidably traverse a second rail (not illustrated).

The bodies **1829a-e** are configured to secure and support one or more activating mechanism. The activating mechanisms described herein are powered or otherwise actuated through the rotation of one or more drive outputs, which correspondingly drive one or more mechanical functions of the surgical tool. For example, and without limitation, an activating mechanism can include intermeshed gearing or a cabling system that causes actuation of a surgical tool function. In the exemplary embodiment of FIG. 18, activating mechanisms are embodied as intermeshed gearing that rotate together with a drive spline coupled to a drive output, as generally described above. More specifically, the first spline **1624a** may be operatively coupled to the first activating mechanism **1638a** such that rotating the first spline **1624a** (via rotation of the second drive input **1636b** of FIGS. 16 and 17B) will correspondingly actuate the first activating mechanism **1638a** and thereby perform a function, such as opening or closing the jaws **1610**, **1612** (FIG. 16A) of the end effector **1604** (FIG. 16A), depending on the rotational direction of the first spline **1624a**.

As illustrated, the first spline **1624a** extends longitudinally through coaxially aligned apertures **1821** (only one visible) defined in the first and second layers **1628a,b** of the carriage **1626**. A drive gear **1838a** may be included with the first spline **1624a** and located between adjacent portions of the first and second layers **1628a,b**. The first spline **1624a** may exhibit a cross-sectional shape matable with a corresponding inner shape passage (described in greater detail below) that extends through the entirety of the body of the drive gear **1838a**. Rotation of the first spline **1624a** correspondingly drives the drive gear **1838a** in rotation. In such embodiments, as the carriage **1626** moves along the longitudinal axis A_1 (FIG. 16A), the drive gear **1838a** will correspondingly move along the length of the first spline

1624a as captured between the first and second layers **1628a,b**. In other embodiments, however, the first spline **1624a** may be shaped and otherwise configured to operate as a drive gear. In such embodiments, the drive gear **1838a** may be omitted to advantageously reduce the number of component parts.

The first activating mechanism **1638a** may include at least one additional gear, e.g., a driven gear (described below as driven gear **2048** of FIG. 20), and the drive gear **1838a** may be positioned on the carriage **1626** to engage or otherwise intermesh with the driven gear. Accordingly, as the first spline **1624a** is rotated, the drive gear **1838a** is able to drive the driven gear in rotation and thereby actuate the first activating mechanism **1638a**.

In accordance with another aspect of the present disclosure, splines that operatively couple drive inputs of an instrument handle to an activating mechanism of a carriage may have various shapes and configurations in order to reduce the mass of the tool and minimize friction while possessing sufficient rigidity to efficiently transfer rotation. FIG. 19A, for example, illustrates a mechanical coupling component embodied as a spline drive gear **1938a** of a structural layer **1928a** rotationally and slidably coupled to a spline **1924a**. The spline gear **1938a** includes a passage **1939a** that extends through the entirety of the gear body. In some embodiments, as illustrated, the passage **1939a** has a shape that is complementary in shape to the cross section of the associated spline **1924a**. In this way, the spline **1924a** is configured to be received within the spline gear **1938a** and allows the spline gear **1938a** to slide along the length of the spline **1924a** in a z-direction while retaining the ability for the spline **1924a** to transfer rotational torque to the spline gear **1938a** at any position along the length of the spline **1924a**. In the illustrated embodiment, the spline **1924a** and passage **1939a** are star-shaped, i.e., the cross-section of each is a 6-pointed star.

In the exemplary embodiment of FIG. 19B, a spline **1924b** is shaped as a half circle, i.e. having a substantially round portion **1940** and a substantially flat edge **1941**. The spline gear **1938b** is secured to or otherwise included in the structural layer **1928b**. Because of the complementary shapes of the spline **1924b** and the spline gear **1938b**, rotating the spline **1924b** will correspondingly drive the spline gear **1938b** in rotation.

While spline shapes are illustrated in FIGS. 19A and 19B as star-shaped and half-circle shaped, the shape of the spline and complementary spline gear passage are not limited to the examples shown. For example, the spline and spline gear passage may have other shapes including but not limited to those illustrated in FIG. 19C, e.g., splines with a rectangular cross section **1924c**, splines with a square cross-section **1924d** and splines with a triangular cross-section **1924e**. Furthermore, splines may be solid, as illustrated in FIG. 19C, or may be hollow, as illustrated in FIGS. 19A and 19B, in order to reduce mass. A reduction of spline mass reduces the rotational inertia of the spline and facilitates changes to the rotational velocity of the drive assembly (drive input, drive output, spline, and spline gear).

The shape of the spline and complementary gear passage, e.g., splines **1924a,b**, and complementary passages **1939a,b** are chosen based on a variety of factors. In some embodiments, it is preferable to reduce the contact surface area between the spline and complementary gear passage to reduce friction between the two components when the carriage, carrying at least one spline gear, is translated along at least one spline. In some embodiments, it is preferable to

have a non-circular cross-section that is less likely to strip or slip during the transfer of rotation from the spline to the spline gear.

The splines may be made of any suitable material with sufficient rigidity to transfer torque from a drive input to a spline gear via a spline gear passage, e.g. spline gear **1938a** and passage **1939a**. Generally, the more stiff or ridged (i.e., geometrically complex) the spline, the more torque the spline can transfer. In some embodiments a spline is composed of a metal material, for example and without limitation, stainless steel. In other embodiments, splines are composed of a plastic material. Splines may be formed from a variety of methods including but not limited to machining, extrusion, injection molding, and 3D printing.

In some embodiments, either the spline or spline passage, e.g. spline **1924a**, spline passage **1939a**, is coated to reduce friction. For example and without limitation, the surface of the spline may be coated with a friction reducing agent, including but not limited to polymer coatings such as baked on Krytox™, polytetrafluoroethylene (Teflon®), Xylon® and the like.

FIG. 20 depicts an example spline coupled to an activating mechanism that may incorporate some or all of the principles of the present disclosure. More specifically, illustrated is a spline **2024**, an idler gear **2038**, and a driven gear **2048**. The idler and driven gears **2038**, **2048** may be rotatably secured to or otherwise housed in a structural layer **2028**. The idler gear **2038** is operatively coupled to the adjacent spline **2024** and an adjacent driven gear **2048** for transferring rotation from the spline **2024** to the driven gear **2048**. More specifically, rather than including a drive gear that slides through a complementary shaped passage of a gear, e.g., passage **1939a,b**, of gears **1938a,b**, the idler gear **2038** defines elongated teeth that intermesh with corresponding elongated teeth of the adjacent spline **2024**. The idler gear **2038** is able to slide along the axial length of the adjacent spline **2024** while retaining the ability to rotate in response to rotation of the spline **2024** at any position along the axial length of the spline **2024**. In some cases, the idler gear **2038** may directly couple to the tool shaft. In other cases, rotation of the idler gear **2038** may correspondingly drive the driven gear **2048**, and the driven gear **2048** may form part of an activating mechanism configured to drive a mechanical function of the surgical tool **1600** (FIG. 16A) Structural Exoskeleton

FIGS. 21A and 21B are exploded isometric and end views, respectively of an alternative embodiment of the handle **1614** of FIG. 16A, according to one or more embodiments. As described herein, the shroud **1640** (FIG. 16A) of the handle **1614** may be configured to add structural benefits to the surgical tool **1600** (FIG. 16A), including but not limited to carriage guidance and stability. As illustrated in FIG. 21A, the handle **1614** is operatively coupled to an instrument driver **2102** at the first end **1618a** of the handle **1614**. The instrument driver **2102** includes one or more drive outputs that extend through the drive interface to mate with drive inputs **2136a-d** provided at the first end **1618a** of the handle **1614**. The drive inputs **2136a-d** are each configured to be operatively coupled to a corresponding spline (analogous to the spline **1624a-c** of FIG. 16A) that extend along the longitudinal axis of the handle **1614** such that rotation of the drive inputs **2136a-d** correspondingly rotate the associated spline. As described above, the central aperture **1632** is configured to receive the shaft **1602** (FIG. 16A) of the surgical tool **1600**.

FIG. 21A also illustrates an alternative embodiment of the shroud **1640** of FIG. 16A, referenced in FIG. 21A at **2140**.

33

The shroud **2140** is characterized as a rigid hollow exoskeleton with a continuous exoskeleton wall **2141** sized to receive a carriage **2126** (FIG. **21B**) of a compatible (e.g., complementary-shaped) cross-section, as well as a lead screw, and splines coupled to drive inputs **2136a-d**. As illustrated, the carriage **2126** defines a plurality of apertures **2124a**, **2124b**, **2124c**, and **2124d** configured to align with the splines. The carriage **2126** may be similar to or the same as the carriage **1626** of FIG. **16A**. In the illustrated embodiment, the shroud **2140** (alternately referred to herein as the “exoskeleton **2140**”) comprises a non-circular tubular structure having a distal end **2142a** mateable with the first end **1618a** of the handle **1614** and a proximal end **2142b** opposite the distal end **2142a**.

The exoskeleton **2140** has a non-circular cross-section that prohibits rotation of a complementary shaped carriage **2126** received within the exoskeleton **2140**. The exoskeleton **2140** may exhibit one or more non-circular features including, but not limited to, indentations **2145**, edges **2146**, corners **2147**, bulges/lobes **2148**, chambers, convexity, concavity, juts, projections, protrusions, and any combination thereof. Such non-circular features may prove advantageous in helping guide the similarly-shaped carriage **2126** as the carriage **2126** traverses between the distal end **2142a** and the proximal end **2142b**. More particularly, the non-circular features function to maintain the angular orientation of the carriage **2126**, and assume any torsional loading applied to the carriage **2126** during operation. Additionally, the non-circular geometry of the exoskeleton **2140** prevents the tool from rolling on a flat surface. For example, if the surgical tool is placed on a planar surface, e.g., a table, on its side, the non-circular geometry prohibits the surgical tool from rolling across the planar surface and potentially falling off and damaging the surgical tool or items/persons nearby.

In some embodiments, the non-circular geometry of the exoskeleton **2140** may exhibit symmetry, i.e. may be symmetric in shape, including but not limited to line symmetry. For example, the exoskeleton **2140** exhibits symmetry in relation to the line of symmetry **S**. That is, the line of symmetry divides the cross-section of the exoskeleton **2140** into two identical halves (bilateral symmetry). While one line of symmetry **S** is illustrated, it is contemplated that an exoskeleton may have multiple lines of symmetry in relation to its cross-section. In other embodiments, however, the non-circular cross-section may be asymmetric in shape, i.e., exhibiting no symmetry. This can be seen, for example, in the non-circular cross-section of the exoskeleton **2540** of FIG. **25B**. In some embodiments, the shape of the non-circular cross-section is designed to minimize the cross-sectional area of the handle **1614**, thus potentially saving on manufacturing costs. In some embodiments, the non-circular cross-section includes indentations **2145** and bulges **2148** in relation to the position and size of the splines (discussed in greater detail below) and/or activating mechanisms of the carriage **2126**.

As described above, the non-circular cross-section of the exoskeleton **2140** helps prevent the carriage **2126** from twisting or rotating within the exoskeleton **2140** upon assuming torsional loads during operation. That is, the configuration of the exoskeleton **2140** may be similar functionally to the guide rails **1644** discussed above in regard to FIGS. **16** and **18**. Rather than including both the exoskeleton **2140** and the guide rails **1644**, which provide substantially the same functionality, the complexity of the surgical tool can be reduced by using only the exoskeleton, thus saving costs, weight, and material. Once the carriage **2126** is aligned and inserted into the shaped hollow exoskeleton

34

2140, the carriage **2126** is rotationally fixed with respect to the exoskeleton **2140** as it advances and retracts between the distal and proximal ends **2142a,b**.

As described above, the carriage **2126** may be coupled to a shaft having an end effector. The shaft and end effector may be similar in some respects to the shaft **1602** and end effector **1604** of FIG. **16A** and, therefore, may be best understood with reference thereto. The translation of the carriage **2126** within the exoskeleton **2140** advances and retracts the end effector relative to the handle **1614** and prohibits unwanted rotation of the shaft and end effector by nature of the complementary geometry of the exoskeleton **2140** and the carriage **2126**. The carriage **2126** is also movably mounted to at least one spline, also housed within the exoskeleton **2140**. In some embodiments, the carriage **2126** may also be movably mounted to a lead screw (e.g., the lead screw **1622** of FIG. **16A**). Torsional loads resulting from rotation of the spline and/or the lead screw is transferred to the exoskeleton wall **2141**, which provides stability to the end effector during functional operation.

In some embodiments, the exoskeleton **2140** includes at least one fastener bore **2150** extending at least partially between the distal and proximal ends **2142a,b**. The fastener bore **2150** is configured to receive a mechanical fastener for securing the exoskeleton **2140** to the first end of the handle **1614**. In some embodiments, the fastener bore **2150** is integrated into or otherwise defined by the exoskeleton wall **2141** of the exoskeleton **2140**. In other embodiments, and as illustrated in the exemplary embodiment of FIG. **21A**, the fastener bore **2150** comprises a through bore defined axially through an alignment feature **2151** of the exoskeleton **2140**.

More specifically, the exoskeleton **2140** may include at least one sidewall projection **2151** that projects outwardly from the exoskeleton wall **2141** and extends longitudinally between the distal and proximal ends **2142a,b**. In some embodiments, the sidewall projection **2151** may include or otherwise comprise a physical alignment feature for mating the exoskeleton **2140** to the first end **1618a** of the handle **1614**. That is, the first end **1618a** of the handle **1614** includes at least one alignment pocket **2152** complementary in shape to the sidewall projection **2151** such that when the exoskeleton **2140** is mated to the first end **1618a**, the alignment feature **2151** nests within the alignment pocket **2152**. In this way, the exoskeleton **2140** and housed carriage **2126** can properly align with and couple to the various drive inputs **2136a,b**, e.g., coupled via splines extending the length of the handle **1614**. In some embodiments, as illustrated, the exoskeleton **2140** may include multiple alignment features **2151**, and each alignment feature **2151** may be configured to fit into a corresponding alignment pocket **2152** of the handle **1614**.

With continued reference to FIG. **21A**, the proximal end **2142b** of the exoskeleton **2140** may be configured to receive a cap **2160**. The cap **2160** may be similar in shape to the cross-section of the exoskeleton **2140**. In some embodiments, the cap **2160** may help provide a sealed interface at the proximal end **2142b** and thereby substantially seal the interior volume **V** of the exoskeleton **2140** from the external environment, which may also prevent dust, debris, and fluid from migrating into the interior and potentially damaging the tool. In some embodiments, a gasket (not illustrated), e.g. a rubber gasket, may be placed between the cap **2160** and the exoskeleton **2140** to enhance the seal quality of the interior volume **V**. The cap **2160** may also help prevent removal of the carriage **2126** from the exoskeleton **2140**. For example, if the tool is turned upside-down, having the cap

2160 secured to the proximal end **2142b** will prevent the carriage **2126** from falling out of the hollow exoskeleton **2140**.

In some embodiments, the cap **2160** may be coupled to the proximal end **2142b** of the exoskeleton **2140** via a snap fit engagement. In other embodiments, or in addition thereto, the cap **2160** may define at least one aperture **2162** located to align with the fastener bore **2150** of the exoskeleton **2140** and configured to receive a mechanical fastener. The mechanical fastener may be used to secure the cap **2160** to the exoskeleton **2140** alone or the mechanical fastener may extend through both the aperture **2162** and the fastener bore **2150** to secure both the cap **2160** and the exoskeleton **2140** to the first end **1618a** of the handle **1614**.

The cap **2160** may also provide ergonomic features to enhance the handling experience of the surgical tool. For example, in some embodiments, the cap **2160** may define or otherwise provide a rounded peripheral edge **2161**. The cap **2160** may include other features, not illustrated, such as a projection or knob for ergonomically accommodating placement of a user's hand.

The exoskeleton **2140** may also include structures and features for facilitating handling and/or enhancing the structural integrity. As depicted in FIG. 21B, for instance, the exoskeleton **2140** may define or otherwise provide a plurality of structural ribs **2155** extending outwardly from the exoskeleton wall **2141**. The ribs **2155** may extend at least partially between the distal and proximal ends **2142a,b** and/or circumferentially around the non-circular cross-section. In some embodiments, the ribs **2155** add support and rigidity to the exoskeleton **2140**. That is, the placement of the ribs **2155** around the perimeter of the exoskeleton wall **2141** allows for the thickness of the exoskeleton wall **2141** to be minimized (saving weight and material) while still maintaining sufficient rigidity to withstand torsional loads applied to the carriage **2126** during translation or activation of a surgical tool function.

In some embodiments, an exterior coating **2156** may be applied to the outer surfaces of the exoskeleton **2140** to enhance the handling properties of the surgical tool. For example, the exoskeleton **2140** may be coated with a material having a high coefficient of friction, which may aid in preventing undesirable slippage when grasped by an operator's hand **2190**. Several suitable materials with a high coefficient of friction may be used for the exterior coating such as, for example, a silicone elastomer. It is also contemplated herein to apply other types of coatings to the outer surfaces of the exoskeleton **2140** including, but not limited to, a hydrophobic coating.

In some embodiments, the exoskeleton **2140** may include a coating applied to its interior surfaces to enhance the sliding relationship between the carriage **2126** and the exoskeleton **2140**. For example, the exoskeleton **2140** may be internally coated with a material having a low coefficient of friction. A low friction material allows to the carriage **2126** to smoothly slide along the guiding cross-section and thereby prevent binding.

The exoskeleton **2140** may be composed of a suitable material that imparts sufficient rigidity for translating the carriage **2126** subjected to forces of varying direction related to the operation of designed surgical tool functions. In some embodiments, the exoskeleton **2140** is made of a metal material such as, but not limited to, steel (e.g., stainless steel), aluminum, any alloy thereof, or any combination thereof. In other embodiments, the exoskeleton **2140** may be made of a polymeric or plastic material such as, but not limited to, urethane, polycarbonate, acrylonitrile butadiene

styrene (ABS), and polyethylene, the polymer or plastic material may include a filled polymer including but not limited to glass or carbon fibers. In yet other embodiments, the exoskeleton **2140** may be made of a composite material, such as fiber glass or carbon fiber.

FIGS. 22A and 22B illustrate alternative embodiments of the handle **1614**, of a surgical tool **1600** according to one or more additional embodiments. As illustrated, the handle **1614** includes a layered carriage **2226** similar in some respects to the carriage **1626** of FIGS. 16 and 18, and the carriage **2226** is movably coupled to at least one spline **2222**. The carriage **2226** is received or otherwise surrounded by an exoskeleton **2240a** (FIG. 22A) or exoskeleton **2240b** (FIG. 22B). The exoskeleton **2240a,b** may be configured to guide the carriage **2226** as it moves axially along the spline **2222** and assume torsional loading applied on the carriage **2226** during operation. Moreover, the exoskeleton **2240a,b** may be configured to help maintain the angular orientation of the carriage **2226** as it translates along the spline **2222**.

In FIG. 22A, the cross-section or geometry of the first exoskeleton **2240a** is non-circular and complementary in shape to the cross-section or geometry of the carriage **2226**, as generally described above with reference to FIGS. 21A-B. In the illustrated embodiment, for example, the geometry of the exoskeleton **2240a** includes multiple (four) lobes extending from a centerline of the exoskeleton **2240a**, and the carriage **2226** similarly defines or otherwise provides multiple (four) lobes that match or otherwise mate with the lobes of the exoskeleton **2240a**. The inner surface of the exoskeleton wall **2241** of the exoskeleton **2240a** may be in sliding contact with the carriage **2226** and thereby operate to absorb any torsional loads assumed by the carriage **2226** during operation.

In FIG. 22B, the second exoskeleton **2240b** is generally circular in shape and a filler material **2243** may be deposited between the carriage **2226** and the inner wall of the exoskeleton **2240b**. In such embodiments, the filler material **2243** may define a bore **2245** configured to accommodate the general cross-sectional shape of the carriage **2226**. Consequently, rather than the carriage **2226** being in sliding contact with the interior surface of the exoskeleton wall **2241** of the exoskeleton **2240b**, the carriage **2226** may instead engage and slide against the filler material **2243**. The filler material **2243** may comprise a variety of rigid or semi-rigid materials. In some embodiments, for instance, the filler material **2243** may comprise, but is not limited to, polymer materials, e.g., polycarbonate, polyurethane, closed or open cell foams, e.g., polyurethane, neoprene, Ethylene propylene diene monomer, chloroprene rubber, styrene-butadiene rubber, or any combination thereof. The filler material **2243** may also have a structure or frame, e.g., honeycomb.

Drive Patterns and Drive Spline Configurations

FIGS. 23A-23C are partial cross-sectional side views of alternative embodiments of the handle **1614** of FIG. 16A, according to one or more additional embodiments. As described herein, the mounting of a carriage to a lead screw for z-axis translation is configured to reduce the rotation of the carriage and minimize binding of the carriage while simultaneously increasing carriage stability. As illustrated, the handle **1614** includes a carriage **2326** similar in some respects to the carriage **1626** of FIG. 16A. For instance, the carriage **2326** is movable between first and second ends **1618a,b** of the handle **1614** along the longitudinal axis A_1 (i.e., z-axis translation), and the shaft **1602** extends distally from the carriage **2326**. Accordingly, as the carriage **2326** moves along the longitudinal axis A_1 , the carriage **2326** is

thereby able to advance or retract an end effector (e.g., the end effector **1604** of FIG. **16A**) attached to the distal end of the shaft **1602** relative to the handle **1614**.

In the illustrated embodiment, the carriage **2326** includes a carriage nut **2334** mounted to a rotatable lead screw **2322**, similar to the lead screw **1622** of FIG. **16A**. The outer surface of the lead screw **2322** defines helical threading and the carriage nut **2334** defines corresponding internal helical threading (not shown) mateable with the outer helical threading of the lead screw **2322**. As a result, rotation of the lead screw **2322** causes the carriage nut **2334** to advance or retract the carriage **2326** along the longitudinal axis **A1** and correspondingly advance or retract the carriage **2326** and the shaft **2302** relative to the handle **1614**.

FIG. **23A** illustrates a single carriage nut **2334** located at or near a distal end **2327** of the carriage **2326**. During operation of the handle **1614**, such as activating various functions of the end effector **1604** (FIG. **16A**), the carriage **2326** may experience various torsional and axial forces **F** that cause the carriage **2326** to rotate or shift in the direction **R**. Shifting the carriage **2326** in the direction **R** can bind or inhibit the movement of the carriage **2326** along the handle **1614**. According to embodiments of the present disclosure, the carriage **2326** may be stabilized and rotation in the direction **R** minimized or eliminated by having portions of the carriage nut **2334** located at or near the distal and proximal ends of the carriage **2326**. In such embodiments, the carriage **2326** may be mounted to the lead screw **2322** at two or more spaced-apart locations or otherwise spanning a substantial length of the carriage **2326**, as described in greater detail below.

In FIG. **23B**, the carriage **2326** includes at least two carriage nuts configured to increase the stability of the carriage **2326**, i.e., minimize twisting and rotation of the carriage **2326** about the lead screw **2322**. More specifically, the carriage **2326** may include a first carriage nut **2334a** and a second carriage nut **2334b**. The first carriage nut **2334a** may be positioned at or near the distal end **2327** of the carriage **2326** and the second carriage nut **2334b** is positioned at or near a proximal end **2329** of the carriage **2326**. The carriage nuts **2334a**, **2334b** are each mounted to the rotatable lead screw **2322** and are each supported by the carriage **2326** in a spaced apart relationship generally located on opposite ends of the carriage **2326**.

In embodiments where the carriage **2326** is composed of a plurality of layers, a carriage nut may be present on at least two layers for facilitating translation of the carriage in response to rotation of the lead screw **2322**. In FIG. **23B**, for example, the carriage **2326** includes four stacked layers, depicted as a first layer **2328a**, a second layer **2328b**, a third layer **2328c**, and a fourth layer **2328d**. While four layers are illustrated it is to be appreciated that the number of layers of the carriage **2326** may be more or less than four, without departing from the scope of the disclosure. The first layer **2328a** may be alternately referred to as the “distal layer **2328a**” and the fourth layer **2328d** may be alternately referred to as the “proximal layer **2328d**”. In such embodiments, the first carriage nut **2334a** may be coupled to the distal layer **2328a** and the second carriage nut **2334b** may be coupled to the proximal layer **2328d**. While not shown, it is contemplated herein to include additional carriage nuts coupled to the other layers, e.g., the second and third layers **2328b,c**.

In some embodiments, as illustrated, the first carriage nut **2334a** may be coupled to or otherwise encompass or span portions of two or more layers of the carriage **2326**. In FIG. **23B**, for instance, the first carriage nut **2334a** is coupled to

or otherwise supported by the distal layer **2328a** and extends into the adjacent second layer **2328b**. Accordingly, in some embodiments, the first carriage nut **2334a** may extend across two layers **2328a**, **2328b** and the second carriage nut **2334b** may be secured to a single layer **2328d**.

In FIG. **23C**, the carriage **2326** includes an elongated carriage nut **2334** that substantially extends from the distal end **2327** to the proximal end **2329** of the carriage **2326**. Moreover, the carriage **2326** includes a platform layer **2328e** that supports the other layers **2328f-h** of the carriage **2326**. In the illustrated embodiment, a portion of the platform layer **2328e** extends generally between the distal and proximal ends **2327**, **2329** of the carriage **2326**, but may alternatively extend only a portion of the distance between the distal and proximal ends **2327**, **2329**, or may extend further than the distance between the distal and proximal ends **2327**, **2329**, without departing from the scope of the disclosure.

The carriage nut **2334** is mounted to the platform layer **2328e** and is thus responsible for the translation of the coupled carriage layers **2328e-h** along the lead screw **2322**. In some embodiments, the carriage nut **2334** may extend along the entire axial length of the platform layer **2328e**, but may alternatively extend along only a portion of the axial length of the platform layer **2328e**. Although not illustrated, it is contemplated that a second layer in a stack of two or more layers may incorporate an elongated carriage nut, similar to the elongated nut **2334**, having a proximal portion **2334p** and a distal portion **2334d** and supporting a first layer distally, and a third layer proximally.

FIGS. **24A** and **24B** are top views of example embodiments of the surgical tool **1600**, according to one or more embodiments of the disclosure. In accordance with another aspect of the present disclosure, the locations of high load splines and activating mechanisms may be optimized with respect to the carriage and exoskeleton to reduce torsional loads and minimize the cross-section of the handle. Certain functions of an associated end effector, triggered by the activating mechanics incorporated within a carriage, require high torsional loads. For example, surgical staplers often experience high torsional loads associated with the firing and closure functions. As described above in regard to FIG. **16A**, the carriage rails **1644** located on the shroud **1640** can assume any torsional loading that would otherwise adversely affect the stability of the carriage **1626**. However, placing high torsional splines and/or activating mechanisms near a load balancing member of the surgical tool **1600** helps reduce twisting loads and sliding friction. That is, the centerline of splines are configured so that high torsional loading spline centerlines are as close to a load balancing member, e.g., translational lead screw centerline and/or the shaft axis, reducing the net torsional force on the system by optimizing spline placement.

This configuration may also provide a slimmer geometry to the surgical tool **1600**, thus minimizing weight and enhancing ergonomic handling. Torsional loading of the carriage is managed by applying torque to the lead screw, other splines, or struts located adjacent to high-torque inputs. This configuration, as discussed below, can result in a non-circular cross-section of the carriage that balances loading and provides a slimmer geometric look to the surgical device.

As illustrated in FIGS. **24A-24B**, carriages **2426a,b**, respectively, are received within the surgical tool **1600**, and each carriage **2462a,b** exhibits a non-circular cross-section. Moreover, each carriage **2462a,b** is movably mounted to a plurality of splines **2424a-c** driven by the outputs of an instrument driver **2402**. The carriages **2426a,b** and the

splines **2424a-c** may be similar in some respects to the carriage **1626** and the splines **1624a-c** of FIG. **16A**, and the instrument driver **2402** may be similar in some respects to the instrument drivers **1102**, **1200**, **1702** of FIGS. **11**, **12**, and **17** respectively, and therefore may be best understood with reference thereto. The shaft **1602** is coupled to and extends distally from the carriage **2426a,b** through the first end of the handle. In the illustrated embodiment, the shaft **1602** penetrates an aperture **1708** of the instrument driver **2402**. The carriage **2426a** includes a plurality of activating mechanisms **2428a-c**, each supported by the carriage **2426a,b** and each associated with a separate function of the end effector. The splines **2424a-c** are each rotationally mounted to the first end of the handle and extend between the handle ends. Each spline **2424a-c** is operatively coupled to one of the activating mechanisms **2428a-c**, respectively, to drive the associated function of the end effector.

In FIG. **24A**, the splines **2424a-c** are arranged such that splines having the highest operating torsional load are located adjacent to a load balancing member **2422**. The load balancing member **2422** comprises an elongated structure extending between the first end and second ends of the handle and has a diameter **D**. The load balancing member **2422** has sufficient stiffness to minimize twisting of the carriage **2426a** during the operation of high load functions, such as closure and firing of the end effector. The load balancing member **2422** may be rod shaped and may exhibit any cross-sectional configuration. In some embodiments, the load balancing member **2422** comprises a strut, similar to the strut **1620** of FIG. **16A**. The arrangement of splines **2424a-c** in relation to a load balancing member **2422** based on their torque lends to design of the carriage **2426a** to have a non-circular cross-section.

The end effector **1604** (FIG. **16A**) arranged at the distal end of the shaft **1602** is capable of articulation, firing, and clamping, as generally described above. The first activating mechanism **2428a** is driven by the first spline **2424a** to perform a high torsional clamping function of the end effector **1604**. The second activating mechanism **2428b** is driven by the second spline **2424b** to perform a high torsional firing function of the end effector **1604**. The third activating mechanism **2428c** is driven by the third spline **2424c** to articulate the end effector **1604**, wherein the articulation function does not require the torque values associated with either the clamping or firing functions, e.g., the torque of the articulation function is less than the torque required for clamping and/or firing. Each spline **2428a**, **2428b** associated with high torsional operations (via activating mechanisms **2428a,b**, respectively) are placed adjacent to the load balancing member **2422** to minimize twisting of the carriage **2426a** during operation.

In some embodiments, the load balancing member **2422** comprises a lead screw, similar in some respects to the lead screws **1622** and **2322**, of FIGS. **16** and **23A-23B**, respectively. Accordingly, the load balancing member **2422** may alternatively be referred to herein as the "lead screw **2422**," which facilitates translation of the carriage **2426a** along the longitudinal axis of the handle ("the insertion axis") e.g., between the first and second ends. In the illustrated embodiment, the shaft **1602** occupies or is otherwise located at the central axis of the instrument driver **2402** as well as the center of the carriage **2426a**. Here, the lead screw **2422** is positioned off-center in relation to the carriage **2426a** and the instrument driver **2402**. The lead screw **2422** is generally configured to have sufficient rigidity to provide the translation function of the carriage **2426a**, wherein the rigidity characteristic may also be leveraged to resist twisting of the

carriage **2426a** during firing and/or clamping functions of the end effector **1604** (FIG. **16A**). That is, due to a potentially larger cross-sectional size and/or stiffer material (e.g., the lead screw **2422** may be composed of a strong metal like stainless steel), placing the high torsional loading splines **2428a,b** as close as possible to the load balancing lead screw **2422** aids in balancing the torsional loading on the carriage **2426a**, and thus minimizing undesirable rotations of the carriage **2426a**.

In FIG. **24B**, a load balancing member may be embodied as an adjacent spline. That is, a first spline **2424a** configured to perform a high torsional clamping function of the end effector **1604** (FIG. **16A**) may be located adjacent to a load balancing member characterized and referred to herein as the second spline **2424b**, which is configured to perform a high torsional firing function of the end effector **1604**. In these embodiments, the first spline **2424a** may be configured to rotate in a first direction (e.g., clockwise) and the load balancing member **2424b** may be configured to rotate in a second direction (e.g., counter-clockwise) opposite the first direction. That is, the splines **2424a,b** are configured such that when one or more are working simultaneously, the torsional load applied by one spline is equal and opposite to the torsional load applied by another spline. The load balancing member **2422b** in FIG. **24B** may have a diameter greater than the diameter of the first spline **2424a** having a high operating torsional load. In other embodiments, the load balancing member **2422b** has a stiffness that is greater than the stiffness of the spline having the highest operating torsional load.

FIGS. **25A** and **25B** illustrate top cross-sectional views of another example embodiment of the surgical tool **1600**, according to one or more additional embodiments. In accordance with another aspect of the present disclosure, the sizes as well as the locations of each spline coupled to the carriage may be optimized to reduce torsional loads experienced by the carriage and minimize the cross-sectional area of the carriage/handle. Rather than having splines of substantially the same diameter, the diameter of a spline may correspond to the torque value required by the associated function of that spline (coupled to an activating mechanism of the carriage). The more torque required to perform a function, the larger the diameter of the spline (imparting increased durability/stiffness), and the less torque required to perform a function, the smaller the diameter of the spline. The multiple sizing of the splines allows for creative compact designs of the carriage and activating mechanisms as well as providing a non-circular cross-section of the tool handle (e.g., carriage and shroud/exoskeleton).

As illustrated, the surgical tool **1600** may include activating mechanisms **2528a-d** and a carriage **2526** substantially similar to the activating mechanisms **1638a-c** and the carriage **1626**, respectively, of FIG. **16A**. Each activating mechanism **2528a-d** is coupled to a separate spline **2524a-d**, respectively, which are driven by the outputs of an instrument driver **2402**. The splines **2524a-d**, are each rotationally mounted to an end of the handle (e.g., the handle **1614** of FIG. **16A**) and extend between the handle ends allowing the carriage **2526** that supports the drive mechanisms **2528a-d** to traverse the length of the handle while activation of the associated end effector functions are decoupled from the z-axis translation.

Each activating mechanism **2428a-d** and the one or more idle gears **2504** are supported by the carriage **2526** and associated with a separate function of the end effector. In the exemplary embodiment of FIGS. **25A-25B** the surgical tool **1600** is a surgical stapler and the first spline **2524a** drives the

41

first activating mechanism **2528a** to perform a high torsional clamping function of the associated end effector. The first spline **2524a** has a first cross-sectional area and a diameter C. The first activating mechanism **2528a** includes a drive gear **2538a** that is directly or indirectly coupled to the shaft **1602**, which extends through an aperture **2512** defined in the instrument driver **2402**. The second activating mechanism **2528b** is driven by the second spline **2524b** to perform a high torsional firing function of the end effector. The second spline **2524b** has a second cross-sectional area and a diameter F. The second activating mechanism **2528b** includes a drive gear **2538b** coupled to the second spline **2524b** for transmitting torque to the second activating mechanism. The third activating mechanism **2528c** is driven by the third spline **2524c** to articulate the end effector, wherein the articulation function does not require the torque values associated with first and second splines **2528a,b**, e.g., the torque of the articulation function is less than the torque required for clamping and/or firing. The third spline **2524c** has a third cross-sectional area and a diameter A. The third activating mechanism **2528c** includes a drive gear **2538c** coupled to the third spline **2524c** for transmitting torque to the third activating mechanism. The fourth activating mechanism **2528d** is driven by the fourth spline **2524d** to axially translate the carriage **2526**. The fourth spline **2528d** has a fourth cross-sectional area and a diameter T. The fourth activating mechanism **2528d** includes a drive gear **2538d** that is coupled to the shaft **1602** via at least one idler gear **2504**.

In some embodiments, the splines **2528a-d** exhibit at least two different cross-sectional areas and/or diameters, e.g., diameters C, F, A, T. That is, the surgical tool **1600** includes at least one spline associated with a high torque function, e.g., clamping, having a large “high torsion” cross-section (e.g., corresponding to diameters C or F). The surgical tool **1600** also includes at least one spline associated with a low torque function, e.g., articulation, having a low torsion cross-section (e.g., corresponding to diameters A or T). In these embodiments, the high torsion cross-section of a spline associated with a high torque function is larger in area than the area of the low torsion cross-section spline associated with a low torque function. In this way, the higher torque functions are driven by stiffer splines with larger cross-sections and/or diameters while the other functions are driven by splines with a smaller cross-section and/or smaller diameter. Sizing the splines **2528a-d** based on input torque requirements allows for a reduction in rotating mass of smaller load splines as well as the overall weight and cross-section of the surgical tool **1600**.

In some embodiments, the cross-section of the tool handle with optimized spline size may be symmetric, i.e., similar to the non-circular cross-sectional shapes of the exoskeleton **2140** and **2240a,b** of FIGS. **21A-22B**, respectively. In other embodiments, and as illustrated in the exemplary embodiment of FIGS. **25A-25B**, the non-circular cross-section of the exoskeleton **2540** with optimized spline size may be asymmetric in shape.

In some embodiments, each spline **2524a-d** has a different cross-sectional area that is related to the input torque requirements of the associated end effector function. For example, the input torque T_c required for clamping is greater than the input torque T_f required for firing, which is greater than the input torque T_t required for translation, which is greater than the input torque T_a required for articulation of the end effector. Thus, $T_c > T_f > T_t > T_a$. Likewise, the cross-sectional area/diameter C of the first spline **2524a** associated with clamping is greater than the cross-sectional area/diam-

42

eter F of the second spline **2524b** associated with firing, which is greater than the cross-sectional area/diameter T of the third spline **2524c** associated with translation, which is greater than the cross-sectional area/diameter A of the fourth spline **2524d** associated with articulation. Thus, in terms of diameter, $C > F > T > A$.

In some embodiments, and as illustrated in FIGS. **25A-25B**, the plurality of splines **2524a-d** and/or associated activating mechanisms **2524a-b** are located in a planetary relationship with respect to the shaft **1602**. In other words, the center of each spline **2524a-d** is located at a distance (radius) r_{a-rd} from the shaft **1602**. In some embodiments, the distance from the shaft to the center of each spline is generally the same for each, such as is depicted in the exemplary embodiment of FIG. **22A**. In other embodiments, and as illustrated in FIG. **25B**, at least two radii extending from the shaft to the center of a spline, e.g., r_a , r_b , are different. In some further embodiments, at least three radii, extending from the shaft to the center of a spline, e.g., r_a , r_b , r_c , are different. In other words, the center of each spline **2524a-d** is configured to be at least the radius of the spline plus the radius of the shaft **1602**. In yet further embodiments, the distance from the shaft **1602** to the center of each spline is different. In each of these embodiments, the carriage **2526** can exhibit a cross-sectional shape that corresponds to the general shape of the spline arrangement.

In some embodiments, the splines **2524a-d** may be arranged such that an outer cross-sectional contour **2541** of the exoskeleton **2540** corresponds with the placement and size of the interior splines **2524a-d** and/or activating mechanisms **2528a-d**. That is, the exoskeleton **2540** is organically shaped to the interior components (i.e., the carriage **2526**, the splines **2524a-d**, and/or the activating mechanisms **2528a-d**) to reduce mass, provide carriage guidance, increase handle rigidity, minimize system twisting, and/or provide an ergonomic form factor for the surgical tool **1600**. While the planetary arrangement is expressly discussed in regard to FIGS. **25A-25B**, it is to be appreciated that other embodiments also illustrate the planetary arrangement including, but not limited to, those embodiments illustrated in FIGS. **21B**, **22A-22B**, and **24A-24B**. FIG. **25C** illustrates a transparent isometric view **2500** of the carriage **2526** of FIG. **25B** spline within an organically shaped exoskeleton. As noted above, the sizing and arrangement of the splines **2524a-d** and associated activating mechanisms **2528a-d**, allows the external contour **2541** of the exoskeleton **2540** to conform to the interior position and sizing of the housed components. That is, the non-circular features of the exoskeleton **2540** correspond to and accommodate the position and sizing of the splines and activating mechanisms. For example, bulge/lobe **2548** relates to the size and position of the first spline **2524a** (FIG. **25A**) and the first activating mechanism **2528a** (FIG. **25A**). Furthermore, indentation **2545** corresponds to a portion of the surgical tool **1600** where internal components have been optimized to reduce the cross-sectional footprint of the handle. These non-circular features also function to maintain the angular position of the carriage **2526**, assuming any torsional loading that would otherwise adversely affect the carriage.

4. Implementing Systems and Terminology.

Implementations disclosed herein provide systems, methods and apparatus for instruments for use with robotic systems. It should be noted that the terms “couple,” “coupling,” “coupled” or other variations of the word couple as used herein may indicate either an indirect connection or a direct connection. For example, if a first component is “coupled” to a second component, the first component may

be either indirectly connected to the second component via another component or directly connected to the second component.

The methods disclosed herein comprise one or more steps or actions for achieving the described method. The method steps and/or actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of steps or actions is required for proper operation of the method that is being described, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the claims.

As used herein, the term “plurality” denotes two or more. For example, a plurality of components indicates two or more components. The term “determining” encompasses a wide variety of actions and, therefore, “determining” can include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database or another data structure), ascertaining and the like. Also, “determining” can include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Also, “determining” can include resolving, selecting, choosing, establishing and the like.

The phrase “based on” does not mean “based only on,” unless expressly specified otherwise. In other words, the phrase “based on” describes both “based only on” and “based at least on.”

As used herein, the terms “generally” and “substantially” are intended to encompass structural or numeral modification which do not significantly affect the purpose of the element or number modified by such term.

The headings used herein are intended as a matter of convenience for readers of this application and any resulting patent and are not intended to limit the scope of the disclosure.

To aid the Patent Office and any readers of this application and any resulting patent in interpreting the claims appended herein, applicants do not intend any of the appended claims or claim elements to invoke 35 U.S.C. 112(f) unless the words “means for” or “step for” are explicitly used in the particular claim.

The foregoing previous description of the disclosed implementations is provided to enable any person skilled in the art to make or use the present invention. Various modifications to these implementations will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other implementations without departing from the scope of the invention. For example, it will be appreciated that one of ordinary skill in the art will be able to employ a number corresponding alternative and equivalent structural details, such as equivalent ways of fastening, mounting, coupling, or engaging tool components, equivalent mechanisms for producing particular actuation motions, and equivalent mechanisms for delivering electrical energy. Thus, the present invention is not intended to be limited to the implementations shown herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

What is claimed is:

1. A surgical tool, comprising:

- a handle having a first end and a second end;
- a lead screw rotatably coupled to the handle and extending between the first and second ends;
- a spline rotatably coupled to the handle and extending between the first and second ends;
- a carriage movably mounted to the lead screw and translatable between the first and second ends as the lead screw rotates, the carriage comprising first, second, and

third layers secured to each other in series, the first layer being located closer to the first end than the second layer and the third layer interposing the first and second layers;

- a carriage nut supported by the carriage and arranged about the lead screw so that rotation of the lead screw causes the carriage nut to traverse the lead screw and thereby translate the carriage, wherein the carriage nut is at least partially disposed in the first and third layers;
- an elongate shaft directly coupled to and extending from the carriage and penetrating the first end;
- an end effector being operatively coupled to an end of the elongate shaft; and
- an activating mechanism supported by the carriage and operatively coupled to the spline so that rotation of the spline operates the activating mechanism and causes the end effector to actuate, which generates torsional loading on the spline,

wherein the spline is arranged adjacent the lead screw so that at least a portion of the torsional loading generated by rotating the spline is transferred to the lead screw.

2. The surgical tool according to claim 1, wherein the spline is a first spline and the activating mechanism is a first activating mechanism, the surgical tool further comprising:

- a second spline rotatably coupled to the handle and extending between the first and second ends, the second spline being operatively coupled to a second activating mechanism supported by the carriage so that rotation of the second spline operates the second activating mechanism and thereby generates torsional loading,

wherein the second spline is arranged adjacent the lead screw so that the lead screw assumes the torsional loading generated by rotating the second spline.

3. The surgical tool according to claim 2, wherein the torsional loading generated by the first spline is greater than the torsional loading generated by the second spline and the first spline is located closer to the lead screw than the second spline.

4. The surgical tool according to claim 2, wherein the first spline rotates in a first direction and the second spline rotates in a second direction opposite the first direction, and wherein a first torsional load generated by rotation of the first spline to actuate the first activating mechanism is canceled by a second torsional load generated by rotation of the second spline to actuate the second activating mechanism.

5. The surgical tool according to claim 1, wherein the spline comprises a first spline and the surgical tool further includes a second spline, the first spline being subjected to a higher torsional load than the second spline and being located closer to the lead screw than the second spline.

6. The surgical tool according to claim 5, wherein the first and second splines exhibit two different cross-sectional areas.

7. A surgical tool, comprising:

- a handle having a shroud, a first end defining a central aperture, and a second end, the shroud extending from the first end to the second end;
- a lead screw disposed within the handle and rotatably coupled to the handle and extending between the first and second ends of the handle;
- first and second splines disposed in the handle and rotatably coupled to the handle, the first and second splines extending between the first and second ends of the handle;
- a carriage disposed in the handle and movably mounted to the lead screw via a carriage nut supported by the carriage and arranged about the lead screw, the carriage

45

being translatable between the first and second ends as the lead screw rotates, wherein the carriage comprises first, second, and third layers removably secured to each other in series, the first layer being located closer to the first end of the handle than the second layer, the third layer interposing the first and second layers, and wherein the carriage nut is partially disposed in the first and third layers;

an elongate shaft extending through the central aperture and having a longitudinal axis that is coaxial with a longitudinal axis of the handle, a first shaft end of the elongate shaft being disposed in the handle and directly coupled to and extending from the carriage, wherein the elongate shaft translates along the longitudinal axis of the handle as the carriage translates, and wherein the first and second splines and the lead screw are radially arranged about the elongate shaft;

an end effector being operatively coupled to a second shaft end of the elongate shaft;

a first activating mechanism supported by the carriage and operatively coupled to the first spline so that rotation of the first spline operates the first activating mechanism and thereby causes the end effector to actuate in a first operation;

a second activating mechanism supported by the carriage and operatively coupled to the second spline so that rotation of the second spline operates the second activating mechanism and thereby causes the end effector to actuate in a second operation;

wherein rotating the first spline in a first direction to actuate the end effector in the first operation generates a first torsional load, and rotating the second spline in a second direction opposite the first direction to actuate the end effector in the second operation generates a second torsional load, and

wherein the first torsional load is balanced by the second torsional load.

8. The surgical tool according to claim 7, wherein the first spline is arranged at a first distance from the lead screw and the second spline is arranged at a second distance from the lead screw, and wherein the lead screw assumes the first and second torsional loads generated by the first and second splines, respectively.

9. The surgical tool according to claim 7, wherein the first spline has a first cross-sectional area and the second spline has a second cross-sectional area smaller than the first cross-sectional area.

10. The surgical tool according to claim 7, wherein the first spline has a first cross-sectional area and the second spline has a second cross-sectional area equal to the first cross-sectional area.

11. A method of minimizing torsional loads on components of a robotic surgical tool that includes:

- a handle having opposing first and second ends;
- a spline and a lead screw rotatably coupled to the handle and extending between the first and second ends;
- a carriage movably mounted to the lead screw via a carriage nut supported by the carriage and arranged about the lead screw, the carriage comprising first, second, and third layers secured to each other in series,

46

the first layer being located closer to the first end than the second layer and the third layer interposing the first and second layers, wherein the carriage nut is at least partially disposed in the first and third layers; and

an elongate shaft directly coupled to and extending from the carriage and penetrating the first end, an end effector being operatively coupled to an end of the elongate shaft,

the method comprising:

- actuating the robotic surgical tool and thereby rotating the spline;
- operating an activating mechanism mounted to the carriage in response to rotation of the spline and thereby actuating the end effector as the spline rotates, which generates a torsional force assumed by the spline;
- transferring the torsional force from the spline to the carriage; and
- assuming the torsional force with the lead screw.

12. The method according to claim 11, wherein the spline comprises a first spline and the robotic surgical tool further includes a second spline, the method further comprising:

- rotating the first spline in a first angular direction; and
- rotating the second spline simultaneously in a second angular direction different from the first angular direction.

13. The method according to claim 11, wherein the spline comprises a first spline and robotic surgical tool further includes a second spline, the first spline being subjected to a higher torsional load than the second spline and being located closer to the lead screw than the second spline.

14. The surgical tool of claim 1, wherein the carriage nut comprises a first carriage nut and a second carriage nut axially spaced from one another along the lead screw.

15. The surgical tool of claim 14, wherein the first carriage nut is disposed in the first and third layers and the second carriage nut is disposed in the second layer.

16. The surgical tool of claim 1, wherein the activating mechanism includes a gear disposed on the spline that is driven by rotation of the spline.

17. The surgical tool of claim 1, wherein the first end includes a central aperture, and the elongate shaft extends through the central aperture and is translatable along a central axis of the surgical tool as the carriage translates.

18. The surgical tool of claim 7, further comprising an instrument driver removably coupled to the first end and including:

- a first drive output operatively coupled to a first drive input rotatably mounted to the first end of the handle and being rotatably coupled to the lead screw so that rotating the first drive output correspondingly rotates the lead screw; and
- a second drive output operatively coupled to a second drive input rotatably mounted to the first end of the handle and being rotatably coupled to the spline so that rotation of the second drive input correspondingly rotates the spline,

wherein the elongate shaft extends through the instrument driver and the instrument driver is arranged between the end effector and the first end of the handle.

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